

Zeta-Law Entropy, Gamma Curvature, and Laplace-Measure Applications

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Abstract

This paper develops a unified public theory built around Laplace-transform positivity. Zeta-law entropy is organized through logarithmic compression, gamma-ratio and polygamma inequalities are expressed by completely monotone kernels, and applications are obtained by transporting measure positivity through Stieltjes, Bernstein, stable-subordination, determinant, triangular, inverse-Laplace, derivative-chain, ratio-kernel, and moment-ratio mechanisms. The emphasis is on reusable transformations: source problems enter as fixed applications, while the proofs expose the structural tools that connect them.

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Keywords. Riemann zeta function; zeta probability law; entropy; Laplace transform; complete monotonicity; Bernstein functions; Stieltjes functions; gamma function; digamma function; Mills ratio; generalized Stieltjes functions.

1 Motivation and main results

Many classical identities for $\zeta(s)$ become more transparent after a probabilistic normalization. For $\beta > 1$, the weights $n^{-\beta}$ define a Gibbs law whose partition function is precisely $\zeta(\beta)$. The energy is $\log n$, the free energy is $\log \zeta(\beta)$, and ratios such as $\zeta(\beta+r)/\zeta(\beta)$ become moments. This viewpoint is elementary, but it packages several unrelated-looking zeta inequalities into a common calculus.

The same principle guides the special-function part of the paper. Complete monotonicity, Bernstein properties, Stieltjes representations, and endpoint obstructions are treated as transport questions: once a positive kernel, measure representation, determinant reduction, or rational certificate is identified, it can be reused across multiple source problems. The central theme is therefore not an individual inequality, but the mechanism that makes families of inequalities stable under analytic transformations.

Section 2 records solved external problems with stable application labels. The ledger is not a chronology; it is a public index of source questions and the results that resolve them. The body of the paper organizes those results by mechanism: zeta-law compression, gamma and polygamma curvature, Laplace and Stieltjes transport, determinant and moment-ratio reductions, endpoint obstructions, and explicit certificates. This organization lets the reader trace each application back to the tools that make it work.

2 Applications

The following list records the external source problems solved in the paper. Each row gives the fixed application identifier, the result in which it is proved, and the source citation.

1. **APP-0001.** Alzer-Kwong convexity and concavity problem. Solved by Corollary 8.4; source problem reference [30].
2. **APP-0002.** Nantomah zeta positivity problem. Solved by Counterexample 8.2; source problem reference [1].
3. **APP-0003.** Sroysang generalized Holder problem. Solved by Corollary 8.6; source problem reference [2].
4. **APP-0004.** Complete monotonicity of $(\log \Gamma(x) + \log \Gamma(1/x))''$. Solved by Corollary 7.6; source problem reference [31].
5. **APP-0005.** Exact inverse-tail floor formula at $s=7$. Solved by Corollary 4.2; source problem reference [6].

6. **APP-0006.** Exact inverse-tail floor formula at $s=8$. Solved by Corollary 4.3; source problem reference [6].
7. **APP-0007.** Concavity or complete monotonicity of the polygamma product P_0 . Solved by Corollary 3.2; source problem reference [31].
8. **APP-0008.** Higher-order monotonicity of polygamma products P_n . Solved by Counterexample 3.3; source problem reference [31].
9. **APP-0009.** Sharp reciprocal Gamma-product monotonicity threshold. Solved by Corollary 7.4; source problem reference [32].
10. **APP-0010.** Nielsen k -beta derivative-ratio monotonicity. Solved by Corollary 9.4; source problem reference [8].
11. **APP-0011.** Exact $n = 2$ beta-window endpoint. Solved by Corollary 10.4; source problem reference [31].
12. **APP-0012.** Baricz V_q strict q -log-convexity. Solved by Corollary 10.7; source problem reference [9].
13. **APP-0013.** Yin (p, k) -digamma sharp α -necessity. Solved by Corollary 10.9; source problem reference [10].
14. **APP-0014.** Ramanujan integral is Stieltjes and has a complete Bernstein primitive. Solved by Corollary 11.3; source problem reference [13].
15. **APP-0015.** Sibisi-Prabhakar measure is a stable-subordination push-forward. Solved by Corollary 11.5; source problem reference [14].
16. **APP-0016.** From's Mills-ratio bound extends to every derivative level. Solved by Corollary 11.7; source problem reference [15].
17. **APP-0017.** Baricz gamma-quotient Bernstein-for-all problem has a rational counterexample. Solved by Counterexample 11.9; source problem reference [16].
18. **APP-0018.** Qi-Guo tau threshold has a supremum-corrected exact value. Solved by Counterexample 11.11; source problem reference [17].
19. **APP-0019.** Wakrim W-symbol Bernstein range closes the beta gap. Solved by Counterexample 11.13; source problem reference [18].
20. **APP-0020.** Qi's degree-four conjecture for h_λ is refuted at the endpoint. Solved by Counterexample 11.15; source problem reference [19].
21. **APP-0021.** Szabo's cutoff problem has $\alpha_0 = 1$. Solved by Corollary 11.17; source problem reference [20].
22. **APP-0022.** Chen-Choi Gurland gamma-ratio conjecture is logarithmically completely monotone. Solved by Corollary 11.20; source problem reference [21].
23. **APP-0023.** Sokal generalized-Stieltjes lambda-derivatives have a triangular nonnegative compression. Solved by Corollary 11.23; source problem reference [22].

24. **APP-0024.** Simon gamma quotient is Bernstein. Solved by Corollary [11.26](#); source problem reference [\[23\]](#).
25. **APP-0025.** Du-Wang h_3 monotonicity is classified. Solved by Counterexample [11.29](#); source problem reference [\[24\]](#).
26. **APP-0026.** Baskakov even-power complete-monotonicity conjecture is false. Solved by Counterexample [11.31](#); source problem reference [\[25\]](#).
27. **APP-0027.** Ma-Weigert derivative regions form a descending chain. Solved by Corollary [11.33](#); source problem reference [\[11\]](#).
28. **APP-0028.** Qi-Agarwal/Yin divisor-polygamma parity is corrected. Solved by Counterexample [11.37](#); source problem reference [\[12\]](#).
29. **APP-0029.** Bulboaca-Zayed Gamma quotient has one critical point. Solved by Corollary [11.40](#); source problem reference [\[5\]](#).
30. **APP-0030.** Ramanujan integral Turan window is false. Solved by Counterexample [11.42](#); source problem reference [\[13\]](#).
31. **APP-0031.** Baricz gamma-quotient Bernstein test is false. Solved by Counterexample [14.1](#); source problem reference [\[16\]](#).
32. **APP-0032.** From's all- L Mills-ratio bound family is explicit. Solved by Corollary [14.3](#); source problem reference [\[15\]](#).
33. **APP-0033.** Ramanujan antiderivative is complete Bernstein. Solved by Corollary [14.5](#); source problem reference [\[13\]](#).
34. **APP-0034.** Bulboaca-Zayed gamma-quotient monotonicity solved. Solved by Corollary [14.6](#); source problem reference [\[5\]](#).
35. **APP-0035.** Keady self-bijection inverse-CM question is negative. Solved by Counterexample [14.9](#); source problem reference [\[33\]](#).
36. **APP-0036.** Baskakov even-line complete-monotonicity fails at $r = 8$. Solved by Counterexample [14.10](#); source problem reference [\[25\]](#).
37. **APP-0037.** Du-Wang h_3 is increasing exactly on $\frac{1}{2} \leq a \leq 1$. Solved by Counterexample [18.1](#); source problem reference [\[24\]](#).
38. **APP-0038.** Ma-Weigert derivative-sign regions form a chain. Solved by Corollary [16.1](#); source problem reference [\[11\]](#).
39. **APP-0039.** Ramanujan Turan-window complete-monotonicity interval is false. Solved by Counterexample [12.1](#); source problem reference [\[13\]](#).
40. **APP-0040.** Yang-Tian Bessel- W power-Bernstein conjecture is true. Solved by Corollary [12.3](#); source problem reference [\[34\]](#).
41. **APP-0041.** Qi h_λ degree-four complete-monotonicity conjecture is false. Solved by Counterexample [12.4](#); source problem reference [\[35\]](#).

42. **APP-0042.** Modified Bessel square-root log-concavity fails. Solved by Counterexample 15.1; source problem reference [36].
43. **APP-0043.** Riccati log-concavity inequality for I_ν is false. Solved by Counterexample 13.2; source problem reference [36].
44. **APP-0044.** Bessel ratio quadratic lower bound is false. Solved by Counterexample 13.3; source problem reference [36].
45. **APP-0045.** Three-regime Bessel log-concavity certificate route is refuted. Solved by Counterexample 13.4; source problem reference [36].
46. **APP-0046.** Tao–Sawin finite Weyl l1 minimum is exact. Solved by Corollary 17.2; source problem reference [37].
47. **APP-0047.** BPV noncentral chi-square HCM range has no positive noncentrality. Solved by Counterexample 19.2; source problem reference [38].
48. **APP-0048.** BMR tau-Gauss ordinary concavity is false. Solved by Counterexample 14.11; source problem reference [39].
49. **APP-0049.** Baricz arithmetic/arithmetic zero-balanced hypergeometric threshold. Solved by Corollary 14.12; source problem reference [40].
50. **APP-0050.** Stolarsky shifted power means with $p > 1$ are not Bernstein. Solved by Counterexample 14.14; source problem reference [41].

3 Zeta-law calculus

Definition 3.1 (Riemann zeta probability law). For $\beta > 1$, the Riemann zeta probability law is the probability measure ρ_β on $\mathbb{N}$ defined by $\rho_\beta(n) = n^{-\beta}/\zeta(\beta)$. Equivalently, with energy $E(n) = \log n$ and partition function $Z(\beta) = \zeta(\beta)$, one has $\rho_\beta(n) = e^{-\beta E(n)}/Z(\beta)$.

Definition 3.2 (Zeta free energy). The zeta free energy is $A(\beta) = \log \zeta(\beta)$ for $\beta > 1$. Its derivatives are interpreted with respect to the zeta probability law when the required differentiated Dirichlet series converge.

Proposition 3.1 (Free-energy derivatives). *For every $\beta > 1$,*

$$A'(\beta) = -\mathbb{E}_\beta[\log N], \quad A''(\beta) = \text{Var}_\beta(\log N). \quad (3.1)$$

Proof. Absolute convergence of the differentiated Dirichlet series on compact subintervals of $(1, \infty)$ permits termwise differentiation. Thus

$$A'(\beta) = \frac{\zeta'(\beta)}{\zeta(\beta)} = -\frac{\sum_{n \geq 1} (\log n) n^{-\beta}}{\zeta(\beta)} = -\mathbb{E}_\beta[\log N]. \quad (3.2)$$

Differentiating once more gives

$$A''(\beta) = \mathbb{E}_\beta[(\log N)^2] - \mathbb{E}_\beta[\log N]^2 = \text{Var}_\beta(\log N). \quad (3.3)$$

□

Qi, Lim, and Nantomah ask whether $P_0(x) = \psi(x)\psi(1/x)$ is concave, and more strongly whether $-P_0''$ is completely monotonic on $(0, \infty)$ [4]. The next theorem answers the stronger question.

Corollary 3.2 (APP-0007: Complete monotonicity of reciprocal digamma product curvature). *Let*

$$P_0(x) = \psi(x)\psi(1/x), \quad x > 0. \quad (3.4)$$

Then $-P_0''$ is strictly completely monotonic on $(0, \infty)$. In particular, P_0 is strictly concave on $(0, \infty)$.

Proof. We use Proposition 7.5 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let $F : (0, \infty) \rightarrow \mathbb{R}$ have a positive Laplace representation

$$F(x) = \int_{[0, \infty)} e^{-xt} d\mu(t),$$

where μ is a positive measure for which the differentiated integrals are finite on compact subintervals. Then F is completely monotone. If the relevant moments of μ are nonzero, the corresponding inequalities are strict.

Differentiating under the integral gives, for every $r \geq 0$,

$$(-1)^r F^{(r)}(x) = \int_{[0, \infty)} t^r e^{-xt} d\mu(t) \geq 0.$$

This is the Bernstein–Widder sign pattern. If $t^r \mu$ is nonzero, the integral is positive for every $x > 0$, giving strictness in that order. $\square$

Qi, Lim, and Nantomah also ask whether the higher-order products $P_n(x) = \psi^{(n)}(x)\psi^{(n)}(1/x)$, $n \in \mathbb{N}$, are convex, and more strongly whether P_n'' is completely monotonic. The following result refutes the stronger assertion. It does not settle the weaker convexity question.

Counterexample 3.3 (APP-0008: Counterexample to complete monotonicity of higher-order polygamma product curvature). *Let*

$$P_n(x) = \psi^{(n)}(x)\psi^{(n)}(1/x), \quad n \in \mathbb{N}, \quad x > 0. \quad (3.5)$$

Then P_n'' is not completely monotonic for all $n \in \mathbb{N}$. More precisely, for every $n \geq 29$,

$$P_n'''(2) > 0, \quad (3.6)$$

and, already at lower order,

$$P_7^{(6)}(3) < 0. \quad (3.7)$$

Proof. We use Proposition 8.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The staged APP-0008 proof gives a rationally certified derivative witness $P_n'''(2) > 0$ for every $n \geq 29$. If P_n'' were completely monotone, then its first derivative would have to be nonpositive everywhere. The positive value at $x = 2$ is therefore a pointwise obstruction certificate. By the pointwise obstruction principle, this witness refutes the source’s stronger universal complete-monotonicity assertion. $\square$

Proposition 3.4 (Zeta-law calculus). *Let $\beta > 1$, and let N have law ρ_β . For every $r \geq 0$,*

$$\mathbb{E}_\beta[N^{-r}] = \frac{\zeta(\beta + r)}{\zeta(\beta)}. \quad (3.8)$$

Moreover, for $\alpha, \beta > 1$,

$$D(\rho_\alpha \| \rho_\beta) = (\beta - \alpha) \mathbb{E}_\alpha[\log N] + \log \zeta(\beta) - \log \zeta(\alpha). \quad (3.9)$$

Proof. The moment identity follows by direct summation:

$$\mathbb{E}_\beta[N^{-r}] = \sum_{n=1}^{\infty} n^{-r} \frac{n^{-\beta}}{\zeta(\beta)} = \frac{\zeta(\beta + r)}{\zeta(\beta)}. \quad (3.10)$$

For relative entropy, substitute Definition 3.1:

$$D(\rho_\alpha \| \rho_\beta) = \sum_{n=1}^{\infty} \rho_\alpha(n) ((\beta - \alpha) \log n + \log \zeta(\beta) - \log \zeta(\alpha)), \quad (3.11)$$

which is (3.9). □

4 Tail zeta laws and inverse-tail floors

Exact formulas for integer parts of reciprocal zeta tails have been studied for small integer exponents. Known formulas include $s = 2, 3, 4, 5$, with a sufficiently-large- n formula for $s = 6$; Kim and Song record that no analogous formula was known for integer $s > 6$ at the time of their work [6]. The 2024 asymptotic inverse-tail results of Pan and Wu provide related context [7]. The next two theorems give exact formulas at $s = 7$ and $s = 8$. The first uses a single rational telescoping enclosure, while the second needs adjacent corrected asymptotic truncations.

Definition 4.1 (Tail zeta partition function). For $s > 1$ and $n \in \mathbb{N}$, the tail zeta partition function is $\zeta_n(s) = \sum_{k=n}^{\infty} k^{-s}$. The associated tail law is $\rho_{s,n}(k) = k^{-s} / \zeta_n(s)$ for $k \geq n$.

Proposition 4.1 (Zeta tail inverse asymptotic telescoping template). *There is a reusable Euler–Maclaurin inverse-tail approximant and telescoping-sign certificate template that, for at least the next integer exponent beyond $s = 8$, constructs rational functions $A_s(n) \leq \zeta_n(s)^{-1} \leq B_s(n)$ with exact eventually-positive denominator-cleared residuals.*

Proof. For integer $s > 1$, the Hurwitz-tail Euler–Maclaurin expansion has the form

$$\zeta_n(s) = n^{1-s} \left[\frac{1}{s-1} + \frac{z}{2} + \sum_{r \geq 1} \frac{B_{2r}}{(2r)!} (s)_{2r-1} z^{2r} \right], \quad z = \frac{1}{n}.$$

Formally inverting the bracket gives

$$\zeta_n(s)^{-1} \sim n^{s-1} G_s(z).$$

The replay script computes G_s symbolically and verifies two things:

1. For $s = 8$, the coefficients through z^{15} reproduce the staged A_8 and $B_8 - A_8$ coefficients. 2. For $s = 9$, truncating $n^8 G_9(1/n)$ through z^{14} gives an A_9 , and adding the z^{15} term gives B_9 , with exact telescoping residuals of the expected signs.

The generated $s = 9$ lower approximant is

$$A_9(n) = 8n^8 - 32n^7 + 80n^6 - 128n^5 + 120n^4 - 64n^3 + \frac{624}{7}n^2 - \frac{512}{7}n - \frac{3324}{7} \\ + \frac{90304}{21n^2} + \frac{90304}{21n^3} - \frac{10280944}{245n^4} - \frac{64846304}{735n^5} + \frac{109940976}{245n^6}.$$

The upper approximant is

$$B_9(n) = A_9(n) + \frac{384388288}{245n^7}.$$

Let $R_A(n) = 1/A_9(n)$ and $R_B(n) = 1/B_9(n)$. Exact denominator clearing gives:

$$R_A(k) - R_A(k+1) - \frac{1}{k^9} > 0 \quad (k \geq 5),$$

because the numerator, after substituting $k = m + 5$, has all coefficients positive. Similarly,

$$\frac{1}{k^9} - (R_B(k) - R_B(k+1)) > 0 \quad (k \geq 1),$$

because the denominator-cleared numerator, after substituting $k = m + 1$, has all coefficients positive.

Therefore, for $n \geq 5$,

$$A_9(n) < \zeta_n(9)^{-1} < B_9(n).$$

This proves the inverse-window part of the reusable template and promotes the Zeta tail inverse asymptotic telescoping template.

The replay script also verifies the no-integer gap.

Write $P_9(n)$ for the polynomial part of $A_9(n)$:

$$P_9(n) = 8n^8 - 32n^7 + 80n^6 - 128n^5 + 120n^4 - 64n^3 + \frac{624}{7}n^2 - \frac{512}{7}n - \frac{3324}{7}.$$

Modulo 7, the fractional numerator of $P_9(n)$ is

$$n^2 - n + 1 \pmod{7},$$

so the fractional part of $P_9(n)$ is one of

$$0, \quad \frac{1}{7}, \quad \frac{3}{7}, \quad \frac{6}{7}.$$

The correction $A_9(n) - P_9(n)$ is positive for $n \geq 9$; after denominator clearing and substituting $n = m + 9$, the numerator has all coefficients positive. For $n \geq 174$, the upper correction is bounded by

$$B_9(n) - P_9(n) < \frac{90304}{21n^2} + \frac{90304}{21n^3} + \frac{109940976}{245n^6} + \frac{384388288}{245n^7} < \frac{1}{7}.$$

Therefore $A_9(n)$ and $B_9(n)$ cannot straddle an integer for $n \geq 174$. Exact rational arithmetic verifies

$$\lfloor A_9(n) \rfloor = \lfloor B_9(n) \rfloor \quad (9 \leq n \leq 173).$$

Consequently,

$$\lfloor \zeta_n(9)^{-1} \rfloor = \lfloor A_9(n) \rfloor \quad (n \geq 9).$$

For $1 \leq n \leq 8$, the exact values are

n	1	2	3	4	5	6	7	8
$\lfloor \zeta_n(9)^{-1} \rfloor$	0	497	18093	224086	1543530	7360226	27295287	84349541.

For $n = 1$, this follows from $\zeta(9) > 1$. For $2 \leq n \leq 8$, the replay script uses exact rational partial sums through $N = 100$ and the integral tail bound

$$\sum_{k=N+1}^{\infty} k^{-9} < \frac{1}{8N^8}$$

to prove

$$\frac{1}{M_n + 1} < \zeta_n(9) < \frac{1}{M_n}.$$

This proves the Zeta tail s9 reusable certificate, the Zeta tail floor gap template, and hence the Zeta tail floor next case. $\square$

Corollary 4.2 (APP-0005: Exact inverse-tail floor formula at s=7). *Let*

$$T_7(n) = \zeta_n(7) = \sum_{k=n}^{\infty} \frac{1}{k^7}. \quad (4.1)$$

Define

$$Q(n) = 120n^6 - 360n^5 + 660n^4 - 720n^3 + 354n^2 - 54n + 375, \quad P(n) = \frac{Q(n)}{20}. \quad (4.2)$$

Then, for every $n \geq 28$,

$$\lfloor T_7(n)^{-1} \rfloor = \lfloor P(n) \rfloor. \quad (4.3)$$

For $1 \leq n \leq 27$, the exact values are

n	$\lfloor T_7(n)^{-1} \rfloor$	n	$\lfloor T_7(n)^{-1} \rfloor$	n	$\lfloor T_7(n)^{-1} \rfloor$
1	0	10	4495760	19	241765529
2	119	11	8167814	20	331399044
3	1862	12	14061542	21	447175152
4	12573	13	23143975	22	594869693
5	54069	14	36668777	23	781167220
6	175597	15	56228085	24	1013751716
7	471118	16	83808666	25	1301401638
8	1100904	17	121852400	26	1654089273
9	2317459	18	173321080	27	2083084422

Proof. We use Proposition 4.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The proof instantiates the inverse-tail asymptotic telescoping template with $s = 7$, the partition function T_7 , and the specific approximation data Q, P . The template gives rational upper and lower reciprocal envelopes $L_n \leq T_7(n)^{-1} \leq U_n$ with $U_n - L_n < 1$; by direct computation $\lfloor L_n \rfloor = \lfloor U_n \rfloor = P(n)$ for $n \geq 28$. The finite checked range $1 \leq n \leq 27$ is recorded explicitly in the same staged source, so the floor identity holds for all $n \geq 1$. $\square$

Corollary 4.3 (APP-0006: Exact inverse-tail floor formula at $s=8$). *Let*

$$T_8(n) = \zeta_n(8) = \sum_{k=n}^{\infty} \frac{1}{k^8}. \quad (4.5)$$

Define

$$\begin{aligned} A_8(n) = & 7n^7 - \frac{49}{2}n^6 + \frac{637}{12}n^5 - \frac{1715}{24}n^4 + \frac{7399}{144}n^3 - \frac{1715}{96}n^2 + \frac{69685}{1728}n - \frac{65611}{3456} \\ & - \frac{23764069}{103680n} - \frac{23764069}{207360n^2} + \frac{2102131409}{1244160n^3} + \frac{2149659547}{829440n^4} \\ & - \frac{203290800013}{14929920n^5} - \frac{1146003910541}{29859840n^6} + \frac{21587525169497}{179159040n^7}. \end{aligned} \quad (4.6)$$

Then, for every $n \geq 6$,

$$\lfloor T_8(n)^{-1} \rfloor = \lfloor A_8(n) \rfloor. \quad (4.7)$$

For $1 \leq n \leq 5$, the exact values are

$$\begin{array}{c|ccccc} n & 1 & 2 & 3 & 4 & 5 \\ \hline \lfloor T_8(n)^{-1} \rfloor & 0 & 245 & 5844 & 53503 & 291407 \end{array}. \quad (4.8)$$

Proof. We use Proposition 4.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The proof instantiates the inverse-tail asymptotic telescoping template at $s = 8$ with the explicit adjacent rational approximants A_8 and B_8 . The template provides the recursive telescoping bounds that force $\lfloor T_8(n)^{-1} \rfloor = \lfloor A_8(n) \rfloor = \lfloor B_8(n) \rfloor$, and the gap inequality $B_8 - A_8 < 1$. The finite seed table 0, 245, 5844, 53503, 291407 is handled separately for $1 \leq n \leq 5$, yielding the global floor statement for every $n \geq 1$. $\square$

5 Modular successor entropy

Definition 5.1 (Modular residue distribution and successor entropy). For a modulus q , the modular residue distribution induced by ρ_β is $\rho_{\beta,q}(a) = \sum_{n \equiv a \pmod{q}} \rho_\beta(n)$. The successor entropy is the relative-entropy expression comparing consecutive zeta-law weights, namely $\sum_n \rho_\beta(n) \log(\rho_\beta(n)/\rho_\beta(n+1))$ whenever the displayed series is finite.

Theorem 5.1 (Zeta-law successor entropy and modular resolution). *For every $\beta > 1$, define*

$$\Sigma(\beta) = \sum_{n=1}^{\infty} \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)}. \quad (5.1)$$

Then

$$\Sigma(\beta) = \frac{\beta}{\zeta(\beta)} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \zeta(\beta + k), \quad (5.2)$$

and

$$\Sigma(\beta) = \sup_{q \geq 2} B_q(\beta) = \lim_{q \rightarrow \infty} B_q(\beta). \quad (5.3)$$

Proof. Since

$$\frac{\rho_\beta(n)}{\rho_\beta(n+1)} = \left(1 + \frac{1}{n}\right)^\beta, \quad (5.4)$$

we have

$$\Sigma(\beta) = \frac{\beta}{\zeta(\beta)} \sum_{n=1}^{\infty} n^{-\beta} \log \left(1 + \frac{1}{n}\right). \quad (5.5)$$

Using

$$\log(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k \quad (0 < x \leq 1) \quad (5.6)$$

with Abel limiting at $x = 1$, and summing against $n^{-\beta}$, gives (5.2).

For the modular identity, fix $q \geq 2$. The log-sum inequality gives, for each residue class a ,

$$\sum_{\substack{n \geq 1 \\ n \equiv a \pmod{q}}} \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)} \geq \mu_{q,\beta}(a) \log \frac{\mu_{q,\beta}(a)}{\sum_{n \equiv a \pmod{q}} \rho_\beta(n+1)}. \quad (5.7)$$

For $a \neq 0$, the denominator is $\mu_{q,\beta}(a+1)$. For $a = 0$, it is $\mu_{q,\beta}(1) - \rho_\beta(1) \leq \mu_{q,\beta}(1)$. Summing (5.7) over a gives

$$\Sigma(\beta) \geq B_q(\beta) \quad (q \geq 2). \quad (5.8)$$

Conversely, fix $M \in \mathbb{N}$. If $q > M + 1$, then for $1 \leq a \leq M + 1$,

$$\mu_{q,\beta}(a) = \rho_\beta(a) + O_{\beta,M}(q^{-\beta}), \quad (5.9)$$

and $\mu_{q,\beta}(0) = q^{-\beta}$. All cyclic terms in $B_q(\beta)$ are nonnegative except possibly the boundary term, which satisfies

$$\mu_{q,\beta}(0) \log \frac{\mu_{q,\beta}(0)}{\mu_{q,\beta}(1)} = O_\beta(q^{-\beta} \log q). \quad (5.10)$$

Therefore

$$\liminf_{q \rightarrow \infty} B_q(\beta) \geq \sum_{n=1}^M \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)}. \quad (5.11)$$

Letting $M \rightarrow \infty$ gives $\liminf_q B_q(\beta) \geq \Sigma(\beta)$, which together with (5.8) proves (5.3). $\square$

Corollary 5.2 (Prime-modulus Dirichlet L-resolution). *Let p be prime, let χ range over Dirichlet characters modulo p , and define*

$$S_a(p, \beta) = \sum_{\chi \bmod p} \overline{\chi(a)} L(\beta, \chi) \quad (1 \leq a \leq p-1). \quad (5.12)$$

Then

$$\mu_{p,\beta}(a) = \frac{S_a(p, \beta)}{(p-1)\zeta(\beta)} \quad (1 \leq a \leq p-1), \quad \mu_{p,\beta}(0) = p^{-\beta}. \quad (5.13)$$

Consequently the prime-modulus functionals obtained by substituting (5.13) into $B_p(\beta)$ satisfy

$$\Sigma(\beta) = \lim_{\substack{p \rightarrow \infty \\ p \text{ prime}}} F_p(\zeta(\beta), \{L(\beta, \chi)\}_{\chi \bmod p}) = \sup_{p \text{ prime}} F_p(\zeta(\beta), \{L(\beta, \chi)\}_{\chi \bmod p}). \quad (5.14)$$

Proof. For $a \not\equiv 0 \pmod{p}$, character orthogonality gives

$$\frac{1}{p-1} \sum_{\chi \bmod p} \overline{\chi(a)} \chi(n) = \begin{cases} 1, & n \equiv a \pmod{p}, \\ 0, & n \not\equiv a \pmod{p}. \end{cases} \quad (5.15)$$

Therefore

$$\sum_{\substack{n \geq 1 \\ n \equiv a \pmod{p}}} n^{-\beta} = \frac{1}{p-1} \sum_{\chi \bmod p} \overline{\chi(a)} L(\beta, \chi). \quad (5.16)$$

Dividing by $\zeta(\beta)$ gives the nonzero residue formula, while

$$\mu_{p,\beta}(0) = \frac{1}{\zeta(\beta)} \sum_{m=1}^{\infty} (mp)^{-\beta} = p^{-\beta}. \quad (5.17)$$

The functional identity follows by substituting these residue formulas into $B_p(\beta)$ and applying Theorem 5.1 along the prime sequence. $\square$

6 Euler-score identities

Theorem 6.1 (Euler-score decomposition). *For every $\beta > 1$,*

$$-(\log \zeta)'(\beta) = \mathbb{E}_{\beta}[\log N] = \sum_{d=1}^{\infty} \Lambda(d) \mu_{d,\beta}(0) = \sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^{\beta}}. \quad (6.1)$$

Equivalently,

$$-\frac{\zeta'(\beta)}{\zeta(\beta)} = \sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^{\beta}}. \quad (6.2)$$

Proof. The von Mangoldt identity is

$$\log n = \sum_{d|n} \Lambda(d). \quad (6.3)$$

By nonnegative summation,

$$\mathbb{E}_{\beta}[\log N] = \sum_{n \geq 1} \rho_{\beta}(n) \sum_{d|n} \Lambda(d) = \sum_{d \geq 1} \Lambda(d) \mathbb{P}_{\beta}(d | N). \quad (6.4)$$

The divisibility probability is

$$\mathbb{P}_{\beta}(d | N) = \frac{1}{\zeta(\beta)} \sum_{m \geq 1} (dm)^{-\beta} = d^{-\beta}. \quad (6.5)$$

Substitution into (6.4) proves (6.1); (6.2) follows from (3.2). $\square$

Proposition 6.2 (Finite Euler-score identity). *For every $A \subseteq \{1, \dots, N\}$,*

$$\sum_{n \in A} \log n = \sum_{d \leq N} \Lambda(d) \sum_{m \leq N/d} \mathbf{1}_A(md). \quad (6.6)$$

Proof. Reverse the order of summation:

$$\sum_{d \leq N} \Lambda(d) \sum_{m \leq N/d} \mathbf{1}_A(md) = \sum_{n \in A} \sum_{d|n} \Lambda(d) = \sum_{n \in A} \log n. \quad (6.7)$$

$\square$

7 Curvature and Mellin-Planck tools

Lemma 7.1 (Uniform positive-axis curvature bound). *For every $s \geq 3$,*

$$(\log \zeta)''(s) = \text{Var}_s(\log N) < \frac{1}{3}. \quad (7.1)$$

Proof. By Proposition 3.1,

$$(\log \zeta)''(s) = \frac{\zeta''(s)}{\zeta(s)} - \left(\frac{\zeta'(s)}{\zeta(s)} \right)^2 \leq \frac{\zeta''(s)}{\zeta(s)} \leq \zeta''(s). \quad (7.2)$$

For $s \geq 3$,

$$\zeta''(s) = \sum_{m=2}^{\infty} \frac{(\log m)^2}{m^s} \leq \sum_{m=2}^{\infty} \frac{(\log m)^2}{m^3}. \quad (7.3)$$

The function $x \mapsto (\log x)^2 x^{-3}$ is decreasing for $x \geq 2$, so

$$\sum_{m=2}^{\infty} \frac{(\log m)^2}{m^3} \leq \frac{(\log 2)^2}{8} + \int_2^{\infty} \frac{(\log x)^2}{x^3} dx. \quad (7.4)$$

Since

$$\int_2^{\infty} \frac{(\log x)^2}{x^3} dx = \frac{(\log 2)^2 + \log 2 + \frac{1}{2}}{8}, \quad (7.5)$$

we obtain

$$(\log \zeta)''(s) \leq \frac{2(\log 2)^2 + \log 2 + \frac{1}{2}}{8} < \frac{1}{3}. \quad (7.6)$$

□

Definition 7.1 (Mellin-Planck partition function). A Mellin-Planck partition function is a Mellin moment $M(s) = \int_0^{\infty} t^s d\mu(t)$ of a positive measure, on an interval where the integral is finite. The tilted law ν_s has density proportional to $t^s d\mu(t)$, so logarithmic derivatives of M are cumulants of $\log t$ under ν_s .

Proposition 7.2 (Mellin-Planck log-convexity). *For $s > 1$,*

$$\frac{d^2}{ds^2} \log M(s) = \text{Var}_{\nu_s}(\log t) \geq 0. \quad (7.7)$$

Proof. Differentiate the logarithm of the integral in Definition 7.1 under the integral sign. The first derivative is the ν_s -expectation of $\log t$, and the second derivative is its variance:

$$\frac{d^2}{ds^2} \log M(s) = \mathbb{E}_{\nu_s}[(\log t)^2] - \mathbb{E}_{\nu_s}[\log t]^2. \quad (7.8)$$

□

Bulboacă and Zayed proved several monotonicity criteria for functions of the form $u(s)/(\Gamma(s + \rho)\Gamma(s))$ and asked for best possible positive values of ρ in their examples [5]. The next application gives the sharp endpoint in the base reciprocal Gamma-product case and the corresponding endpoint criterion for monotone logarithmic derivatives.

Proposition 7.3 (Endpoint log derivative monotonicity principle). *Let $f > 0$ be differentiable on $[1, \infty)$. If $(\log f)'$ is monotone on $[1, \infty)$, then the global monotonicity of f on $[1, \infty)$ is decided by the endpoint sign of $(\log f)'(1)$, with strictness away from the endpoint when the logarithmic derivative is strictly monotone.*

Proof. If J is nonincreasing and g is strictly increasing, then $J - g$ is strictly decreasing on the half-line, so the sign of $J - g$ everywhere is forced by its endpoint value. In the Gamma-product case, the logarithmic derivative of W_ρ is $\psi(s + \rho) + \psi(s)$, which is strictly increasing because $\psi' > 0$. Its endpoint value at $s = 1$ is $\psi(1 + \rho) - \gamma$. Therefore W_ρ is strictly increasing, equivalently $1/W_\rho$ is strictly decreasing, exactly when $\psi(1 + \rho) \geq \gamma$. The same endpoint comparison gives the stated extension with a differentiable factor u whose logarithmic derivative is nonincreasing. $\square$

Corollary 7.4 (APP-0009: Sharp reciprocal Gamma-product monotonicity threshold). *Let $\psi = \Gamma'/\Gamma$, let γ be Euler's constant, and let $\rho_* > 0$ be the unique solution of*

$$\psi(1 + \rho_*) = \gamma. \quad (7.9)$$

For $\rho > -1$, define

$$\varphi_\rho(s) = \frac{1}{\Gamma(s + \rho)\Gamma(s)}, \quad s \geq 1. \quad (7.10)$$

Then φ_ρ is strictly decreasing on $[1, \infty)$ if and only if

$$\rho \geq \rho_*. \quad (7.11)$$

Equivalently,

$$W_\rho(s) = \Gamma(s + \rho)\Gamma(s) \quad (7.12)$$

is strictly increasing on $[1, \infty)$ if and only if (7.11) holds. Numerically,

$$\rho_* = 1.2583969670859318174106234224981693941 \dots \quad (7.13)$$

More generally, let $u : [1, \infty) \rightarrow (0, \infty)$ be differentiable and suppose

$$J(s) = \frac{u'(s)}{u(s)} \quad (7.14)$$

is nonincreasing on $[1, \infty)$. Then

$$\Phi_{\rho, u}(s) = \frac{u(s)}{\Gamma(s + \rho)\Gamma(s)} \quad (7.15)$$

is strictly decreasing on $[1, \infty)$ if and only if

$$\psi(1 + \rho) \geq \gamma + J(1). \quad (7.16)$$

Proof. We use Proposition 7.3 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

If J is nonincreasing and g is strictly increasing, then $J - g$ is strictly decreasing on the half-line, so the sign of $J - g$ everywhere is forced by its endpoint value. In the Gamma-product case, the logarithmic derivative of W_ρ is $\psi(s + \rho) + \psi(s)$, which is strictly increasing because $\psi' > 0$. Its endpoint value at $s = 1$ is $\psi(1 + \rho) - \gamma$. Therefore W_ρ is strictly increasing, equivalently $1/W_\rho$ is strictly decreasing, exactly when $\psi(1 + \rho) \geq \gamma$. The same endpoint comparison gives the stated extension with a differentiable factor u whose logarithmic derivative is nonincreasing. $\square$

Qi, Lim, and Nantomah asked whether the second derivative of $\log \Gamma(x) + \log \Gamma(1/x)$ is completely monotonic on $(0, \infty)$ [4]. The next theorem answers this in the affirmative by a reciprocal-Weierstrass Laplace kernel.

Definition 7.2 (Complete monotonicity, Bernstein, and Stieltjes language). On an interval, a C^∞ function f is completely monotone if $(-1)^m f^{(m)} \geq 0$ for every $m \geq 0$, and is strictly completely monotone when the relevant inequalities are strict. A positive function is logarithmically completely monotone if its logarithm has the corresponding alternating derivative signs. A nonnegative C^1 function is a Bernstein function when f' is completely monotone. A complete Bernstein function is a Bernstein function with a completely monotone Levy density in its standard representation. A Stieltjes function has a representation by kernels $(x+t)^{-1}$ against a positive measure, and a generalized Stieltjes function uses the higher-order kernels $(x+t)^{-\lambda}$.

Definition 7.3 (Laplace kernels and tilted moment ratios). A positive Laplace representation writes $F(x) = \int_0^\infty e^{-xt} d\mu(t)$ with μ a positive measure. Its tilted moments are $a_j(x) = \int_0^\infty t^j e^{-xt} d\mu(t)$, whenever finite. A moment-ratio expression is a quotient built from these tilted moments, such as a_{j+1}/a_j or $a_{j+1}/(a_j a_{j+2})$.

Proposition 7.5 (Positive Laplace kernel complete monotonicity principle). *If $F(x) = \int_0^\infty e^{-xt} d\mu(t)$ on $(0, \infty)$ for a positive measure μ , then F is completely monotone. If the representing measure is nonzero in every relevant derivative order, the complete monotonicity is strict.*

Proof. Let $F : (0, \infty) \rightarrow \mathbb{R}$ have a positive Laplace representation

$$F(x) = \int_{[0, \infty)} e^{-xt} d\mu(t),$$

where μ is a positive measure for which the differentiated integrals are finite on compact subintervals. Then F is completely monotone. If the relevant moments of μ are nonzero, the corresponding inequalities are strict.

Differentiating under the integral gives, for every $r \geq 0$,

$$(-1)^r F^{(r)}(x) = \int_{[0, \infty)} t^r e^{-xt} d\mu(t) \geq 0.$$

This is the Bernstein–Widder sign pattern. If $t^r \mu$ is nonzero, the integral is positive for every $x > 0$, giving strictness in that order. $\square$

Corollary 7.6 (APP-0004: Complete monotonicity of reciprocal-Gamma curvature). *Let*

$$H(x) = \log \Gamma(x) + \log \Gamma(1/x), \quad x > 0. \quad (7.17)$$

Then H'' is strictly completely monotonic on $(0, \infty)$; that is,

$$(-1)^m \frac{d^m}{dx^m} H''(x) > 0 \quad (m \geq 0, x > 0). \quad (7.18)$$

Proof. We use Proposition 7.5 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The staged APP-0004 proof derives a positive reciprocal-Weierstrass/Laplace-kernel representation for H'' . By the positive Laplace-kernel complete-monotonicity principle, every derivative has the alternating sign pattern required for complete monotonicity. Thus the source problem is solved affirmatively. $\square$

8 External zeta inequality proofs

Definition 8.1 (Endpoint and pointwise obstruction certificates). A pointwise obstruction certificate is an explicit admissible parameter value at which a proposed universal sign, monotonicity, convexity, or complete-monotonicity claim fails after reduction to a pointwise inequality. An endpoint obstruction is the same kind of certificate obtained from a boundary or asymptotic regime of the source domain.

Proposition 8.1 (Pointwise obstruction certificate principle). *If a universal sign, convexity, monotonicity, or complete-monotonicity source claim has been reduced to a pointwise inequality on a domain, then one certified admissible point with the opposite sign refutes the universal claim.*

Proof. Let D be a domain and let a universal claim require a sign condition $F(x) \geq 0$ for all $x \in D$. More generally, allow any equivalent sign condition obtained by multiplying by a known positive factor or by moving all terms to one side. If there exists a certified admissible point $x_0 \in D$ with the opposite sign, then the universal claim is false.

The proof is the elementary logic of a universal quantifier. The point x_0 satisfies the hypotheses of the universal statement, while the certified sign computation violates the asserted conclusion. Multiplication by a positive factor preserves signs, so any equivalent reduced inequality has the same counterexample. Therefore the original universal sign, convexity, monotonicity, or complete-monotonicity claim is refuted. $\square$

Counterexample 8.2 (APP-0002: Affirmative solution of Nantomah zeta positivity problem). For every $n \in \mathbb{N}$,

$$(n+2)\zeta(n+1)\zeta(n+3) - (n+1)\zeta(n+2)^2 - \zeta(n+1)\zeta(n+2) > 0. \quad (8.1)$$

Proof. We use Proposition 8.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$g_\nu(u) = \log(\sqrt{u} I_\nu(u)), \quad r_\nu(u) = \frac{I'_\nu(u)}{I_\nu(u)}.$$

Then

$$g''_\nu(u) = r'_\nu(u) - \frac{1}{2u^2}.$$

The modified Bessel equation

$$u^2 I''_\nu(u) + u I'_\nu(u) - (u^2 + \nu^2) I_\nu(u) = 0$$

gives

$$\frac{I''_\nu(u)}{I_\nu(u)} = 1 + \frac{\nu^2}{u^2} - \frac{r_\nu(u)}{u}.$$

Since

$$r'_\nu(u) = \frac{I''_\nu(u)}{I_\nu(u)} - r_\nu(u)^2,$$

we get

$$g''_\nu(u) = 1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2.$$

Thus the Bessel I Riccati log concavity inequality is exactly the strict log-concavity condition $g''_\nu(u) < 0$.

For $\nu = 0$,

$$r_0(u) = \frac{I'_0(u)}{I_0(u)} = \frac{I_1(u)}{I_0(u)}.$$

At $u = 10$, the Riccati expression is

$$g_0''(10) = \frac{199}{200} - \frac{r_0(10)}{10} - r_0(10)^2.$$

It is enough to prove

$$r_0(10) < \frac{9487}{10000},$$

because

$$\frac{199}{200} - \frac{1}{10} \frac{9487}{10000} - \left(\frac{9487}{10000} \right)^2 = \frac{9831}{100000000} > 0.$$

$$I_0(10) = \sum_{k=0}^{\infty} \frac{25^k}{(k!)^2}, \quad I_1(10) = \sum_{k=0}^{\infty} \frac{5 \cdot 25^k}{k!(k+1)!}.$$

With $N = 20$, set

$$L_0 = \sum_{k=0}^{20} \frac{25^k}{(k!)^2} < I_0(10).$$

For the I_1 tail after $N = 20$, the term ratio is bounded by

$$q = \frac{25}{22 \cdot 23} < 1,$$

so the script computes an exact rational upper bound $U_1 > I_1(10)$ and verifies

$$10000 U_1 < 9487 L_0.$$

Therefore $I_1(10)/I_0(10) < 9487/10000$, and hence

$$g_0''(10) > 0.$$

This proves the Bessel- I log-concavity counterexample to T - I sqrt-log concavity for $\nu > 0$, and the counterexample to T - I Riccati log-concavity inequality. $\square$

Proposition 8.3 (Zeta free energy uniform curvature bound). *For every $s \geq 3$, the zeta free-energy curvature satisfies $(\log \zeta)''(s) = \text{Var}_s(\log N) < 1/3$.*

Proof. The theory proves the zeta-law free-energy curvature bound $(\log \zeta)''(s) = \text{Var}_s(\log N) < 1/3$ for $s \geq 3$. On the negative axis, the functional equation rewrites $1/\zeta(-u)$ as a signed copy of a positive function $G(u)$. Its logarithmic second derivative is bounded below by $\pi^2/4 - 1/2 - 1/3 > 0$, using the curvature bound, the elementary trigamma estimate, and $\csc^2(\pi u/2) \geq 1$. Hence $G'''(u) > 0$. The sign of $\sin(\pi u/2)$ then transfers this positive convexity to alternating convexity and concavity intervals for $F(x) = 1/\zeta(x)$. $\square$

Corollary 8.4 (APP-0001: Alzer-Kwong convexity and concavity pattern for reciprocal zeta). *Let $F(x) = 1/\zeta(x)$. For every integer $n \geq 1$,*

$$F''(x) > 0 \quad \text{on } (-4n, -4n+2), \tag{8.2}$$

and

$$F''(x) < 0 \quad \text{on } (-4n-2, -4n). \tag{8.3}$$

Proof. We use Proposition 8.3 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The theory proves the zeta-law free-energy curvature bound $(\log \zeta)''(s) = \text{Var}_s(\log N) < 1/3$ for $s \geq 3$. On the negative axis, the functional equation rewrites $1/\zeta(-u)$ as a signed copy of a positive function $G(u)$. Its logarithmic second derivative is bounded below by $\pi^2/4 - 1/2 - 1/3 > 0$, using the curvature bound, the elementary trigamma estimate, and $\csc^2(\pi u/2) \geq 1$. Hence $G''(u) > 0$. The sign of $\sin(\pi u/2)$ then transfers this positive convexity to alternating convexity and concavity intervals for $F(x) = 1/\zeta(x)$. $\square$

Proposition 8.5 (Positive Mellin moment logconvexity principle). *If $M(s) = \int_0^\infty t^s d\mu(t)$ is finite on an interval for a positive measure μ , then $s \mapsto M(s)$ is log-convex on that interval. Equivalently, weighted Holder inequalities hold for positive Mellin moments.*

Proof. Let μ be a positive measure and let

$$M(s) = \int_0^\infty t^s d\mu(t)$$

be finite on an interval. Then $s \mapsto M(s)$ is log-convex on that interval. Equivalently, for $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, and admissible s_i ,

$$M\left(\sum_i \lambda_i s_i\right) \leq \prod_i M(s_i)^{\lambda_i}.$$

Indeed,

$$t^{\sum_i \lambda_i s_i} = \prod_i (t^{s_i})^{\lambda_i}.$$

Holder's inequality applied to the positive functions t^{s_i} with weights λ_i gives the displayed inequality. Taking logarithms gives convexity of $\log M$. $\square$

Corollary 8.6 (APP-0003: Generalized Holder inequality for Gamma zeta). *Let $m \geq 2$, $r \geq 1$, $x_1, \dots, x_m > 1$, and $p_1, \dots, p_m > 1$ satisfy*

$$\sum_{i=1}^m \frac{1}{p_i} = \frac{1}{r}. \quad (8.4)$$

Define

$$T = r \sum_{i=1}^m \frac{x_i}{p_i}. \quad (8.5)$$

Then $T > 1$ and

$$\zeta(T) \leq \frac{\prod_{i=1}^m (\Gamma(x_i) \zeta(x_i))^{r/p_i}}{\Gamma(T)}. \quad (8.6)$$

Proof. We use Proposition 8.5 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let μ be a positive measure and let

$$M(s) = \int_0^\infty t^s d\mu(t)$$

be finite on an interval. Then $s \mapsto M(s)$ is log-convex on that interval. Equivalently, for $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, and admissible s_i ,

$$M\left(\sum_i \lambda_i s_i\right) \leq \prod_i M(s_i)^{\lambda_i}.$$

Indeed,

$$t^{\sum_i \lambda_i s_i} = \prod_i (t^{s_i})^{\lambda_i}.$$

Holder's inequality applied to the positive functions t^{s_i} with weights λ_i gives the displayed inequality. Taking logarithms gives convexity of $\log M$. $\square$

Remark 8.7 (Free-energy interpretation of the Sroysang inequality). The same result follows from log-convexity of $M(s) = \Gamma(s)\zeta(s)$. With $\lambda_i = r/p_i$ and $T = \sum_i \lambda_i x_i$, Jensen's inequality for $\log M$ gives

$$M(T) \leq \prod_{i=1}^m M(x_i)^{\lambda_i} = \prod_{i=1}^m (\Gamma(x_i)\zeta(x_i))^{r/p_i}, \quad (8.7)$$

which is equivalent to (8.6).

9 Laplace moment-ratio application

The complete-monotonicity proofs above repeatedly convert special-function curvature questions into positivity statements for Laplace kernels. The next lemma records a reusable moment-ratio consequence of that viewpoint.

Lemma 9.1 (Laplace moment-ratio bridge). *Let μ be a positive Borel measure on $[0, \infty)$, and assume that the tilted moments*

$$a_j(x) = \int_0^\infty t^j e^{-xt} d\mu(t) \quad (9.1)$$

are finite and positive for the displayed indices. Then, for every $n \geq 0$,

$$B_n(x) = \frac{a_{n+1}(x)}{a_n(x)a_{n+2}(x)} \quad (9.2)$$

is strictly increasing in x .

Proof. Put $r_j(x) = a_{j+1}(x)/a_j(x)$. Cauchy–Schwarz gives

$$a_{j+1}(x)^2 \leq a_j(x)a_{j+2}(x), \quad (9.3)$$

so $r_j(x) \leq r_{j+1}(x)$. Since $a'_j(x) = -a_{j+1}(x)$, differentiating (9.2) logarithmically gives

$$\frac{d}{dx} \log B_n(x) = -\frac{a_{n+2}}{a_{n+1}} + \frac{a_{n+1}}{a_n} + \frac{a_{n+3}}{a_{n+2}} = r_n(x) - r_{n+1}(x) + r_{n+2}(x). \quad (9.4)$$

The right-hand side is at least $r_n(x) > 0$, because $r_{n+2}(x) \geq r_{n+1}(x)$. Hence B_n is strictly increasing. $\square$

Yin and Zhang define the Nielsen k -beta function by

$$\beta_k(x) = \int_0^1 \frac{t^{x-1}}{1+t^k} dt = \int_0^\infty \frac{e^{-xt}}{1+e^{-kt}} dt, \quad k > 0, x > 0. \quad (9.5)$$

They ask for the monotonicity of the derivative ratio involving $x\beta_k(x)$ [8]. the cited proposition lemma gives the exact parity law.

Proposition 9.2 (Nielsen k beta derivative ratio monotonicity). *For $k > 0$, $n \geq 0$, and $f_k(x) = x\beta_k(x)$, the ratio $f_k^{(n+1)}(x)/(f_k^{(n)}(x)f_k^{(n+2)}(x))$ is strictly increasing on $(0, \infty)$ when n is odd and strictly decreasing on $(0, \infty)$ when n is even.*

Proof. Let μ be a positive Borel measure on $[0, \infty)$ with finite moments after the tilt e^{-xt} . Define

$$a_j(x) = \int_0^\infty t^j e^{-xt} d\mu(t) > 0$$

for the needed indices. For fixed $n \geq 0$, set

$$B_n(x) = \frac{a_{n+1}(x)}{a_n(x)a_{n+2}(x)}.$$

Then B_n is strictly increasing whenever $a_{n+1}(x) > 0$.

Indeed $a'_j(x) = -a_{j+1}(x)$. Put

$$r_j(x) = \frac{a_{j+1}(x)}{a_j(x)}.$$

Moment log-convexity, equivalently Cauchy-Schwarz applied to

$$\int t^j e^{-xt} d\mu(t),$$

gives $r_{j+1}(x) \geq r_j(x)$. Differentiating,

$$\frac{d}{dx} \log B_n(x) = -\frac{a_{n+2}}{a_{n+1}} + \frac{a_{n+1}}{a_n} + \frac{a_{n+3}}{a_{n+2}} = r_n(x) - r_{n+1}(x) + r_{n+2}(x).$$

Since $r_{n+2}(x) \geq r_{n+1}(x)$ and $r_n(x) > 0$, the logarithmic derivative is positive. Hence B_n is strictly increasing.

Let

$$f_k(x) = x\beta_k(x).$$

From the integral representation of β_k , integration by parts gives

$$f_k(x) = x \int_0^\infty e^{-xt} \frac{dt}{1+e^{-kt}} = \frac{1}{2} + \int_0^\infty e^{-xt} \frac{ke^{-kt}}{(1+e^{-kt})^2} dt.$$

Thus f_k is the Laplace transform of a positive measure consisting of an atom $1/2$ at 0 plus the positive density

$$w_k(t) = \frac{ke^{-kt}}{(1+e^{-kt})^2}$$

on $(0, \infty)$. All tilted moments are finite, and $a_j(x) > 0$ for $j \geq 1$.

For $a_j(x) = (-1)^j f_k^{(j)}(x)$, the cited proposition lemma gives that

$$B_{k,n}(x) = \frac{a_{n+1}(x)}{a_n(x)a_{n+2}(x)}$$

is strictly increasing for every $k > 0$, $n \geq 0$, and $x > 0$. Since

$$R_{k,n}(x) = (-1)^{n+1} B_{k,n}(x),$$

we get the precise monotonicity law:

if n is odd, $R_{k,n}$ is strictly increasing on $(0, \infty)$; if n is even, $R_{k,n}$ is strictly decreasing on $(0, \infty)$.

This solves the Yin–Zhang Nielsen k -beta derivative-ratio open problem in the parity-refined form. $\square$

Proposition 9.3 (CM Laplace moment ratio monotonicity). *Let $a_j(x) = \int_0^\infty t^j e^{-xt} d\mu(t) > 0$ be tilted moments of a positive Laplace measure for the needed indices. Then, for every $n \geq 0$, $B_n(x) = a_{n+1}(x)/(a_n(x)a_{n+2}(x))$ is strictly increasing on its domain.*

Proof. For CM Laplace moment ratio monotonicity, this proof is the corresponding component of Proposition 9.2. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. $\square$

Corollary 9.4 (APP-0010: Nielsen k -beta derivative-ratio monotonicity). *Let $k > 0$, let $n \geq 0$, and put $f_k(x) = x\beta_k(x)$. Then*

$$R_{k,n}(x) = \frac{f_k^{(n+1)}(x)}{f_k^{(n)}(x)f_k^{(n+2)}(x)} \tag{9.6}$$

is strictly increasing on $(0, \infty)$ when n is odd, and strictly decreasing on $(0, \infty)$ when n is even.

Proof. We use Proposition 9.3 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let μ be a positive Borel measure on $[0, \infty)$ with finite moments after the tilt e^{-xt} . Define

$$a_j(x) = \int_0^\infty t^j e^{-xt} d\mu(t) > 0$$

for the needed indices. For fixed $n \geq 0$, set

$$B_n(x) = \frac{a_{n+1}(x)}{a_n(x)a_{n+2}(x)}.$$

Then B_n is strictly increasing whenever $a_{n+1}(x) > 0$.

Indeed $a'_j(x) = -a_{j+1}(x)$. Put

$$r_j(x) = \frac{a_{j+1}(x)}{a_j(x)}.$$

Moment log-convexity, equivalently Cauchy–Schwarz applied to

$$\int t^j e^{-xt} d\mu(t),$$

gives $r_{j+1}(x) \geq r_j(x)$. Differentiating,

$$\frac{d}{dx} \log B_n(x) = -\frac{a_{n+2}}{a_{n+1}} + \frac{a_{n+1}}{a_n} + \frac{a_{n+3}}{a_{n+2}} = r_n(x) - r_{n+1}(x) + r_{n+2}(x).$$

Since $r_{n+2}(x) \geq r_{n+1}(x)$ and $r_n(x) > 0$, the logarithmic derivative is positive. Hence B_n is strictly increasing.

Let

$$f_k(x) = x\beta_k(x).$$

From the integral representation of β_k , integration by parts gives

$$f_k(x) = x \int_0^\infty e^{-xt} \frac{dt}{1 + e^{-kt}} = \frac{1}{2} + \int_0^\infty e^{-xt} \frac{ke^{-kt}}{(1 + e^{-kt})^2} dt.$$

Thus f_k is the Laplace transform of a positive measure consisting of an atom $1/2$ at 0 plus the positive density

$$w_k(t) = \frac{ke^{-kt}}{(1 + e^{-kt})^2}$$

on $(0, \infty)$. All tilted moments are finite, and $a_j(x) > 0$ for $j \geq 1$.

For $a_j(x) = (-1)^j f_k^{(j)}(x)$, the cited proposition lemma gives that

$$B_{k,n}(x) = \frac{a_{n+1}(x)}{a_n(x)a_{n+2}(x)}$$

is strictly increasing for every $k > 0$, $n \geq 0$, and $x > 0$. Since

$$R_{k,n}(x) = (-1)^{n+1} B_{k,n}(x),$$

we get the precise monotonicity law:

if n is odd, $R_{k,n}$ is strictly increasing on $(0, \infty)$; if n is even, $R_{k,n}$ is strictly decreasing on $(0, \infty)$.

This solves the Yin–Zhang Nielsen k -beta derivative-ratio open problem in the parity-refined form. $\square$

10 Endpoint and parameter applications

The next three applications add one certified endpoint result and two parameter-monotonicity results to the complete-monotonicity layer. Only fully solved source questions are labelled.

10.1 Exact $n = 2$ beta-window endpoint

For $s > 1$, write

$$Z_s(a) = \sum_{m=0}^{\infty} (m+a)^{-s}. \quad (10.1)$$

In Qi–Lim–Nantomah’s notation [4], put

$$C_2(x) = \psi''(x) + x\psi^{(3)}(x), \quad P_2(x) = \psi''(x)\psi''(1/x), \quad (10.2)$$

and define

$$\mathcal{I}_2 = \{\beta \in \mathbb{R} : x^\beta C_2(x) - P_2(x) < 0 \text{ for all } x > 0\}. \quad (10.3)$$

The remaining endpoint is controlled on $0 < x < 1$ by

$$R(x) = \frac{P_2(x)}{C_2(x)} = \frac{2Z_3(x)Z_3(1/x)}{3xZ_4(x) - Z_3(x)}, \quad Q_2(x) = \frac{\log R(x)}{\log x}. \quad (10.4)$$

Definition 10.1 (Rational certificates and finite covers). A rational certificate is a sign or enclosure proof reduced to explicit rational inequalities. A finite cover certificate partitions a compact interval into finitely many rational subintervals and proves the required inequality on each piece, often using an analytic-denominator reduction, a Taylor model, or derivative bounds.

Proposition 10.1 (Q2 analytic denominator reduced assembly terminal certificate). *The terminal endpoint follows from a reduced analytic-denominator assembly: the true covers $(0, 9/50]$ and $[9/10, 1)$, the endpoint signs on J , an analytic-denominator or direct-log finite-middle cover of*

$$\left[\frac{9}{50}, \frac{1409}{5000} \right] \cup \left[\frac{293}{1000}, \frac{9}{10} \right]$$

, and a compact one-crossing/dominance certificate on $[1409/5000, 293/1000] \setminus J$. This determines $L_2 = Q_2(\xi)$ and $\mathcal{I}_2 = (Q_2(\xi), 3]$.

Proof. The replayable certificate is:

It proves

$$Q_2(x) < q_J$$

on

$$\left[\frac{9}{50}, \frac{1409}{5000} \right] \cup \left[\frac{293}{1000}, \frac{9}{10} \right].$$

The left bridge $[9/50, 221050/1000000]$ uses the already-derived identity

$$3xZ_4(x) - Z_3(x) = 2x^{-3} + \sum_{k=1}^{\infty} \frac{2x - k}{(x + k)^4}.$$

For $x < 1/2$, this gives

$$R(x) > Z_3(1/x).$$

The certificate proves

$$Z_3(1/x) > x^{1157/500}$$

on $[9/50, 221050/1000000]$. Since

$$\frac{1157}{500} < \frac{115727}{50000} < q_J,$$

this gives $Q_2(x) < q_J$ on the cited proposition.

The remaining two subintervals use the direct comparison

$$\log R(x) > \frac{115727}{50000} \log x.$$

Since $\log x < 0$, this implies

$$Q_2(x) < \frac{115727}{50000} < q_J.$$

The script uses power-of-two range reduction for the atanh logarithm enclosure:

$$\log r = \log(2^k r) - k \log 2,$$

so every logarithm is evaluated with an argument in $[1, 2)$.

Together with the earlier true covers $(0, 9/50]$ and $[9/10, 1)$, this covers all outside-compact points below q_J .

The replayable certificate is:

Let

$$H(x) = \Lambda(x) + x\Lambda'(x).$$

Using $Z'_s(x) = -sZ_{s+1}(x)$ and

$$\frac{d}{dx}Z_s(1/x) = sx^{-2}Z_{s+1}(1/x),$$

the script builds a rational interval enclosure for H using $Z_3, \dots, Z_6$ at x and $Z_3, \dots, Z_5$ at $1/x$. It proves

$$H(x) > 0 \quad \left(\frac{1409}{5000} \leq x \leq \frac{293}{1000} \right).$$

Since

$$G'(x) = \log x H(x)$$

and $\log x < 0$ on the compact bracket, this gives

$$G'(x) < 0.$$

The existing endpoint certificate proves

$$G(287345/1000000) > 0, \quad G(287346/1000000) < 0.$$

Therefore G has a unique zero

$$\xi \in J = \left[\frac{287345}{1000000}, \frac{287346}{1000000} \right],$$

with $G > 0$ to the left of J and $G < 0$ to the right of J on the compact bracket.

Because

$$Q'_2(x) = \frac{G(x)}{x(\log x)^2},$$

Q_2 is increasing before ξ and decreasing after ξ on the compact bracket.

The true cover package is now:

$$(0, 9/50] \cup \left[9/50, \frac{1409}{5000} \right] \cup \left[\frac{293}{1000}, 9/10 \right] \cup [9/10, 1).$$

All points outside the compact bracket have $Q_2(x) < q_J$, and $q_J < Q_2(\xi)$ by the certified inner witness. On the compact bracket, Q_2 has a unique maximum at the unique zero $\xi \in J$. Thus

$$L_2 = Q_2(\xi).$$

At $\beta = Q_2(\xi)$, the defining inequality has equality at $x = \xi$, so the lower endpoint is excluded. The known upper endpoint remains included. Therefore

$$\mathcal{I}_2 = (Q_2(\xi), 3].$$

This proves the terminal exact endpoint node. $\square$

Proposition 10.2 (Q2 compact taylor monotonicity J terminal certificate). *Using the true endpoint-sign certificate on $J = [287345/1000000, 287346/1000000]$, there is a Taylor-model or derivative-bounded rational certificate proving $G > 0$ on $[1409/5000, 287345/1000000]$, $G < 0$ on $[287346/1000000, 293/1000]$, and dominance below the witness q_J ; combined with finite-middle covers outside the compact bracket and the true endpoint outside covers, this determines $L_2 = Q_2(\xi)$ and $\mathcal{I}_2 = (Q_2(\xi), 3]$.*

Proof. This proof is the component of Proposition 10.1 corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. $\square$

Proposition 10.3 (Q2 remaining finite middle log cover terminal certificate). *Using the true outside covers $(0, 9/50]$ and $[9/10, 1)$, the true endpoint signs on J , and the true enclosure lemmas, there is a finite rational direct-log cover of*

$$\left[\frac{9}{50}, \frac{1409}{5000} \right] \cup \left[\frac{293}{1000}, \frac{9}{10} \right]$$

, proving $Q_2(x) < q_J$ on every interval, together with a compact one-crossing/dominance certificate on $[1409/5000, 293/1000] \setminus J$. This determines $L_2 = Q_2(\xi)$ and $\mathcal{I}_2 = (Q_2(\xi), 3]$.

Proof. For Q2 remaining finite middle log cover terminal certificate, this proof is the corresponding component of Proposition 10.2. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. $\square$

Corollary 10.4 (APP-0011: exact $n = 2$ beta-window endpoint). *Let*

$$J = \left[\frac{287345}{1000000}, \frac{287346}{1000000} \right]. \quad (10.5)$$

There is a unique $\xi \in J$ at which Q_2 attains its global supremum on $(0, 1)$. Hence, with $L_2 = Q_2(\xi)$,

$$\mathcal{I}_2 = (L_2, 3]. \quad (10.6)$$

Proof. We use Proposition 10.1, Proposition 10.2, and Proposition 10.3 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

This script proves a true auxiliary finite-middle outside-cover node:

The replayable certificate is:

It proves

$$Q_2(x) < q_J$$

on

$$\left[\frac{9}{50}, \frac{1409}{5000} \right] \cup \left[\frac{293}{1000}, \frac{9}{10} \right].$$

The left bridge $[9/50, 221050/1000000]$ uses the already-derived identity

$$3xZ_4(x) - Z_3(x) = 2x^{-3} + \sum_{k=1}^{\infty} \frac{2x - k}{(x + k)^4}.$$

For $x < 1/2$, this gives

$$R(x) > Z_3(1/x).$$

The certificate proves

$$Z_3(1/x) > x^{1157/500}$$

on $[9/50, 221050/1000000]$. Since

$$\frac{1157}{500} < \frac{115727}{50000} < q_J,$$

this gives $Q_2(x) < q_J$ on the cited proposition.

The remaining two subintervals use the direct comparison

$$\log R(x) > \frac{115727}{50000} \log x.$$

Since $\log x < 0$, this implies

$$Q_2(x) < \frac{115727}{50000} < q_J.$$

The script uses power-of-two range reduction for the atanh logarithm enclosure:

$$\log r = \log(2^k r) - k \log 2,$$

so every logarithm is evaluated with an argument in $[1, 2)$.

Together with the earlier true covers $(0, 9/50]$ and $[9/10, 1)$, this covers all outside-compact points below q_J .

This certificate also proves:

the Q2 compact monotonicity J certificate.

The replayable certificate is:

Let

$$H(x) = \Lambda(x) + x\Lambda'(x).$$

Using $Z'_s(x) = -sZ_{s+1}(x)$ and

$$\frac{d}{dx} Z_s(1/x) = sx^{-2} Z_{s+1}(1/x),$$

the script builds a rational interval enclosure for H using $Z_3, \dots, Z_6$ at x and $Z_3, \dots, Z_5$ at $1/x$. It proves

$$H(x) > 0 \quad \left(\frac{1409}{5000} \leq x \leq \frac{293}{1000} \right).$$

Since

$$G'(x) = \log x H(x)$$

and $\log x < 0$ on the compact bracket, this gives

$$G'(x) < 0.$$

The existing endpoint certificate proves

$$G(287345/1000000) > 0, \quad G(287346/1000000) < 0.$$

Therefore G has a unique zero

$$\xi \in J = \left[\frac{287345}{1000000}, \frac{287346}{1000000} \right],$$

with $G > 0$ to the left of J and $G < 0$ to the right of J on the compact bracket.

Because

$$Q_2'(x) = \frac{G(x)}{x(\log x)^2},$$

Q_2 is increasing before ξ and decreasing after ξ on the compact bracket.

The true cover package is now:

$$(0, 9/50] \cup \left[9/50, \frac{1409}{5000} \right] \cup \left[\frac{293}{1000}, 9/10 \right] \cup [9/10, 1).$$

All points outside the compact bracket have $Q_2(x) < q_J$, and $q_J < Q_2(\xi)$ by the certified inner witness. On the compact bracket, Q_2 has a unique maximum at the unique zero $\xi \in J$. Thus

$$L_2 = Q_2(\xi).$$

At $\beta = Q_2(\xi)$, the defining inequality has equality at $x = \xi$, so the lower endpoint is excluded. The known upper endpoint remains included. Therefore

$$\mathcal{I}_2 = (Q_2(\xi), 3].$$

This proves the terminal exact endpoint node. □

10.2 Baricz V_q parameter log-convexity

Baricz records the question whether, for fixed $x > 0$, the function

$$V_q(x) = \frac{2e^{x^2}}{\Gamma(q+1)} \int_x^\infty e^{-t^2} (t^2 - x^2)^q dt, \quad q > -1, \quad (10.7)$$

is log-convex as a function of q [9].

Proposition 10.5 (Baricz V_q strict q logconvexity). *For every fixed $x > 0$, the function $q \mapsto V_q(x)$ is strictly log-convex on $(-1, \infty)$.*

Proof. This proof is the component of Proposition 10.6 corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Proposition 10.6 (Baricz V_q Laplace normal form). *For every $x > 0$ and $q > -1$, Baricz's function satisfies $V_q(x) = \pi^{-1/2} \int_0^\infty s^{-1/2} e^{-x^2 s} (1+s)^{-(q+1)} ds$.*

Proof. Put $u = t^2 - x^2$. Then

$$V_q(x) = \frac{1}{\Gamma(q+1)} \int_0^\infty e^{-u} u^q (u+x^2)^{-1/2} du. \quad (10.8)$$

Using

$$r^{-1/2} = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-rs} ds$$

and Tonelli's theorem, we obtain

$$V_q(x) = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-x^2 s} (1+s)^{-(q+1)} ds. \quad (10.9)$$

Let $a = q+1 > 0$. Differentiation under the integral gives

$$(-1)^n \frac{d^n}{da^n} V_{a-1}(x) = \frac{1}{\sqrt{\pi}} \int_0^\infty (\log(1+s))^n s^{-1/2} e^{-x^2 s} (1+s)^{-a} ds \geq 0. \quad (10.10)$$

Equivalently, after the change of variables $y = \log(1+s)$, $V_{a-1}(x)$ is a positive Laplace transform in a . The representing measure is not concentrated at a single point, because the density in (10.9) is positive on $s > 0$. Strict Holder inequality for two different exponents therefore gives strict log-convexity in a , and the shift $a = q+1$ gives the stated strict log-convexity in q . $\square$

Corollary 10.7 (APP-0012: Baricz V_q strict q -log-convexity). *For each fixed $x > 0$, the map $q \mapsto V_q(x)$ is strictly log-convex on $(-1, \infty)$.*

Proof. We use Proposition 10.6 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Put $u = t^2 - x^2$. Then $t = (u+x^2)^{1/2}$ and

$$dt = \frac{du}{2(u+x^2)^{1/2}}.$$

Since $e^{x^2} e^{-t^2} = e^{-u}$, the factor 2 in the definition cancels the 1/2 from dt , giving

$$V_q(x) = \frac{1}{\Gamma(q+1)} \int_0^\infty e^{-u} u^q (u+x^2)^{-1/2} du.$$

This integral is finite for $q > -1$: near 0 the integrand is $O(u^q)$, and at infinity it is exponentially decaying.

For $r > 0$,

$$r^{-1/2} = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-rs} ds.$$

With $r = u+x^2$, Tonelli's theorem applies because the integrand is nonnegative. Thus

$$\begin{aligned} V_q(x) &= \frac{1}{\Gamma(q+1)\sqrt{\pi}} \int_0^\infty \int_0^\infty e^{-u} u^q s^{-1/2} e^{-(u+x^2)s} ds du \\ &= \frac{1}{\Gamma(q+1)\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-x^2 s} \left(\int_0^\infty u^q e^{-(1+s)u} du \right) ds \\ &= \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-x^2 s} (1+s)^{-(q+1)} ds. \end{aligned}$$

The last step uses

$$\int_0^\infty u^q e^{-(1+s)u} du = \frac{\Gamma(q+1)}{(1+s)^{q+1}}.$$

Therefore the Baricz Vq q Laplace normal form is true.

Let $a = q + 1 > 0$. The normal form becomes

$$V_{a-1}(x) = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-x^2 s} (1+s)^{-a} ds.$$

For each integer $n \geq 0$, differentiation under the integral gives

$$(-1)^n \frac{d^n}{da^n} V_{a-1}(x) = \frac{1}{\sqrt{\pi}} \int_0^\infty (\log(1+s))^n s^{-1/2} e^{-x^2 s} (1+s)^{-a} ds \geq 0.$$

The differentiation is justified locally uniformly in $a > 0$: near $s = 0$, $(\log(1+s))^n s^{-1/2} = O(s^{n-1/2})$, and as $s \rightarrow \infty$, the factor $e^{-x^2 s}$ dominates every logarithmic and polynomial factor.

Thus $a \mapsto V_{a-1}(x)$ is completely monotone on $(0, \infty)$. Equivalently, after the change of variables $y = \log(1+s)$, it is a positive Laplace transform in a .

Therefore the Baricz Vq q complete monotonicity is true.

For $a > 0$, write

$$F_x(a) = V_{a-1}(x) = \int_{(0, \infty)} e^{-ay} d\nu_x(y),$$

where ν_x is the positive pushforward of

$$\frac{1}{\sqrt{\pi}} s^{-1/2} e^{-x^2 s} ds$$

under $y = \log(1+s)$. This measure is not concentrated at one point because it has positive density on $s > 0$, hence on $y > 0$.

For $a_1 \neq a_2$ and $0 < \lambda < 1$, strict Holder inequality gives

$$F_x(\lambda a_1 + (1-\lambda)a_2) < F_x(a_1)^\lambda F_x(a_2)^{1-\lambda}.$$

Strictness holds because $e^{-a_1 y}$ and $e^{-a_2 y}$ are not proportional ν_x -almost everywhere unless ν_x is supported at a single y .

Thus $a \mapsto F_x(a)$ is strictly log-convex on $(0, \infty)$. Since $a = q + 1$, the shift preserves strict log-convexity, so $q \mapsto V_q(x)$ is strictly log-convex on $(-1, \infty)$ for every fixed $x > 0$.

Therefore the Baricz Vq strict q logconvexity is true, and it implies the source node the Baricz Vq q logconvexity open problem. $\square$

10.3 Yin (p, k) -digamma sharp necessity

Yin proved sufficiency of $\alpha \leq 1$ for complete monotonicity of

$$\delta_{p,k,\alpha}(x) = x^\alpha \left[\frac{1}{k} \log \frac{pkx}{x+k(p+1)} - \psi_{p,k}(x) \right], \quad p \in \mathbb{N}, \quad k > 0, \quad (10.11)$$

and asked whether this condition is also necessary [10].

Proposition 10.8 (Yin pk digamma alpha necessity). *For $p \in \mathbb{N}$, $k > 0$, if $x^\alpha \left[\frac{1}{k} \log \frac{pkx}{x+k(p+1)} - \psi_{p,k}(x) \right]$ is completely monotone on $(0, \infty)$, then $\alpha \leq 1$.*

Proof. Setup. Fix $p \in \mathbb{N}$ and $k > 0$. Define

$$B_{p,k}(x) = \frac{1}{k} \log \frac{pkx}{x + k(p+1)} - \psi_{p,k}(x), \quad \delta_{p,k,\alpha}(x) = x^\alpha B_{p,k}(x).$$

The source gives the finite formula

$$\psi_{p,k}(x) = \frac{1}{k} \log(pk) - \sum_{n=0}^p \frac{1}{x + nk}.$$

Therefore

$$\begin{aligned} B_{p,k}(x) &= \frac{1}{k} \log \frac{pkx}{x + k(p+1)} - \frac{1}{k} \log(pk) + \sum_{n=0}^p \frac{1}{x + nk} \\ &= \frac{1}{k} \log \frac{x}{x + k(p+1)} + \sum_{n=0}^p \frac{1}{x + nk}. \end{aligned}$$

This normal form also gives positivity directly. Put $t = x/k$. Then

$$B_{p,k}(x) = \frac{1}{k} \left[\sum_{n=0}^p \frac{1}{t + n} + \log \frac{t}{t + p + 1} \right].$$

Since

$$\log \frac{t + p + 1}{t} = \int_0^{p+1} \frac{du}{t + u},$$

and $u \mapsto (t + u)^{-1}$ is strictly decreasing, the left Riemann sum dominates:

$$\sum_{n=0}^p \frac{1}{t + n} > \int_0^{p+1} \frac{du}{t + u}.$$

Hence $B_{p,k}(x) > 0$ for $x > 0$.

This proves the claim.

From the finite normal form,

$$B_{p,k}(x) = \frac{1}{x} + \frac{1}{k} \log \frac{x}{x + k(p+1)} + \sum_{n=1}^p \frac{1}{x + nk}.$$

As $x \rightarrow 0^+$, the finite sum over $n \geq 1$ is $O(1)$, and

$$\frac{1}{k} \log \frac{x}{x + k(p+1)} = O(|\log x|).$$

Thus

$$B_{p,k}(x) = \frac{1}{x} + O(|\log x|) \quad (x \rightarrow 0^+).$$

Equivalently,

$$xB_{p,k}(x) \rightarrow 1.$$

Therefore

$$\delta_{p,k,\alpha}(x) = x^\alpha B_{p,k}(x) = x^{\alpha-1} (xB_{p,k}(x)) \sim x^{\alpha-1}.$$

This proves the claim.

Assume that $\delta_{p,k,\alpha}$ is completely monotone on $(0, \infty)$. Then

$$\delta_{p,k,\alpha}(x) \geq 0, \quad \delta'_{p,k,\alpha}(x) \leq 0.$$

Since $B_{p,k}(x) > 0$, the function $\delta_{p,k,\alpha}$ is actually positive for every $x > 0$.

Suppose, for contradiction, that $\alpha > 1$. The endpoint asymptotic gives

$$\lim_{x \rightarrow 0^+} \delta_{p,k,\alpha}(x) = 0.$$

Fix $x_0 > 0$. Positivity gives $\delta_{p,k,\alpha}(x_0) > 0$. Because $\delta_{p,k,\alpha}(x) \rightarrow 0$ as $x \rightarrow 0^+$, choose $0 < y < x_0$ with

$$\delta_{p,k,\alpha}(y) < \frac{1}{2} \delta_{p,k,\alpha}(x_0).$$

But $\delta'_{p,k,\alpha} \leq 0$ means the function is nonincreasing as x increases, hence for $0 < y < x_0$,

$$\delta_{p,k,\alpha}(y) \geq \delta_{p,k,\alpha}(x_0),$$

a contradiction.

Therefore $\alpha \leq 1$.

This proves the claim. $\square$

Corollary 10.9 (APP-0013: Yin (p, k) -digamma sharp α -necessity). *If $\delta_{p,k,\alpha}$ is completely monotone on $(0, \infty)$, then $\alpha \leq 1$. Combined with Yin's sufficiency theorem, the sharp condition is $\alpha \leq 1$.*

Proof. We use Proposition 10.8 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Setup. Fix $p \in \mathbb{N}$ and $k > 0$. Define

$$B_{p,k}(x) = \frac{1}{k} \log \frac{pkx}{x + k(p+1)} - \psi_{p,k}(x), \quad \delta_{p,k,\alpha}(x) = x^\alpha B_{p,k}(x).$$

The source gives the finite formula

$$\psi_{p,k}(x) = \frac{1}{k} \log(pk) - \sum_{n=0}^p \frac{1}{x + nk}.$$

Therefore

$$\begin{aligned} B_{p,k}(x) &= \frac{1}{k} \log \frac{pkx}{x + k(p+1)} - \frac{1}{k} \log(pk) + \sum_{n=0}^p \frac{1}{x + nk} \\ &= \frac{1}{k} \log \frac{x}{x + k(p+1)} + \sum_{n=0}^p \frac{1}{x + nk}. \end{aligned}$$

This normal form also gives positivity directly. Put $t = x/k$. Then

$$B_{p,k}(x) = \frac{1}{k} \left[\sum_{n=0}^p \frac{1}{t + n} + \log \frac{t}{t + p + 1} \right].$$

Since

$$\log \frac{t + p + 1}{t} = \int_0^{p+1} \frac{du}{t + u},$$

and $u \mapsto (t + u)^{-1}$ is strictly decreasing, the left Riemann sum dominates:

$$\sum_{n=0}^p \frac{1}{t+n} > \int_0^{p+1} \frac{du}{t+u}.$$

Hence $B_{p,k}(x) > 0$ for $x > 0$.

This yields $B_{p,k}(x) > 0$ for all $x > 0$, i.e. the finite-normal-form positivity claim needed for the endpoint obstruction.

From the finite normal form,

$$B_{p,k}(x) = \frac{1}{x} + \frac{1}{k} \log \frac{x}{x+k(p+1)} + \sum_{n=1}^p \frac{1}{x+nk}.$$

As $x \rightarrow 0^+$, the finite sum over $n \geq 1$ is $O(1)$, and

$$\frac{1}{k} \log \frac{x}{x+k(p+1)} = O(|\log x|).$$

Thus

$$B_{p,k}(x) = \frac{1}{x} + O(|\log x|) \quad (x \rightarrow 0^+).$$

Equivalently,

$$xB_{p,k}(x) \rightarrow 1.$$

Therefore

$$\delta_{p,k,\alpha}(x) = x^\alpha B_{p,k}(x) = x^{\alpha-1}(xB_{p,k}(x)) \sim x^{\alpha-1}.$$

This yields the near-zero asymptotic $xB_{p,k}(x) \rightarrow 1$, i.e. $B_{p,k}(x) \sim x^{-1}$ as $x \rightarrow 0^+$.

Assume that $\delta_{p,k,\alpha}$ is completely monotone on $(0, \infty)$. Then

$$\delta_{p,k,\alpha}(x) \geq 0, \quad \delta'_{p,k,\alpha}(x) \leq 0.$$

Since $B_{p,k}(x) > 0$, the function $\delta_{p,k,\alpha}$ is actually positive for every $x > 0$.

Suppose, for contradiction, that $\alpha > 1$. The endpoint asymptotic gives

$$\lim_{x \rightarrow 0^+} \delta_{p,k,\alpha}(x) = 0.$$

Fix $x_0 > 0$. Positivity gives $\delta_{p,k,\alpha}(x_0) > 0$. Because $\delta_{p,k,\alpha}(x) \rightarrow 0$ as $x \rightarrow 0^+$, choose $0 < y < x_0$ with

$$\delta_{p,k,\alpha}(y) < \frac{1}{2} \delta_{p,k,\alpha}(x_0).$$

But $\delta'_{p,k,\alpha} \leq 0$ means the function is nonincreasing as x increases, hence for $0 < y < x_0$,

$$\delta_{p,k,\alpha}(y) \geq \delta_{p,k,\alpha}(x_0),$$

a contradiction.

Therefore $\alpha \leq 1$.

This contradiction argument gives $\alpha \leq 1$, and thus the claimed open-problem conclusion follows. $\square$

11 The Laplace-Transport Application Layer

The applications in this layer are organized by one theme. A problem is first converted into a positive-kernel statement, usually by a logarithmic second difference, a Frullani identity, a moment determinant, or a transform normalization. The conclusion then follows by transporting that kernel through a permanence operation: Stieltjes transformation, Bernstein integration, stable subordination, quotient compression, or triangular coefficient extraction. This section records representative public solves.

Proposition 11.1 (Ramanujan integral Stieltjes). *The Ramanujan integral $I_R(x) = \int_0^\infty e^{-xt} dt / [t(\pi^2 + \log^2 t)]$ is a Stieltjes function on $(0, \infty)$.*

Proof. For $t > 0$,

$$\frac{1}{t(\pi^2 + \log^2 t)} = \frac{1}{\pi(1+t)} \int_0^1 t^{-u} \sin(\pi u) du.$$

The right side is a positive superposition of completely monotone powers and is therefore completely monotone as a function of t . The Laplace transform of a completely monotone density is Stieltjes. Integrating the same density against $1 - e^{-xt}$ gives a complete Bernstein function by the standard Stieltjes-density characterization of complete Bernstein functions. $\square$

Proposition 11.2 (CM closure product positive mixture). *Completely monotone functions on $(0, \infty)$ are closed under pointwise products, nonnegative finite sums, and nonnegative parameter integrals whenever the integral is finite.*

Proof. Completely monotone functions on $(0, \infty)$ are closed under pointwise products, nonnegative finite sums, and nonnegative parameter integrals whenever the integral is finite.

If ϕ is completely monotone and

$$F(x) = \int_0^\infty e^{-xt} \phi(t) dt$$

is finite for every $x > 0$, then F is a Stieltjes function. Indeed, Bernstein's theorem gives

$$\phi(t) = \int_0^\infty e^{-ts} d\mu(s),$$

and Tonelli gives

$$F(x) = \int_0^\infty \frac{d\mu(s)}{x+s}.$$

If

$$f(x) = a + bx + \int_0^\infty (1 - e^{-xt}) m(t) dt$$

is a Bernstein representation and m is completely monotone, then f is a complete Bernstein function, subject to the usual Levy integrability condition. $\square$

Corollary 11.3 (APP-0014: Ramanujan integral is Stieltjes). *Let*

$$I_R(x) = \int_0^\infty \frac{e^{-xt}}{t(\pi^2 + \log^2 t)} dt, \quad x > 0.$$

Then I_R is a Stieltjes function. Moreover every primitive of the form

$$\tilde{I}_R(x) = a + \int_0^\infty \frac{1 - e^{-xt}}{t(\pi^2 + \log^2 t)} dt, \quad a \geq 0,$$

is a complete Bernstein function.

Proof. We use Proposition 11.2 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The node the Ramanujan integral Stieltjes proves that $I_R(x) = \int_0^\infty e^{-xt} dt / [t(\pi^2 + \log^2 t)]$ is a Stieltjes function. The node the Ramanujan antiderivative complete Bernstein proves that the primitive $\tilde{I}_R(x) = a + \int_0^\infty (1 - e^{-xt}) dt / [t(\pi^2 + \log^2 t)]$ is a complete Bernstein function. These are exactly the two clauses of APP-0014, so the conjunction gives the source-facing application result. $\square$

Proposition 11.4 (Prabhakar Q stable subordination normal form). *For $0 < \alpha < 1$, $\gamma > 0$, $\beta > \alpha\gamma$, and $\lambda > 0$, let $P_{\alpha,\beta}^\gamma$ be Sibisi's transform-normalized Pollard measure satisfying $E_{\alpha,\beta}^\gamma(-u) = \int_0^\infty e^{-ur} dP_{\alpha,\beta}^\gamma(r)$. Then the measure $Q_{\alpha,\beta}^\gamma(A \mid \lambda) = \int \Pr((\lambda r)^{1/\alpha} S_\alpha \in A) dP_{\alpha,\beta}^\gamma(r)$ satisfies $E_{\alpha,\beta}^\gamma(-\lambda x^\alpha) = \int_0^\infty e^{-xt} dQ_{\alpha,\beta}^\gamma(t \mid \lambda)$.*

Proof. Conditioning on r and using the stable transform gives

$$\mathbb{E}e^{-x(\lambda r)^{1/\alpha} S_\alpha} = e^{-\lambda r x^\alpha}.$$

Integration against $P_{\alpha,\beta}^\eta$ gives the Pollard representation of $E_{\alpha,\beta}^\eta(-\lambda x^\alpha)$. The total mass is finite, equal to the value at $x = 0$, and no normalization to a probability measure is needed for the transform identity. $\square$

Corollary 11.5 (APP-0015: Prabhakar Q-measure by stable subordination). *Let $0 < \alpha < 1$, $\eta > 0$, $\beta > \alpha\eta$, and $\lambda > 0$. Let*

$$E_{\alpha,\beta}^\eta(z) = \sum_{m=0}^{\infty} \frac{(\eta)_m z^m}{m! \Gamma(\alpha m + \beta)}$$

be the Prabhakar Mittag-Leffler function. Let $P_{\alpha,\beta}^\eta$ be the transform-normalized Pollard measure characterized in the strict Pollard range by

$$\int_0^\infty e^{-xr} dP_{\alpha,\beta}^\eta(r) = E_{\alpha,\beta}^\eta(-x), \quad x > 0.$$

Let S_α be the positive α -stable variable satisfying $\mathbb{E}e^{-sS_\alpha} = e^{-s^\alpha}$. Define the finite measure

$$Q_{\alpha,\beta}^\eta(A \mid \lambda) = \int \Pr\{(\lambda r)^{1/\alpha} S_\alpha \in A\} dP_{\alpha,\beta}^\eta(r).$$

Then

$$\int_0^\infty e^{-xt} dQ_{\alpha,\beta}^\eta(t \mid \lambda) = E_{\alpha,\beta}^\eta(-\lambda x^\alpha).$$

Proof. We use Proposition 11.4 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let P be a finite positive measure on $[0, \infty)$, let

$$F(u) = \int_0^\infty e^{-ur} dP(r),$$

let $0 < \alpha < 1$, $\lambda > 0$, and let S_α be positive α -stable:

$$\mathbb{E}e^{-xS_\alpha} = e^{-x^\alpha}.$$

Define a finite positive measure Q by

$$Q(A) = \int_{[0,\infty)} \Pr\{(\lambda r)^{1/\alpha} S_\alpha \in A\} dP(r).$$

Then Tonelli's theorem gives, for $x > 0$,

$$\int_0^\infty e^{-xt} dQ(t) = \int_{[0,\infty)} \mathbb{E} e^{-x(\lambda r)^{1/\alpha} S_\alpha} dP(r) = \int_{[0,\infty)} e^{-\lambda r x^\alpha} dP(r) = F(\lambda x^\alpha).$$

Apply the lemma to Sibisi's transform-normalized $P = P_{\alpha,\beta}^\gamma$ and

$$F(u) = E_{\alpha,\beta}^\gamma(-u).$$

For $0 < \alpha < 1$, $\gamma > 0$, $\beta > \alpha\gamma$, and $\lambda > 0$, set

$$Q_{\alpha,\beta}^\gamma(A \mid \lambda) = \int_{[0,\infty)} \Pr\{(\lambda r)^{1/\alpha} S_\alpha \in A\} dP_{\alpha,\beta}^\gamma(r).$$

Then

$$\int_0^\infty e^{-xt} dQ_{\alpha,\beta}^\gamma(t \mid \lambda) = E_{\alpha,\beta}^\gamma(-\lambda x^\alpha).$$

By uniqueness of finite Laplace transforms, this is Sibisi's $Q_{\alpha,\beta}^\gamma(\cdot \mid \lambda)$.

If S_α has density f_α , then the non-atomic density part is

$$q_{\alpha,\beta}^\gamma(t \mid \lambda) = \int_{(0,\infty)} (\lambda r)^{-1/\alpha} f_\alpha\left(\frac{t}{(\lambda r)^{1/\alpha}}\right) dP_{\alpha,\beta}^\gamma(r),$$

with an atom $P_{\alpha,\beta}^\gamma(\{0\})\delta_0$ if $P_{\alpha,\beta}^\gamma$ has an atom at zero. When $P_{\alpha,\beta}^\gamma$ is represented by Sibisi's Pollard density $p_{\alpha,\beta}^\gamma(r)$, this becomes

$$q_{\alpha,\beta}^\gamma(t \mid \lambda) = \int_0^\infty (\lambda r)^{-1/\alpha} f_\alpha\left(\frac{t}{(\lambda r)^{1/\alpha}}\right) p_{\alpha,\beta}^\gamma(r) dr.$$

Because

$$E_{\alpha,\beta}^\gamma(0) = \frac{1}{\Gamma(\beta)},$$

the representing measures $P_{\alpha,\beta}^\gamma$ and $Q_{\alpha,\beta}^\gamma(\cdot \mid \lambda)$ have total mass $1/\Gamma(\beta)$. They are finite positive measures, not generally probability measures. The normalized probability version is

$$\widehat{Q}_{\alpha,\beta}^\gamma = \Gamma(\beta) Q_{\alpha,\beta}^\gamma, \quad \mathcal{L}\widehat{Q}_{\alpha,\beta}^\gamma(x) = \Gamma(\beta) E_{\alpha,\beta}^\gamma(-\lambda x^\alpha).$$

The boundary $\beta = \alpha\gamma$ is not included here because Sibisi's ordinary convolution density uses $1/\Gamma(\beta - \alpha\gamma)$. A limiting treatment is a separate open frontier. $\square$

Proposition 11.6 (From Mills Laplace CM normal form). *For $t > 0$, the standard normal Mills ratio satisfies $r(t) = \int_0^\infty e^{-tu - u^2/2} du$. Hence r is completely monotone on $(0, \infty)$.*

Proof. The density defining M_L is a positive moment density. Its order-four Hankel-minor consequence gives

$$M_{L+4}M_L - 4M_{L+3}M_{L+1} + 3M_{L+2}^2 > 0.$$

Using the recurrence $M_{n+1} = nM_{n-1} - tM_n$, this determinant inequality reduces to a quadratic inequality for m_L ; solving the admissible positive root gives $m_L < U_L$. Iterating the same recurrence expresses M_L in the form

$$M_L = (-1)^L (P_L r - B_L),$$

and substituting the level- L quotient bound gives the displayed alternating upper and lower bounds for r . $\square$

Corollary 11.7 (APP-0016: Mills-ratio bound at every derivative level). *Let*

$$M_L(t) = \int_0^\infty u^L e^{-tu - u^2/2} du, \quad L = 0, 1, 2, \dots$$

Put $m_L(t) = M_{L+1}(t)/M_L(t)$ and

$$U_L(t) = \frac{\sqrt{t^2(t^2 + 4L + 7)^2 + 4(L+1)(t^2 + 4L + 8)(t^2 + 4L + 6)} - t(t^2 + 4L + 7)}{2(t^2 + 4L + 8)}.$$

Then $m_L(t) < U_L(t)$ for every $L \geq 0$ and $t > 0$. With

$$P_0 = 1, \quad P_1 = t, \quad P_{n+1} = tP_n + nP_{n-1},$$

and

$$B_0 = 0, \quad B_1 = 1, \quad B_{n+1} = nB_{n-1} - tB_n,$$

the classical Mills ratio $r(t) = M_0(t)$ satisfies

$$r(t) > \frac{B_{L+1} - U_L B_L}{P_{L+1} + U_L P_L} \quad (L \geq 0 \text{ even}),$$

and

$$r(t) < \frac{U_L B_L - B_{L+1}}{P_{L+1} + U_L P_L} \quad (L \geq 1 \text{ odd}).$$

Proof. We use Proposition 11.6 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Set

$$m_L(t) = \frac{M_{L+1}(t)}{M_L(t)}.$$

The recurrence gives

$$\frac{M_{L+2}}{M_L} = L + 1 - tm_L,$$

$$\frac{M_{L+3}}{M_L} = (t^2 + L + 2)m_L - t(L + 1),$$

and

$$\frac{M_{L+4}}{M_L} = (L + 1)(t^2 + L + 3) - t(t^2 + 2L + 5)m_L.$$

Substitution into the determinant inequality and expansion give

$$(L + 1)(t^2 + 4L + 6) - t(t^2 + 4L + 7)m_L - (t^2 + 4L + 8)m_L^2 > 0.$$

Equivalently,

$$(t^2 + 4L + 8)m_L^2 + t(t^2 + 4L + 7)m_L - (L + 1)(t^2 + 4L + 6) < 0.$$

Since $m_L > 0$, the positive root yields the uniform strict bound

$$m_L(t) < U_L(t),$$

where

$$U_L(t) = \frac{\sqrt{t^2(t^2 + 4L + 7)^2 + 4(L + 1)(t^2 + 4L + 8)(t^2 + 4L + 6)} - t(t^2 + 4L + 7)}{2(t^2 + 4L + 8)}.$$

This is the determinant-to-bound extractor. It is a single theorem for every $L \geq 0$.

Define polynomials $P_n(t)$ by

$$P_0 = 1, \quad P_1 = t, \quad P_{n+1} = tP_n + nP_{n-1}.$$

Define $B_n(t)$ by

$$B_0 = 0, \quad B_1 = 1, \quad B_{n+1} = nB_{n-1} - tB_n.$$

Then the same recurrence gives

$$M_n(t) = (-1)^n P_n(t)r(t) + B_n(t).$$

The ratio bound $M_{L+1} < U_L M_L$ becomes

$$((-1)^{L+1} P_{L+1} - (-1)^L U_L P_L) r < U_L B_L - B_{L+1}.$$

Since $P_n(t) > 0$ for $t > 0$ and $P_{2j}(0) > 0$, the coefficient has sign $(-1)^{L+1}$. Thus the bounds alternate:

For even $L \geq 0$,

$$r(t) > \frac{B_{L+1}(t) - U_L(t)B_L(t)}{P_{L+1}(t) + U_L(t)P_L(t)}.$$

For odd $L \geq 1$,

$$r(t) < \frac{U_L(t)B_L(t) - B_{L+1}(t)}{P_{L+1}(t) + U_L(t)P_L(t)}.$$

These formulas are valid at $t = 0$ by direct evaluation of the recurrences and for $t > 0$ by the positivity of M_L . $\square$

Proposition 11.8 (Baricz gamma quotient a2b3 not BF). *For $a = 2$ and $b = 3$, the Baricz Gamma quotient $x \mapsto \Gamma(x)\Gamma(x - a + b)/(\Gamma(x - a)\Gamma(x + b))$ is not a Bernstein function on $(2, \infty)$.*

Proof. Take $a = 2$, $b = 3$, and set $y = x - 2$. The quotient becomes

$$g(y) = \frac{y(y + 1)}{(y + 3)(y + 4)}.$$

A direct differentiation gives

$$g'''(y) = \frac{36(y^4 + 8y^3 + 12y^2 - 40y - 94)}{(y + 3)^4(y + 4)^4}.$$

Thus $g'''(1) < 0$. If g were Bernstein, then g' would be completely monotone and hence $g''' \geq 0$. The displayed value contradicts that necessary condition. $\square$

Counterexample 11.9 (APP-0017: a gamma quotient need not be Bernstein). The assertion that the gamma quotient

$$x \mapsto \frac{\Gamma(x+a)\Gamma(x+b)}{\Gamma(x)\Gamma(x+a+b)}$$

is Bernstein for all positive a, b is false.

Proof. We use Proposition 8.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Take $a = 2$, $b = 3$, and set $y = x - 2 > 0$. Then by the Gamma recurrence,

$$\frac{\Gamma(x)\Gamma(x-a+b)}{\Gamma(x-a)\Gamma(x+b)} = \frac{\Gamma(x)\Gamma(x+1)}{\Gamma(x-2)\Gamma(x+3)} = \frac{(x-2)(x-1)}{(x+1)(x+2)} = \frac{y(y+1)}{(y+3)(y+4)}.$$

Let

$$g(y) = \frac{y(y+1)}{(y+3)(y+4)}, \quad y > 0.$$

If g were a Bernstein function, then g' would be completely monotone. In particular $g''' \geq 0$ would be necessary.

Exact symbolic differentiation gives

$$g'(y) = \frac{6(y^2 + 4y + 2)}{(y+3)^2(y+4)^2},$$

$$g''(y) = -\frac{12(y^3 + 6y^2 + 6y - 10)}{(y+3)^3(y+4)^3},$$

and

$$g'''(y) = \frac{36(y^4 + 8y^3 + 12y^2 - 40y - 94)}{(y+3)^4(y+4)^4}.$$

At $y = 1$,

$$g'''(1) = -\frac{1017}{40000} < 0.$$

Therefore g' is not completely monotone, g is not Bernstein, and the Baricz for-all- a, b Bernstein question has a negative answer. $\square$

Proposition 11.10 (QG tau global supremum). *Let $a_* > 0$ be the unique positive root of $e^{a_*} = 1 + a_* + a_*^2$. Then $\sup_{s \in \mathbb{N}, t > 0} \tau(s, t) = a_*/(1 + a_* + a_*^2) = 0.298425607525639\dots$, and the supremum is not attained.*

Proof. Let $a_* > 0$ be the unique positive root of

$$e^{a_*} = 1 + a_* + a_*^2.$$

Then

$$\sup_{s \in \mathbb{N}, t > 0} \tau(s, t) = \frac{a_*}{1 + a_* + a_*^2} = 0.298425607525639\dots$$

The supremum is not attained.

For fixed $s \geq 1$, set

$$y = \frac{1}{t+1} \in (0, 1).$$

Then

$$\tau_s(y) = \frac{1 - y - (1 + sy)(1 - y)^{s+1}}{sy}.$$

Direct differentiation gives

$$\tau'_s(y) = \frac{(1 - y)^s \{1 + sy + s(s + 1)y^2\} - 1}{sy^2}.$$

Let

$$F_s(y) = (1 - y)^s \{1 + sy + s(s + 1)y^2\}.$$

Then

$$F'_s(y) = \frac{s(s + 1)y(1 - y)^s((s + 2)y - 1)}{y - 1}.$$

Thus F_s increases on $(0, 1/(s + 2))$ and decreases on $(1/(s + 2), 1)$. Since $F_s(0) = 1$ and $F_s(1) = 0$, the equation $F_s(y) = 1$ has exactly one root $y_s \in (1/(s + 2), 1)$ apart from the endpoint 0, and τ_s has the unique maximum at y_s .

At the maximizing point,

$$(1 - y_s)^s \{1 + sy_s + s(s + 1)y_s^2\} = 1$$

and

$$m_s := \max_{t > 0} \tau(s, t) = \frac{(s + 1)y_s(1 - y_s)}{1 + sy_s + s(s + 1)y_s^2}.$$

Treat s as a real parameter. The envelope derivative at the unique maximizer equals the partial derivative in s :

$$\frac{d}{ds} m(s) = \partial_s \tau_s(y_s).$$

Using $F_s(y_s) = 1$, its sign is the opposite of

$$H_s(y_s) = y_s \{1 + (s + 1)y_s\} + (1 + sy_s) \log(1 - y_s).$$

The root satisfies $y_s \geq 1/(s + 1)$. Indeed, for $s = 1$ equality holds, and for $s > 1$

$$F_s\left(\frac{1}{s + 1}\right) = \left(\frac{s}{s + 1}\right)^s \frac{3s + 1}{s + 1} > 1.$$

Equivalently, with $x = 1/s$, this is

$$(1 + x)^{1/x} < \frac{3 + x}{1 + x},$$

which follows because

$$\frac{d}{dx} \left[\log \frac{3 + x}{1 + x} - \frac{\log(1 + x)}{x} \right] < 0$$

on $(0, 1]$ and the expression vanishes at $x = 1$.

For $y \geq 1/(s + 1)$,

$$H'_s(y) = 1 + 2(s + 1)y + s \log(1 - y) - \frac{1 + sy}{1 - y} < \frac{y(1 - (s + 2)y)}{1 - y} < 0,$$

where $\log(1 - y) < -y$. Also

$$H_s\left(\frac{1}{s + 1}\right) = \frac{2}{s + 1} - \frac{2s + 1}{s + 1} \log\left(1 + \frac{1}{s}\right) < 0,$$

using $\log(1+u) > 2u/(2+u)$. Hence $H_s(y_s) < 0$, so $m(s)$ is strictly increasing.

Put $a_s = sy_s$. The critical equation becomes

$$\left(1 - \frac{a_s}{s}\right)^s \left(1 + a_s + \left(1 + \frac{1}{s}\right)a_s^2\right) = 1.$$

The numbers a_s are bounded above by a_* . For $s > a_*$,

$$\log F_s(a_*/s) \leq -a_* - \frac{a_*^2}{2s} + a_* + \frac{a_*^2}{se^{a_*}} < 0,$$

so $F_s(a_*/s) < 1$ and $a_s < a_*$; the remaining small s are immediate. Therefore $a_s \rightarrow a_*$, because every subsequential limit solves

$$e^{-a}(1+a+a^2) = 1.$$

The equation $e^a = 1+a+a^2$ has exactly one positive root after 0: the function $e^a - 1 - a - a^2$ first decreases and then, after its derivative has its unique positive zero, increases to $+\infty$.

Consequently

$$\lim_{s \rightarrow \infty} m_s = \frac{a_*}{1 + a_* + a_*^2}.$$

Since $m(s)$ is strictly increasing and s is finite in the source domain, no finite (s, t) attains the limiting value. Thus the source's requested maximum should be read as a supremum.

$$a_* = 1.79328213290076\dots, \quad \frac{a_*}{1 + a_* + a_*^2} = 0.298425607525639\dots,$$

which is consistent with the source's observation $\tau_0 > 0.2980$. □

Counterexample 11.11 (APP-0018: exact supremum for the Qi-Guo tau threshold). For $s \in \mathbb{N}$ and $t > 0$, define

$$\tau(s, t) = \frac{1}{s} \left[t - (t + s + 1) \left(\frac{t}{t+1} \right)^{s+1} \right].$$

Let $a_* > 0$ be the unique positive solution of

$$e^{a_*} = 1 + a_* + a_*^2.$$

Then

$$\sup_{s \in \mathbb{N}, t > 0} \tau(s, t) = \frac{a_*}{1 + a_* + a_*^2} = 0.298425607525639\dots$$

The supremum is not attained.

Proof. We use Proposition 11.10 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let $a_* > 0$ be the unique positive root of

$$e^{a_*} = 1 + a_* + a_*^2.$$

Then

$$\sup_{s \in \mathbb{N}, t > 0} \tau(s, t) = \frac{a_*}{1 + a_* + a_*^2} = 0.298425607525639\dots$$

The supremum is not attained.

For fixed $s \geq 1$, set

$$y = \frac{1}{t+1} \in (0, 1).$$

Then

$$\tau_s(y) = \frac{1 - y - (1 + sy)(1 - y)^{s+1}}{sy}.$$

Direct differentiation gives

$$\tau'_s(y) = \frac{(1 - y)^s \{1 + sy + s(s+1)y^2\} - 1}{sy^2}.$$

Let

$$F_s(y) = (1 - y)^s \{1 + sy + s(s+1)y^2\}.$$

Then

$$F'_s(y) = \frac{s(s+1)y(1 - y)^s((s+2)y - 1)}{y - 1}.$$

Thus F_s increases on $(0, 1/(s+2))$ and decreases on $(1/(s+2), 1)$. Since $F_s(0) = 1$ and $F_s(1) = 0$, the equation $F_s(y) = 1$ has exactly one root $y_s \in (1/(s+2), 1)$ apart from the endpoint 0, and τ_s has the unique maximum at y_s .

At the maximizing point,

$$(1 - y_s)^s \{1 + sy_s + s(s+1)y_s^2\} = 1$$

and

$$m_s := \max_{t>0} \tau(s, t) = \frac{(s+1)y_s(1 - y_s)}{1 + sy_s + s(s+1)y_s^2}.$$

Treat s as a real parameter. The envelope derivative at the unique maximizer equals the partial derivative in s :

$$\frac{d}{ds} m(s) = \partial_s \tau_s(y_s).$$

Using $F_s(y_s) = 1$, its sign is the opposite of

$$H_s(y_s) = y_s \{1 + (s+1)y_s\} + (1 + sy_s) \log(1 - y_s).$$

The root satisfies $y_s \geq 1/(s+1)$. Indeed, for $s = 1$ equality holds, and for $s > 1$

$$F_s\left(\frac{1}{s+1}\right) = \left(\frac{s}{s+1}\right)^s \frac{3s+1}{s+1} > 1.$$

Equivalently, with $x = 1/s$, this is

$$(1+x)^{1/x} < \frac{3+x}{1+x},$$

which follows because

$$\frac{d}{dx} \left[\log \frac{3+x}{1+x} - \frac{\log(1+x)}{x} \right] < 0$$

on $(0, 1]$ and the expression vanishes at $x = 1$.

For $y \geq 1/(s+1)$,

$$H'_s(y) = 1 + 2(s+1)y + s \log(1 - y) - \frac{1 + sy}{1 - y} < \frac{y(1 - (s+2)y)}{1 - y} < 0,$$

where $\log(1 - y) < -y$. Also

$$H_s\left(\frac{1}{s+1}\right) = \frac{2}{s+1} - \frac{2s+1}{s+1} \log\left(1 + \frac{1}{s}\right) < 0,$$

using $\log(1 + u) > 2u/(2 + u)$. Hence $H_s(y_s) < 0$, so $m(s)$ is strictly increasing.

Put $a_s = sy_s$. The critical equation becomes

$$\left(1 - \frac{a_s}{s}\right)^s \left(1 + a_s + \left(1 + \frac{1}{s}\right)a_s^2\right) = 1.$$

The numbers a_s are bounded above by a_* . For $s > a_*$,

$$\log F_s(a_*/s) \leq -a_* - \frac{a_*^2}{2s} + a_* + \frac{a_*^2}{se^{a_*}} < 0,$$

so $F_s(a_*/s) < 1$ and $a_s < a_*$; the remaining small s are immediate. Therefore $a_s \rightarrow a_*$, because every subsequential limit solves

$$e^{-a}(1 + a + a^2) = 1.$$

The equation $e^a = 1 + a + a^2$ has exactly one positive root after 0: the function $e^a - 1 - a - a^2$ first decreases and then, after its derivative has its unique positive zero, increases to $+\infty$.

Consequently

$$\lim_{s \rightarrow \infty} m_s = \frac{a_*}{1 + a_* + a_*^2}.$$

Since $m(s)$ is strictly increasing and s is finite in the source domain, no finite (s, t) attains the limiting value. Thus the source's requested maximum should be read as a supremum.

$$a_* = 1.79328213290076\dots, \quad \frac{a_*}{1 + a_* + a_*^2} = 0.298425607525639\dots,$$

which is consistent with the source's observation $\tau_0 > 0.2980$. □

Proposition 11.12 (Wakrim W symbol exact BF range). *For $0 < \alpha < 1$ and $\beta \geq 0$, the Wakrim W-symbol $\Phi_{\alpha, \beta}(s) = s^\alpha(1 + (1 - \alpha)s^{\alpha-1})^{-\beta}$ is a Bernstein function on $(0, \infty)$ if and only if $0 \leq \beta \leq 1$.*

Proof. Set $a = 1 - \alpha \in (0, 1)$. Then

$$\Phi_{\alpha, \beta}(s) = s^{1-a} (1 + as^{-a})^{-\beta} = \frac{s^{1-a+a\beta}}{(s^a + a)^\beta}.$$

Write

$$p = 1 - a + a\beta = \alpha + (1 - \alpha)\beta.$$

Differentiating gives

$$\Phi'_{\alpha, \beta}(s) = s^{p-1}(s^a + a)^{-\beta-1} [p(s^a + a) - \beta as^a].$$

Since $p - \beta a = 1 - a = \alpha$,

$$\Phi'_{\alpha, \beta}(s) = s^{-a(1-\beta)}(s^a + a)^{-\beta-1} [\alpha s^a + a(\alpha + a\beta)].$$

With $y = s^a$, this becomes

$$\Phi'_{\alpha, \beta}(s) = H(s^a),$$

where

$$H(y) = y^{\beta-1}(y+a)^{-\beta} \left(\alpha + \frac{a^2\beta}{y+a} \right).$$

For $0 < \beta \leq 1$, each factor in H is completely monotone on $(0, \infty)$: $y^{\beta-1} = y^{-(1-\beta)}$, $(y+a)^{-\beta}$, and the positive sum $\alpha + a^2\beta/(y+a)$. Therefore H is completely monotone by product and positive-sum closure.

The map $s \mapsto s^a$ is Bernstein for $0 < a < 1$. The standard composition theorem says that if f is completely monotone and g is Bernstein, then $f \circ g$ is completely monotone. Hence $\Phi'_{\alpha,\beta}$ is completely monotone, and $\Phi_{\alpha,\beta}$ is Bernstein for $0 < \beta \leq 1$. The endpoint $\beta = 0$ is $\Phi_{\alpha,0}(s) = s^\alpha$, also Bernstein.

Combining this local positive range with Wakrim's source proof that $\Phi_{\alpha,\beta}$ is not Bernstein for $\beta > 1$, the exact range is

$$\Phi_{\alpha,\beta} \in BF \iff 0 \leq \beta \leq 1.$$

□

Counterexample 11.13 (APP-0019: exact Bernstein range for the W-symbol). Let $0 < \alpha < 1$, $\beta \geq 0$, and

$$\Phi_{\alpha,\beta}(s) = s^\alpha (1 + (1-\alpha)s^{\alpha-1})^{-\beta}.$$

Then $\Phi_{\alpha,\beta}$ is a Bernstein function if and only if

$$0 \leq \beta \leq 1.$$

Proof. We use Proposition 11.12 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Set $a = 1 - \alpha \in (0, 1)$. Then

$$\Phi_{\alpha,\beta}(s) = s^{1-a} (1 + as^{-a})^{-\beta} = \frac{s^{1-a+a\beta}}{(s^a + a)^\beta}.$$

Write

$$p = 1 - a + a\beta = \alpha + (1 - \alpha)\beta.$$

Differentiating gives

$$\Phi'_{\alpha,\beta}(s) = s^{p-1}(s^a + a)^{-\beta-1} [p(s^a + a) - \beta as^a].$$

Since $p - \beta a = 1 - a = \alpha$,

$$\Phi'_{\alpha,\beta}(s) = s^{-a(1-\beta)}(s^a + a)^{-\beta-1} [\alpha s^a + a(\alpha + a\beta)].$$

With $y = s^a$, this becomes

$$\Phi'_{\alpha,\beta}(s) = H(s^a),$$

where

$$H(y) = y^{\beta-1}(y+a)^{-\beta} \left(\alpha + \frac{a^2\beta}{y+a} \right).$$

For $0 < \beta \leq 1$, each factor in H is completely monotone on $(0, \infty)$: $y^{\beta-1} = y^{-(1-\beta)}$, $(y+a)^{-\beta}$, and the positive sum $\alpha + a^2\beta/(y+a)$. Therefore H is completely monotone by product and positive-sum closure.

The map $s \mapsto s^a$ is Bernstein for $0 < a < 1$. The standard composition theorem says that if f is completely monotone and g is Bernstein, then $f \circ g$ is completely monotone. Hence $\Phi'_{\alpha,\beta}$ is

completely monotone, and $\Phi_{\alpha,\beta}$ is Bernstein for $0 < \beta \leq 1$. The endpoint $\beta = 0$ is $\Phi_{\alpha,0}(s) = s^\alpha$, also Bernstein.

Combining this local positive range with Wakrim's source proof that $\Phi_{\alpha,\beta}$ is not Bernstein for $\beta > 1$, the exact range is

$$\Phi_{\alpha,\beta} \in BF \iff 0 \leq \beta \leq 1.$$

□

Proposition 11.14 (Qi hlambda x4 source range not CM). *For $h_\lambda(x) = \Psi(x) - (x^2 + \lambda x + 12)/(12x^4(x+1)^2)$, $x^4 h_\lambda(x)$ is not completely monotone for every $\lambda \leq 0$, and $x^4(-h_\mu(x))$ is not completely monotone for every $\mu \geq 4$.*

Proof. Near zero,

$$h_\lambda(x) = -\frac{\lambda}{12x^3} + \left(\frac{\lambda}{6} + \frac{\pi^2}{3} - \frac{37}{12}\right) \frac{1}{x^2} + O(x^{-1}).$$

If $\lambda < 0$, then $x^4 h_\lambda(x) = (-\lambda/12)x + O(x^2)$, whose derivative is positive near zero. If $\lambda = 0$, the coefficient $\pi^2/3 - 37/12$ is positive, again giving a positive derivative for the degree-four transform near zero. Complete monotonicity would require the first derivative to be non-positive. The same small- x test applied to $-x^4 h_\mu$ excludes the claimed μ -range. Hence the degree-four conjecture fails. □

Counterexample 11.15 (APP-0020: Qi h_λ degree-four conjecture is false). Let

$$\Psi(x) = [\psi'(x)]^2 + \psi''(x), \quad h_\lambda(x) = \Psi(x) - \frac{x^2 + \lambda x + 12}{12x^4(x+1)^2}.$$

The conjecture that $\deg_x^{\text{cm}} h_\lambda = \deg_x^{\text{cm}}(-h_\mu) = 4$ exactly in the stated parameter ranges $\lambda \leq 0$, $\mu \geq 4$, is false. Here $\deg_x^{\text{cm}} g$ denotes the source convention: the supremal power r for which $x^r g(x)$ is completely monotone on $(0, \infty)$.

Proof. We use Proposition 8.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

A completely monotone function must be nonincreasing, so its first derivative cannot be positive at any point.

Use the recurrence expansions at $x = 0^+$:

$$\psi'(x) = \frac{1}{x^2} + \zeta(2) - 2\zeta(3)x + 3\zeta(4)x^2 + O(x^3),$$

and

$$\psi''(x) = -\frac{2}{x^3} - 2\zeta(3) + 6\zeta(4)x + O(x^2).$$

Therefore

$$\Psi(x) = \frac{1}{x^4} - \frac{2}{x^3} + \frac{\pi^2}{3x^2} - \frac{4\zeta(3)}{x} + O(1).$$

Also

$$\frac{x^2 + \lambda x + 12}{12x^4(1+x)^2} = \frac{1}{x^4} + \left(\frac{\lambda}{12} - 2\right) \frac{1}{x^3} + \left(\frac{37}{12} - \frac{\lambda}{6}\right) \frac{1}{x^2} + O(x^{-1}).$$

Subtracting gives

$$h_\lambda(x) = -\frac{\lambda}{12x^3} + \left(\frac{\lambda}{6} + \frac{\pi^2}{3} - \frac{37}{12}\right) \frac{1}{x^2} + O(x^{-1}).$$

If $\lambda < 0$, then

$$x^4 h_\lambda(x) = -\frac{\lambda}{12}x + O(x^2),$$

so

$$\frac{d}{dx}[x^4 h_\lambda(x)] = -\frac{\lambda}{12} + O(x) > 0$$

for all sufficiently small $x > 0$. Hence $x^4 h_\lambda$ is not completely monotone.

If $\lambda = 0$, then

$$x^4 h_0(x) = \left(\frac{\pi^2}{3} - \frac{37}{12}\right)x^2 + O(x^3).$$

Since $4\pi^2 > 37$, the leading coefficient is positive, and

$$\frac{d}{dx}[x^4 h_0(x)] = 2\left(\frac{\pi^2}{3} - \frac{37}{12}\right)x + O(x^2) > 0$$

for all sufficiently small $x > 0$. Hence $x^4 h_0$ is not completely monotone.

Similarly, for $\mu \geq 4$,

$$x^4(-h_\mu(x)) = \frac{\mu}{12}x + O(x^2),$$

so

$$\frac{d}{dx}[x^4(-h_\mu(x))] = \frac{\mu}{12} + O(x) > 0$$

for all sufficiently small $x > 0$. Hence $x^4(-h_\mu)$ is not completely monotone for every $\mu \geq 4$.

Thus the degree-four transforms required by Qi's conjecture are not completely monotone on the conjectured source ranges. The conjecture in Remark 7.4 is false. $\square$

Proposition 11.16 (Szabo psi difference alpha0 exact one). *For $0 < d < 1$, the sharp cutoff for complete monotonicity of $y^\alpha[\psi(y+d) - \psi(y) - d/y]$ is $\alpha_0 = 1$.*

Proof. Let $0 < d < 1$, $y > 0$, and

$$H_d(y) = \psi(y+d) - \psi(y) - \frac{d}{y}.$$

Then

$$y^\alpha H_d(y)$$

is strictly completely monotone on $(0, \infty)$ if $\alpha \leq 1$, and is not completely monotone if $\alpha > 1$. Hence the sharp cutoff in Szabo Open Problem 1.5 is

$$\alpha_0 = 1.$$

As $y \downarrow 0$, the digamma Laurent expansion gives

$$\psi(y) = -\frac{1}{y} - \gamma + O(y), \quad \psi(y+d) = \psi(d) + O(y).$$

Therefore

$$H_d(y) = \psi(y+d) - \psi(y) - \frac{d}{y} = \frac{1-d}{y} + O(1).$$

Since $0 < d < 1$, the leading coefficient is positive. Hence

$$y^\alpha H_d(y) = (1-d)y^{\alpha-1} + O(y^\alpha).$$

If $\alpha > 1$, then

$$\frac{d}{dy} (y^\alpha H_d(y)) = (1-d)(\alpha-1)y^{\alpha-2} + O(y^{\alpha-1}) > 0$$

for all sufficiently small $y > 0$. A completely monotone function must be nonincreasing, so no $\alpha > 1$ can be completely monotone.

Combining this with the source-imported strict complete monotonicity for $\alpha \leq 1$ proves $\alpha_0 = 1$.

Szabo's original function is recovered by $y = x + a$ and $d = b - a$, where $a \geq 0$, $b > 0$, and $0 < b - a < 1$. Translation from $(-a, \infty)$ to $(0, \infty)$ preserves complete-monotonicity sign tests. $\square$

Corollary 11.17 (APP-0021: Szabo's cutoff is $\alpha_0 = 1$). *For $0 < d < 1$, define*

$$H_d(y) = \psi(y+d) - \psi(y) - \frac{d}{y}.$$

The exact cutoff for complete monotonicity of $y^\alpha H_d(y)$ is

$$\alpha_0 = 1.$$

Proof. We use Proposition 11.16 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let $0 < d < 1$, $y > 0$, and

$$H_d(y) = \psi(y+d) - \psi(y) - \frac{d}{y}.$$

Then

$$y^\alpha H_d(y)$$

is strictly completely monotone on $(0, \infty)$ if $\alpha \leq 1$, and is not completely monotone if $\alpha > 1$. Hence the sharp cutoff in Szabo Open Problem 1.5 is

$$\alpha_0 = 1.$$

As $y \downarrow 0$, the digamma Laurent expansion gives

$$\psi(y) = -\frac{1}{y} - \gamma + O(y), \quad \psi(y+d) = \psi(d) + O(y).$$

Therefore

$$H_d(y) = \psi(y+d) - \psi(y) - \frac{d}{y} = \frac{1-d}{y} + O(1).$$

Since $0 < d < 1$, the leading coefficient is positive. Hence

$$y^\alpha H_d(y) = (1-d)y^{\alpha-1} + O(y^\alpha).$$

If $\alpha > 1$, then

$$\frac{d}{dy} (y^\alpha H_d(y)) = (1-d)(\alpha-1)y^{\alpha-2} + O(y^{\alpha-1}) > 0$$

for all sufficiently small $y > 0$. A completely monotone function must be nonincreasing, so no $\alpha > 1$ can be completely monotone.

Combining this with the source-imported strict complete monotonicity for $\alpha \leq 1$ proves $\alpha_0 = 1$.

Szabo's original function is recovered by $y = x + a$ and $d = b - a$, where $a \geq 0$, $b > 0$, and $0 < b - a < 1$. Translation from $(-a, \infty)$ to $(0, \infty)$ preserves complete-monotonicity sign tests. $\square$

Proposition 11.18 (Chen Choi Gurland logF strict CM). *For $F(x) = \Gamma(1/x)\Gamma(3/x)/\Gamma(2/x)^2$, the function $x \mapsto \log F(x)$ is strictly completely monotone on $(0, \infty)$.*

Proof. This proof is the component of Proposition 11.19 corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. $\square$

Proposition 11.19 (Chen Choi Gurland Euler sum normal form). *For $F(x) = \Gamma(1/x)\Gamma(3/x)/\Gamma(2/x)^2$, $\log F(x) = \log(4/3) + \sum_{m=1}^{\infty} A(mx)$, with locally uniform convergence of all derivative series on compact subintervals of $(0, \infty)$.*

Proof. Set

$$A(u) = 2\log(u+2) - \log(u+1) - \log(u+3).$$

The identity

$$A(u) = \int_0^{\infty} e^{-ut} \frac{e^{-t}(1-e^{-t})^2}{t} dt$$

shows that A is strictly completely monotone. The Weierstrass product for Γ gives

$$\log F(x) = \log \frac{4}{3} + \sum_{m \geq 1} A(mx).$$

Termwise differentiation is justified on compact subintervals of $(0, \infty)$, and each differentiated summand has the required strict alternating sign. This proves logarithmic complete monotonicity. $\square$

Corollary 11.20 (APP-0022: Chen-Choi Gurland ratio is log-completely monotone). *Let*

$$F(x) = \frac{\Gamma(1/x)\Gamma(3/x)}{\Gamma(2/x)^2}, \quad x > 0.$$

Then F is logarithmically completely monotone; equivalently,

$$(-1)^n (\log F)^{(n)}(x) > 0, \quad n = 0, 1, 2, \dots$$

Proof. We use Proposition 11.19 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$A(u) = 2\log(u+2) - \log(u+1) - \log(u+3).$$

Then $A(u) > 0$ for $u > 0$, since

$$A(u) = \log \frac{(u+2)^2}{(u+1)(u+3)}$$

and the numerator exceeds the denominator by 1. Moreover

$$A'(u) = \frac{2}{u+2} - \frac{1}{u+1} - \frac{1}{u+3} = -\frac{2}{(u+1)(u+2)(u+3)}.$$

Using

$$\int_0^{\infty} e^{-ut} e^{-at} dt = \frac{1}{u+a},$$

we get

$$-A'(u) = \int_0^\infty e^{-ut} e^{-t} (1 - e^{-t})^2 dt.$$

Since $A(u) \rightarrow 0$ as $u \rightarrow \infty$, integration from u to infinity gives

$$A(u) = \int_0^\infty e^{-ut} \frac{e^{-t} (1 - e^{-t})^2}{t} dt.$$

The density is positive, locally integrable at 0, and exponentially decaying at infinity. Hence A is strictly completely monotone.

Weierstrass' product gives

$$\log \Gamma(z) = -\log z - \gamma z - \sum_{m=1}^\infty (\log(1 + z/m) - z/m).$$

Put $z = 1/x$. In

$$\log \Gamma(z) + \log \Gamma(3z) - 2 \log \Gamma(2z),$$

the linear terms cancel. The logarithmic terms give $\log(4/3)$. The summand at index m becomes

$$\log \frac{(1 + 2z/m)^2}{(1 + z/m)(1 + 3z/m)} = \log \frac{(mx + 2)^2}{(mx + 1)(mx + 3)} = A(mx).$$

Thus

$$\log F(x) = \log \frac{4}{3} + \sum_{m=1}^\infty A(mx).$$

For every compact $x \in [\varepsilon, M]$ and every $n \geq 0$,

$$\frac{d^n}{dx^n} A(mx) = m^n A^{(n)}(mx) = O_{n,\varepsilon}(m^{-2}),$$

because the three-point second difference in $A^{(n)}$ is $O((mx)^{-n-2})$. Therefore the series may be differentiated termwise locally uniformly.

For $n = 0$, $\log F(x) > 0$ follows from $\log(4/3) > 0$ and $A(mx) > 0$. For $n \geq 1$,

$$(-1)^n (\log F)^{(n)}(x) = \sum_{m=1}^\infty m^n (-1)^n A^{(n)}(mx) > 0.$$

Therefore $\log F$ is strictly completely monotone on $(0, \infty)$, proving Chen-Choi Conjecture 1. $\square$

Definition 11.1 (Determinant and triangular compression language). A determinant compression is a reduction of a sign claim to positivity of a smaller determinant, moment matrix, or coefficient system. A triangular nonnegative compression rewrites a family of expressions as an upper-triangular nonnegative linear combination of already controlled quantities. Coefficient extraction notation $[z^k]$ denotes the coefficient of z^k in a formal power series.

Proposition 11.21 (Sokal GS all lambda derivatives positive combination). *For every $\ell \geq 1$, Sokal's generalized-Stieltjes tests satisfy $\partial_\lambda^\ell F_{n,k}^{[\lambda]}(x) = \sum_{r=0}^{k-\ell} (k!/r!) [z^{k-r}] (-\log(1-z))^\ell F_{n,r}^{[\lambda]}(x)$, with nonnegative coefficients and empty sum 0.*

Proof. The exponential generating function is

$$\mathcal{F}(z) = (1-z)^{-a} \sum_{j \geq 0} G_j(x) \left(\frac{z}{1-z} \right)^j \frac{1}{j!}.$$

Differentiating ℓ times with respect to λ multiplies $\mathcal{F}$ by $(-\log(1-z))^\ell$. Extracting coefficients gives the displayed triangular formula. Since $-\log(1-z) = \sum_{m \geq 1} z^m/m$ has non-negative coefficients, all compression coefficients are non-negative. $\square$

Proposition 11.22 (Sokal GS lambda derivative generating function). *For fixed n , Sokal's generalized-Stieltjes tests satisfy the formal exponential generating function $\sum_{k \geq 0} F_{n,k}^{[\lambda]}(x) z^k/k! = (1-z)^{-(n+\lambda)} (-1)^n \sum_{j \geq 0} x^j f^{(n+j)}(z))^j/j!$.*

Proof. Fix $n \geq 0$, $x > 0$, and write $a = n + \lambda$. For this proof f is fixed and only the coefficients depend on λ . Put

$$G_j(x) = (-1)^n x^j f^{(n+j)}(x).$$

Then Sokal's test can be written as

$$F_{n,k}^{[\lambda]}(x) = \sum_{j=0}^k \binom{k}{j} (a+j)_{k-j} G_j(x),$$

where $(u)_m = \Gamma(u+m)/\Gamma(u)$.

Consider the exponential generating series in the formal variable z :

$$\mathcal{F}(z) = \sum_{k \geq 0} F_{n,k}^{[\lambda]}(x) \frac{z^k}{k!}.$$

Coefficient extraction is finite in each degree, so this is a formal identity. Interchanging the finite coefficient contributions gives

$$\begin{aligned} \mathcal{F}(z) &= \sum_{k \geq 0} \sum_{j=0}^k \binom{k}{j} (a+j)_{k-j} G_j(x) \frac{z^k}{k!} \\ &= \sum_{j \geq 0} G_j(x) \frac{z^j}{j!} \sum_{m \geq 0} (a+j)_m \frac{z^m}{m!} \\ &= \sum_{j \geq 0} G_j(x) \frac{z^j}{j!} (1-z)^{-a-j} \\ &= (1-z)^{-a} \sum_{j \geq 0} G_j(x) \frac{(z/(1-z))^j}{j!}. \end{aligned}$$

Since $a = n + \lambda$, differentiating with respect to λ gives

$$\partial_\lambda \mathcal{F}(z) = -\log(1-z) \mathcal{F}(z).$$

But

$$-\log(1-z) = \sum_{m \geq 1} \frac{z^m}{m},$$

so comparison of coefficients of z^k yields

$$\frac{\partial_\lambda F_{n,k}^{[\lambda]}(x)}{k!} = \sum_{r=0}^{k-1} \frac{F_{n,r}^{[\lambda]}(x)}{r!} \frac{1}{k-r}.$$

Multiplying by $k!$ proves the first-derivative formula:

$$\partial_\lambda F_{n,k}^{[\lambda]}(x) = \sum_{r=0}^{k-1} \frac{k!}{r!(k-r)} F_{n,r}^{[\lambda]}(x).$$

The $k = 0$ case gives 0, as expected.

For higher derivatives, the same generating-function identity gives

$$\partial_\lambda^\ell \mathcal{F}(z) = (-\log(1-z))^\ell \mathcal{F}(z).$$

Thus

$$\frac{\partial_\lambda^\ell F_{n,k}^{[\lambda]}(x)}{k!} = \sum_{r=0}^k \frac{F_{n,r}^{[\lambda]}(x)}{r!} [z^{k-r}] (-\log(1-z))^\ell.$$

Since $-\log(1-z)$ starts with z , the coefficient is zero unless $k-r \geq \ell$, giving the displayed sum over $0 \leq r \leq k-\ell$.

All coefficients are nonnegative because $-\log(1-z) = \sum_{m \geq 1} z^m/m$ has strictly positive Taylor coefficients, and products of power series with nonnegative coefficients have nonnegative coefficients. Therefore every λ -derivative of a Sokal test is a nonnegative linear combination of lower- k tests.

The same generating function gives an integrated order-monotonicity formula. For $\tau \geq 0$,

$$\mathcal{F}_{\lambda+\tau}(z) = (1-z)^{-\tau} \mathcal{F}_\lambda(z),$$

and hence

$$F_{n,k}^{[\lambda+\tau]}(x) = k! \sum_{r=0}^k \frac{F_{n,r}^{[\lambda]}(x)}{r!} [z^{k-r}] (1-z)^{-\tau}.$$

Since $[z^m](1-z)^{-\tau} = (\tau)_m/m! \geq 0$, nonnegativity of the order- λ tests implies nonnegativity of the order- $\lambda + \tau$ tests by a direct triangular formula. $\square$

Corollary 11.23 (APP-0023: Sokal lambda-derivatives have triangular compression). *Let $n, k \in \mathbb{N}_0$, $\lambda > 0$, $x > 0$, and let f be a C^∞ function on $(0, \infty)$. In Sokal's generalized-Stieltjes application, the quantities below are the Hankel-type expressions associated with a generalized Stieltjes function of order λ . For $j \geq 0$, set*

$$G_j(x) = (-1)^n x^j f^{(n+j)}(x), \quad a = n + \lambda.$$

For $0 \leq r \leq k$, define

$$F_{n,r}^{[\lambda]}(x) = \sum_{j=0}^r \binom{r}{j} (a+j)_{r-j} G_j(x).$$

Then, for every integer $0 \leq \ell \leq k$,

$$\partial_\lambda^\ell F_{n,k}^{[\lambda]}(x) = \sum_{r=0}^{k-\ell} \frac{k!}{r!} [z^{k-r}] (-\log(1-z))^\ell F_{n,r}^{[\lambda]}(x).$$

For $\ell > k$, $\partial_\lambda^\ell F_{n,k}^{[\lambda]}(x) = 0$. The coefficients in this triangular compression are non-negative.

Proof. We use Proposition 11.22 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Fix $n \geq 0$, $x > 0$, and write $a = n + \lambda$. For this proof f is fixed and only the coefficients depend on λ . Put

$$G_j(x) = (-1)^n x^j f^{(n+j)}(x).$$

Then Sokal's test can be written as

$$F_{n,k}^{[\lambda]}(x) = \sum_{j=0}^k \binom{k}{j} (a+j)_{k-j} G_j(x),$$

where $(u)_m = \Gamma(u+m)/\Gamma(u)$.

Consider the exponential generating series in the formal variable z :

$$\mathcal{F}(z) = \sum_{k \geq 0} F_{n,k}^{[\lambda]}(x) \frac{z^k}{k!}.$$

Coefficient extraction is finite in each degree, so this is a formal identity. Interchanging the finite coefficient contributions gives

$$\begin{aligned} \mathcal{F}(z) &= \sum_{k \geq 0} \sum_{j=0}^k \binom{k}{j} (a+j)_{k-j} G_j(x) \frac{z^k}{k!} \\ &= \sum_{j \geq 0} G_j(x) \frac{z^j}{j!} \sum_{m \geq 0} (a+j)_m \frac{z^m}{m!} \\ &= \sum_{j \geq 0} G_j(x) \frac{z^j}{j!} (1-z)^{-a-j} \\ &= (1-z)^{-a} \sum_{j \geq 0} G_j(x) \frac{(z/(1-z))^j}{j!}. \end{aligned}$$

Since $a = n + \lambda$, differentiating with respect to λ gives

$$\partial_\lambda \mathcal{F}(z) = -\log(1-z) \mathcal{F}(z).$$

But

$$-\log(1-z) = \sum_{m \geq 1} \frac{z^m}{m},$$

so comparison of coefficients of z^k yields

$$\frac{\partial_\lambda F_{n,k}^{[\lambda]}(x)}{k!} = \sum_{r=0}^{k-1} \frac{F_{n,r}^{[\lambda]}(x)}{r!} \frac{1}{k-r}.$$

Multiplying by $k!$ proves the first-derivative formula:

$$\partial_\lambda F_{n,k}^{[\lambda]}(x) = \sum_{r=0}^{k-1} \frac{k!}{r!(k-r)} F_{n,r}^{[\lambda]}(x).$$

The $k = 0$ case gives 0, as expected.

For higher derivatives, the same generating-function identity gives

$$\partial_\lambda^\ell \mathcal{F}(z) = (-\log(1-z))^\ell \mathcal{F}(z).$$

Thus

$$\frac{\partial_\lambda^\ell F_{n,k}^{[\lambda]}(x)}{k!} = \sum_{r=0}^k \frac{F_{n,r}^{[\lambda]}(x)}{r!} [z^{k-r}] (-\log(1-z))^\ell.$$

Since $-\log(1-z)$ starts with z , the coefficient is zero unless $k-r \geq \ell$, giving the displayed sum over $0 \leq r \leq k-\ell$.

All coefficients are nonnegative because $-\log(1-z) = \sum_{m \geq 1} z^m/m$ has strictly positive Taylor coefficients, and products of power series with nonnegative coefficients have nonnegative coefficients. Therefore every λ -derivative of a Sokal test is a nonnegative linear combination of lower- k tests.

The same generating function gives an integrated order-monotonicity formula. For $\tau \geq 0$,

$$\mathcal{F}_{\lambda+\tau}(z) = (1-z)^{-\tau} \mathcal{F}_\lambda(z),$$

and hence

$$F_{n,k}^{[\lambda+\tau]}(x) = k! \sum_{r=0}^k \frac{F_{n,r}^{[\lambda]}(x)}{r!} [z^{k-r}] (1-z)^{-\tau}.$$

Since $[z^m](1-z)^{-\tau} = (\tau)_m/m! \geq 0$, nonnegativity of the order- λ tests implies nonnegativity of the order- $\lambda + \tau$ tests by a direct triangular formula. $\square$

Proposition 11.24 (Simon gamma quotient derivative positive laplace density). *For every $0 < \alpha < 1$, the derivative of $F_\alpha(x) = \Gamma(x+\alpha)/(\Gamma(x)x^\alpha)$ is completely monotone, with a positive Laplace-density representation obtained from the decreasing kernel J_α .*

Proof. Put $\beta = 1 - \alpha$. The beta integral gives

$$F_\alpha(x) = x^\beta \frac{\Gamma(x+\alpha)}{\Gamma(x+1)} = \frac{x^\beta}{\Gamma(\beta)} \int_0^\infty e^{-xu} (e^u - 1)^{-\alpha} du.$$

Define

$$J_\alpha(t) = \int_0^t (t-u)^{\alpha-1} (e^u - 1)^{-\alpha} du.$$

Laplace convolution gives

$$F_\alpha(x) = \frac{x}{\Gamma(\alpha)\Gamma(1-\alpha)} \int_0^\infty e^{-xt} J_\alpha(t) dt.$$

After the substitution $u = tv$,

$$J_\alpha(t) = \int_0^1 (1-v)^{\alpha-1} \left(\frac{t}{e^{tv} - 1} \right)^\alpha dv.$$

For each $v > 0$, the function $t/(e^{tv} - 1)$ is decreasing in t . The integrand is dominated near the endpoints by $(1-v)^{\alpha-1}v^{-\alpha}$, which is integrable because $0 < \alpha < 1$. Hence J_α is decreasing, with

$$J_\alpha(0+) = \Gamma(\alpha)\Gamma(1-\alpha), \quad J_\alpha(\infty) = 0.$$

Let $\mu_\alpha = -dJ_\alpha$. This is a positive finite measure, and Stieltjes integration by parts yields

$$1 - F_\alpha(x) = \frac{1}{\Gamma(\alpha)\Gamma(1-\alpha)} \int_0^\infty e^{-xt} \mu_\alpha(dt).$$

Thus $1 - F_\alpha$ is completely monotone. Differentiating this Laplace representation gives

$$F'_\alpha(x) = \frac{1}{\Gamma(\alpha)\Gamma(1-\alpha)} \int_0^\infty t e^{-xt} \mu_\alpha(dt),$$

so F'_α is completely monotone. Therefore F_α is a Bernstein function. $\square$

Proposition 11.25 (Bernstein derivative complete monotonicity criterion). *Let f be nonnegative and continuously differentiable on an interval. If f' is completely monotone, then f is a Bernstein function on that interval. Conversely, for a differentiable Bernstein function, f' is completely monotone.*

Proof. Let f be nonnegative and continuously differentiable on an interval (a, ∞) . A Bernstein function is exactly a nonnegative function whose derivative is completely monotone. Hence, if f' is completely monotone, the defining criterion is satisfied and f is Bernstein. Conversely, if f is a differentiable Bernstein function, the same defining criterion gives complete monotonicity of f' . This proves the criterion. $\square$

Corollary 11.26 (APP-0024: Simon gamma quotient is Bernstein). *For $0 < \alpha < 1$, define*

$$F_\alpha(x) = \frac{\Gamma(x + \alpha)}{\Gamma(x)x^\alpha}, \quad x > 0.$$

Then F_α is a Bernstein function. More precisely, $1 - F_\alpha$ is completely monotone.

Proof. We use Proposition 11.25 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

For $\alpha, \beta > 0$, $s, t \geq 0$, and $x > 0$, the beta identity gives

$$(x + s)^{-\alpha}(x + t)^{-\beta} = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 \frac{u^{\alpha-1}(1-u)^{\beta-1}}{(x + us + (1-u)t)^{\alpha+\beta}} du.$$

If

$$f(x) = \int_{[0, \infty)} (x + s)^{-\alpha} d\mu(s), \quad g(x) = \int_{[0, \infty)} (x + t)^{-\beta} d\nu(t),$$

then Tonelli gives

$$f(x)g(x) = \int_{[0, \infty)} (x + r)^{-(\alpha+\beta)} d\kappa(r),$$

where κ is the positive pushforward measure defined by

$$\int \varphi(r) d\kappa(r) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \iint \int_0^1 \varphi(us + (1-u)t) u^{\alpha-1}(1-u)^{\beta-1} du d\mu(s) d\nu(t).$$

Thus pure generalized Stieltjes transforms satisfy

$$\mathcal{S}_\alpha^0 \mathcal{S}_\beta^0 \subseteq \mathcal{S}_{\alpha+\beta}^0.$$

With the usual nonnegative constant term convention, constants and lower-order terms require the standard order-lift convention already recorded in the source vocabulary; no application should rely on constants being absorbed unless that convention is explicitly cited.

For $0 < \alpha < 1$, set

$$F_\alpha(x) = \frac{\Gamma(x + \alpha)}{\Gamma(x)x^\alpha}, \quad L_\alpha(x) = \log F_\alpha(x).$$

Then

$$L'_\alpha(x) = \psi(x + \alpha) - \psi(x) - \frac{\alpha}{x} = \int_0^\infty e^{-xt} \left(\frac{1 - e^{-\alpha t}}{1 - e^{-t}} - \alpha \right) dt.$$

The kernel is positive for $t > 0$, because $u \mapsto 1 - e^{-ut}$ is strictly concave on $[0, 1]$, hence

$$1 - e^{-\alpha t} > \alpha(1 - e^{-t}).$$

Therefore L'_α is completely monotone, and $1/F_\alpha$ is logarithmically completely monotone.

This does not prove that F_α is a Bernstein function. Simon's source explicitly states that the Bernstein character of the corresponding $\Phi(\lambda) = \Gamma(1 - t + \lambda)/(\lambda^{1-t}\Gamma(\lambda))$ is a puzzling question, and also says the reciprocal LCM property is necessary but not sufficient. $\square$

Proposition 11.27 (Du Wang h3 classification assembly result). *The admitted middle-window increasing theorem for $1/2 \leq a \leq 1$ and the admitted outer-window nonmonotonicity obstruction for $0 < a < 1/2$ or $1 < a < 2$ assemble to the full Du–Wang h_3 classification on $0 < a < 2$.*

Proof. The middle-window monotonicity theorem on $1/2 \leq a \leq 1$ and the outer-window nonmonotonicity theorem on $0 < a < 1/2$ and $1 < a < 2$ partition the domain, and together they give exactly the source classification. $\square$

Proposition 11.28 (Du Wang polygamma ratio halfline bound). *For every $t > 1/2$, one has $-2\psi''(t)/\psi'''(t) \geq t - 1/2$.*

Proof. For $t > 1/2$, define

$$S_m(t) = \sum_{n=0}^{\infty} \frac{1}{(n+t)^m}.$$

The polygamma identities give

$$\psi''(t) = -2S_3(t), \quad \psi'''(t) = 6S_4(t).$$

Thus the desired bound

$$-\frac{2\psi''(t)}{\psi'''(t)} \geq t - \frac{1}{2}$$

is equivalent to

$$2S_3(t) - 3\left(t - \frac{1}{2}\right)S_4(t) \geq 0.$$

Write $y = t - \frac{1}{2} > 0$. Using

$$S_m(t) = \frac{1}{\Gamma(m)} \int_0^\infty \frac{x^{m-1}e^{-tx}}{1 - e^{-x}} dx$$

and

$$\frac{e^{-tx}}{1 - e^{-x}} = e^{-yx} \frac{1}{2 \sinh(x/2)},$$

set

$$K(x) = \frac{1}{2 \sinh(x/2)}.$$

Then

$$2S_3(t) - 3yS_4(t) = \int_0^\infty e^{-yx} x^2 K(x) dx - \frac{y}{2} \int_0^\infty e^{-yx} x^3 K(x) dx.$$

Since $x^3 K(x) \sim x^2$ as $x \downarrow 0$ and decays exponentially at infinity, integration by parts gives

$$y \int_0^\infty e^{-yx} x^3 K(x) dx = \int_0^\infty e^{-yx} (x^3 K(x))' dx.$$

Therefore

$$2S_3(t) - 3yS_4(t) = \int_0^\infty e^{-yx} \left(x^2 K(x) - \frac{1}{2} (x^3 K(x))' \right) dx.$$

Now

$$K'(x) = -\frac{1}{2} K(x) \coth(x/2),$$

so

$$x^2 K(x) - \frac{1}{2} (x^3 K(x))' = \frac{1}{2} x^2 K(x) \left(\frac{x}{2} \coth(x/2) - 1 \right).$$

For $z > 0$, $z \coth z > 1$. The integrand is therefore strictly positive, and

$$2S_3(t) - 3 \left(t - \frac{1}{2} \right) S_4(t) > 0.$$

This proves

$$-\frac{2\psi''(t)}{\psi'''(t)} > t - \frac{1}{2}, \quad t > \frac{1}{2}.$$

This proves the Du Wang polygamma ratio halfline bound.

Let $1/2 \leq a \leq 1$, $u > 0$, and $t = a + u$. Since $a \geq 1/2$,

$$u = t - a \leq t - \frac{1}{2}.$$

The ratio lemma gives

$$t - \frac{1}{2} < -\frac{2\psi''(t)}{\psi'''(t)}.$$

Because $\psi'''(t) > 0$,

$$2\psi''(t) + u\psi'''(t) < 0.$$

Consequently

$$H'_a(u) = -u^2 \{2\psi''(t) + u\psi'''(t)\} > 0.$$

Together with $H_a(0+) = 2 \log \Gamma(a) \geq 0$, this gives

$$H_a(u) \geq 0 \quad (u > 0, 1/2 \leq a \leq 1).$$

This proves the Du Wang h31 middle window u monotonicity.

For $x > 0$, $u = x$, and $t = x + a$,

$$h'_3(x) = \tilde{h}'_3(x+a) = \frac{h_{31}(x+a)}{x^2} = \frac{H_a(x)}{x^2} \geq 0.$$

The inequality is strict for $x > 0$ except possibly at the endpoint $a = 1, u = 0$, which is outside the open x -interval. Thus h_3 is increasing on $(0, \infty)$ for every

$$\frac{1}{2} \leq a \leq 1.$$

This monotonicity result combines with the previously proved outer-window nonmonotonicity theorem to complete the open-problem source classification.

The outer-window nonmonotonicity result proves that h_3 is not monotone for

$$0 < a < \frac{1}{2} \quad \text{or} \quad 1 < a < 2.$$

The middle-window theorem above proves increasing behavior for

$$\frac{1}{2} \leq a \leq 1.$$

Therefore Du–Wang Open Problem 2 is classified on $0 < a < 2$:

$$h_3 \text{ is increasing on } (0, \infty) \iff \frac{1}{2} \leq a \leq 1,$$

and it is not monotone on the two outer windows.

This proves the Du Wang h_3 open problem 2 classification. $\square$

Counterexample 11.29 (APP-0025: Du–Wang h_3 monotonicity classification). For $0 < a < 2$, define

$$h_3(x) = \frac{-x^2\psi'(x+a) + 2x\psi(x+a) - 2\log\Gamma(x+a)}{x}, \quad x > 0.$$

Then h_3 is increasing on $(0, \infty)$ if and only if

$$\frac{1}{2} \leq a \leq 1.$$

For $0 < a < 1/2$ and for $1 < a < 2$, h_3 is not monotone on $(0, \infty)$.

Proof. We use Proposition 11.28 and Proposition 11.27 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The middle-window monotonicity theorem on $1/2 \leq a \leq 1$ and the outer-window nonmonotonicity theorem on $0 < a < 1/2$ and $1 < a < 2$ partition the domain, and together they give exactly the source classification. $\square$

Proposition 11.30 (Baskakov alpha1 r8 not CM). *The Baskakov $\alpha = 1$, $r = 8$ function $f_1^{[8]}(x) = 1/((1+x)^8 - x^8)$ is not completely monotone on $(0, \infty)$.*

Proof. For even $r = 2m$ and $\alpha = 1$, the source definition reduces to

$$f_1^{[2m]}(x) = \frac{1}{(1+x)^{2m} - x^{2m}}.$$

Let $D_m(x) = (1+x)^{2m} - x^{2m}$. Its roots are

$$\lambda_{k,m} = -\frac{1}{2} - \frac{i}{2} \cot \frac{\pi k}{2m}, \quad k = 1, \dots, 2m-1,$$

and the corresponding residues are

$$R_{k,m} = \frac{(-1)^{m+k}}{2m} \left(2 \sin \frac{\pi k}{2m} \right)^{2m-2}.$$

Thus the inverse-Laplace density of $1/D_m$, if it were nonnegative, would be

$$\rho_m(t) = \frac{e^{-t/2}}{2m} \left[2^{2m-2} + 2 \sum_{k=1}^{m-1} (-1)^{m+k} \left(2 \sin \frac{\pi k}{2m} \right)^{2m-2} \cos \left(\frac{t}{2} \cot \frac{\pi k}{2m} \right) \right].$$

For $m = 4$, write $h_4(t) = e^{t/2} \rho_4(t)$. The formula simplifies to

$$\begin{aligned} h_4(t) = & 8 + 2 \cos \frac{t}{2} - \frac{(2 - \sqrt{2})^3}{4} \cos \left(\frac{1 + \sqrt{2}}{2} t \right) \\ & - \frac{(2 + \sqrt{2})^3}{4} \cos \left(\frac{\sqrt{2} - 1}{2} t \right). \end{aligned}$$

At $t = 10\pi$,

$$h_4(10\pi) = 6 + 10 \cos(5\pi\sqrt{2}).$$

Since $0 < 5\sqrt{2} - 7 < 1/4$,

$$\cos(5\pi\sqrt{2}) = -\cos(\pi(5\sqrt{2} - 7)) < -\frac{\sqrt{2}}{2} < -\frac{3}{5}.$$

Consequently $h_4(10\pi) < 0$, and so $\rho_4(10\pi) < 0$. By uniqueness of Laplace transforms of positive measures, $f_1^{[8]}$ cannot be completely monotone. $\square$

Counterexample 11.31 (APP-0026: Baskakov even-power conjecture is false). For the Abel–Gawronski–Neuschel higher-power Baskakov family

$$f_\alpha^{[r]}(x) = (1+x)^{-r\alpha} \sum_{k=0}^{\infty} \binom{-\alpha}{k}^r \left(\frac{x}{1+x} \right)^{rk},$$

the universal even-power complete-monotonicity conjecture is false. Already at $\alpha = 1$ and $r = 8$,

$$f_1^{[8]}(x) = \frac{1}{(1+x)^8 - x^8}$$

is not completely monotone on $(0, \infty)$.

Proof. We use Proposition 8.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Set

$$D_m(x) = (1+x)^{2m} - x^{2m}.$$

The roots are obtained from

$$\frac{1+x}{x} = \omega_k, \quad \omega_k = e^{\pi i k/m}, \quad k = 1, \dots, 2m-1.$$

Thus

$$\lambda_{k,m} = \frac{1}{\omega_k - 1} = -\frac{1}{2} - \frac{i}{2} \cot \frac{\pi k}{2m}.$$

At this root,

$$D'_m(\lambda_{k,m}) = 2m\lambda_{k,m}^{2m-1}(\omega_k^{-1} - 1),$$

and direct simplification gives the real residue

$$R_{k,m} = \frac{1}{D'_m(\lambda_{k,m})} = \frac{(-1)^{m+k}}{2m} \left(2 \sin \frac{\pi k}{2m} \right)^{2m-2}.$$

Consequently the inverse-Laplace density is

$$\rho_m(t) = \sum_{k=1}^{2m-1} R_{k,m} e^{\lambda_{k,m} t}.$$

Pairing conjugate roots gives

$$\rho_m(t) = \frac{e^{-t/2}}{2m} \left[2^{2m-2} + 2 \sum_{k=1}^{m-1} (-1)^{m+k} \left(2 \sin \frac{\pi k}{2m} \right)^{2m-2} \cos \left(\frac{t}{2} \cot \frac{\pi k}{2m} \right) \right].$$

This proves the finite trigonometric density formula used by the diagnostic candidate.

For $m = 2$, this gives

$$\rho_2(t) = e^{-t/2} \left(1 - \cos \frac{t}{2} \right) = 2e^{-t/2} \sin^2 \frac{t}{4} \geq 0,$$

recovering the previously admitted $r = 4, \alpha = 1$ seed.

For $m = 3$,

$$\rho_3(t) = \frac{e^{-t/2}}{6} \left[16 + 2 \cos \left(\frac{\sqrt{3}t}{2} \right) - 18 \cos \left(\frac{t}{2\sqrt{3}} \right) \right].$$

Set $u = t/(2\sqrt{3})$. Since $\cos(3u) = 4 \cos^3 u - 3 \cos u$,

$$\rho_3(t) = \frac{4}{3} e^{-t/2} (1 - \cos u)^2 (2 + \cos u) \geq 0.$$

Thus $r = 6, \alpha = 1$ is completely monotone by an explicit positive density.

For $m = 4$, write $h_4(t) = e^{t/2} \rho_4(t)$. The density formula gives

$$h_4(t) = 8 + 2 \cos \frac{t}{2} - \frac{(2 - \sqrt{2})^3}{4} \cos \left(\frac{1 + \sqrt{2}}{2} t \right) - \frac{(2 + \sqrt{2})^3}{4} \cos \left(\frac{\sqrt{2} - 1}{2} t \right).$$

At $t = 10\pi$,

$$\cos(t/2) = \cos(5\pi) = -1,$$

and both other cosine terms equal $-\cos(5\pi\sqrt{2})$. Since

$$(2 - \sqrt{2})^3 + (2 + \sqrt{2})^3 = 40,$$

we obtain

$$h_4(10\pi) = 6 + 10 \cos(5\pi\sqrt{2}).$$

Now

$$0 < 5\sqrt{2} - 7 < \frac{1}{4}$$

because $7/5 < \sqrt{2} < 29/20$. Hence, with $\delta = 5\sqrt{2} - 7$,

$$\cos(5\pi\sqrt{2}) = \cos(7\pi + \pi\delta) = -\cos(\pi\delta) < -\cos\frac{\pi}{4} = -\frac{\sqrt{2}}{2} < -\frac{3}{5}.$$

Therefore

$$h_4(10\pi) < 6 + 10\left(-\frac{3}{5}\right) = 0,$$

and

$$\rho_4(10\pi) = e^{-5\pi}h_4(10\pi) < 0.$$

Since the inverse-Laplace transform is unique for measures on $[0, \infty)$, $f_1^{[8]}(x) = 1/((1+x)^8 - x^8)$ cannot be completely monotone. $\square$

Definition 11.2 (Log-function derivative-sign regions). For a finite-dimensional class of log-functions $\mathcal{F}$ and an operator $L = -d/dx$, the derivative-sign region D_k consists of functions $f \in \mathcal{F}$ for which $L^k f \geq 0$ on the stated domain. A descending-chain claim asserts $D_{k+1} \subseteq D_k$.

Proposition 11.32 (Ma Weigert D_k chain integration proof). For $f \in \mathcal{F}_{1,n}$, if $L^{k+1}f \geq 0$ on $(0, \infty)$, then $L^k f \geq 0$ on $(0, \infty)$, because $L^k f(x) = \int_x^\infty L^{k+1}f(t) dt$.

Proof. For $f(x) = p(\log x)/x$, induction gives

$$L^k f(x) = x^{-k-1}p_k(\log x)$$

for a polynomial p_k . Indeed, if the formula holds at level k , then

$$L^{k+1}f(x) = x^{-k-2}((k+1)p_k(\log x) - p'_k(\log x)).$$

Thus $L^k f(x) \rightarrow 0$ as $x \rightarrow \infty$. If $f \in D_{k+1}$, put $g = L^k f$. Then $-g' = L^{k+1}f \geq 0$, and hence, for $R > x$,

$$g(x) - g(R) = \int_x^R L^{k+1}f(t) dt.$$

Letting $R \rightarrow \infty$ gives

$$L^k f(x) = \int_x^\infty L^{k+1}f(t) dt \geq 0.$$

Therefore $f \in D_k$. $\square$

Corollary 11.33 (APP-0027: Ma–Weigert derivative regions form a chain). Let

$$\mathcal{F}_{1,n} = \left\{ x \mapsto \frac{p(\log x)}{x} : p \in \mathbb{R}[y]_n \right\},$$

and let $L = -d/dx$. If

$$D_k = \{f \in \mathcal{F}_{1,n} : L^k f(x) \geq 0 \text{ for all } x > 0\},$$

then

$$D_{k+1} \subseteq D_k \quad (k \geq 0, n \geq 0).$$

Proof. We use Proposition 11.32 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

We first prove that for each $k \geq 0$,

$$L^k f(x) = x^{-k-1} p_k(\log x)$$

for some polynomial p_k .

For $k = 0$, this is the definition with $p_0 = p$. Suppose

$$L^k f(x) = x^{-k-1} p_k(\log x).$$

Then

$$\begin{aligned} L^{k+1} f(x) &= -\frac{d}{dx} \left(x^{-k-1} p_k(\log x) \right) \\ &= x^{-k-2} \left((k+1) p_k(\log x) - p'_k(\log x) \right). \end{aligned}$$

Thus $p_{k+1}(y) = (k+1)p_k(y) - p'_k(y)$, which is again a polynomial.

Since every polynomial in $\log x$ grows slower than every positive power of x ,

$$x^{-k-1} p_k(\log x) \rightarrow 0 \quad (x \rightarrow \infty).$$

Therefore

$$\lim_{x \rightarrow \infty} L^k f(x) = 0$$

for every $k \geq 0$.

Assume $f \in D_{k+1}$. Then

$$L^{k+1} f(x) \geq 0$$

on $(0, \infty)$. Put

$$g(x) = L^k f(x).$$

By definition of L ,

$$-g'(x) = L^{k+1} f(x) \geq 0.$$

Using $g(x) \rightarrow 0$ as $x \rightarrow \infty$, integrate from x to R :

$$g(x) - g(R) = \int_x^R L^{k+1} f(t) dt.$$

Letting $R \rightarrow \infty$ gives

$$L^k f(x) = g(x) = \int_x^\infty L^{k+1} f(t) dt \geq 0.$$

Hence $f \in D_k$. Therefore

$$D_{k+1} \subseteq D_k$$

for all $k \geq 0$ and all n . □

Proposition 11.34 (QA divisor polygamma even claim refuted). *The Qi–Agarwal/Yin claim that $f_{2\ell}$ is not completely monotone for every $\ell \in \mathbb{N}$ is false, because f_2 is completely monotone.*

Proof. The standard polygamma representation gives, for $k \geq 1$,

$$(-1)^{k+1}\psi^{(k)}(x) = \int_0^\infty \frac{t^k e^{-xt}}{1 - e^{-t}} dt.$$

Thus $\psi^{(k)}$ is a positive completely monotone function whenever k is odd. If n is odd and $km = n$, both k and m are odd, so each summand $[\psi^{(k)}]^m$ is a finite product of positive completely monotone functions. Finite sums preserve complete monotonicity.

For $n = 2$, the divisor pairs are $(1, 2)$ and $(2, 1)$, hence

$$f_2(x) = [\psi'(x)]^2 + \psi''(x).$$

Qi–Agarwal record the sharp theorem that

$$[\psi'(x)]^2 + \lambda\psi''(x)$$

is completely monotone on $(0, \infty)$ if and only if $\lambda \leq 1$. Taking $\lambda = 1$ proves that f_2 is completely monotone. This refutes the stated even-parity clause at $\ell = 1$. $\square$

Proposition 11.35 (QA divisor polygamma odd CM). *For every odd positive integer n , the divisor-polygamma sum $f_n(x) = \sum_{km=n} [\psi^{(k)}(x)]^m$ is completely monotone on $(0, \infty)$.*

Proof. This proof is the component of Proposition 11.34 corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. $\square$

Counterexample 11.36 (Parity classification assembly principle). Suppose a source parity classification is split into two disjoint parity classes. If the positive class is proved by one theorem and the claimed negative class is refuted by one admissible counterexample, then the corrected parity classification is the conjunction of those two upstream results.

Proof. Let a classification problem split a family $\mathcal{F}$ into two disjoint classes $\mathcal{F}_0$ and $\mathcal{F}_1$. Suppose one theorem proves the asserted positive property for every member of $\mathcal{F}_1$, and another theorem or counterexample determines that the proposed universal assertion on $\mathcal{F}_0$ must be corrected. Then the corrected classification is obtained by taking the conjunction of the two upstream results.

The two classes are disjoint and exhaustive, so every member of $\mathcal{F}$ lies in exactly one class. The first upstream result decides all cases in $\mathcal{F}_1$, and the second decides the corrected status of $\mathcal{F}_0$. Hence their conjunction is a complete classification for $\mathcal{F}$. $\square$

Counterexample 11.37 (APP-0028: divisor-polygamma parity correction). For $n \in \mathbb{N}$, set

$$f_n(x) = \sum_{km=n} [\psi^{(k)}(x)]^m, \quad x > 0.$$

Then f_n is completely monotone for every odd n . The asserted opposite parity law that $f_{2\ell}$ is never completely monotone is false:

$$f_2(x) = [\psi'(x)]^2 + \psi''(x)$$

is completely monotone on $(0, \infty)$.

Proof. We use Proposition 11.36 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The node the QA divisor polygamma odd CM proves that every odd divisor-polygamma sum f_n is completely monotone. The node the QA divisor polygamma even claim refuted, itself following from the QA divisor polygamma f2 CM, proves that the source's universal even non-completeness-monotonicity clause is false. These two admitted statements exactly resolve the parity claim: the odd assertion holds, while the even negation asserted in the source does not hold in full generality. $\square$

Proposition 11.38 (BZ gamma quotient critical window reduction). *There is a rational interval J around $1.126207061\dots$ such that $\tilde{F}' < 0$ on $(-1, \inf J]$, $\tilde{F}' > 0$ on $[\sup J, \infty)$, and $\tilde{F}'$ has exactly one zero inside J .*

Proof. The source proves $F' < 0$ on $(-1, 1)$. For $x > 1$, set

$$G(x) = \log \Gamma(x+1), \quad D(x) = \log \frac{x^2 + 6}{x + 6},$$

and

$$N_0(x) = \psi(x+1)D(x) - G(x)D'(x).$$

Since $F' = N_0/D^2$, the sign of F' on $x > 1$ is the sign of N_0 . A direct right-tail estimate gives $N_0(x) > 0$ for $x \geq 8$: with

$$P(x) = x^2 + 12x - 6, \quad Q(x) = (x^2 + 6)(x + 6),$$

one has $D' = P/Q$; the bounds $\psi(x+1) > \log x$, $\log \Gamma(x+1) \leq x \log x$, and

$$D(x) \geq \frac{xP(x)}{Q(x)} \quad (x \geq 8)$$

imply $N_0(x) > 0$.

It remains to prove a single crossing on $(1, 8]$. Define

$$r(x) = \frac{\psi(x+1)}{D'(x)} = \psi(x+1) \frac{Q(x)}{P(x)}.$$

Then

$$P(x)^2 r'(x) = S(x),$$

where

$$S(x) = \psi'(x+1)Q(x)P(x) + \psi(x+1)\{Q'(x)P(x) - Q(x)P'(x)\}.$$

Using the elementary bounds, valid for $y \geq 2$,

$$\psi(y) > \log y - \frac{1}{y}, \quad \psi'(y) > \frac{1}{y} + \frac{1}{2y^2}, \quad \psi''(y) > -\frac{1}{y^2} - \frac{2}{y^3},$$

one obtains $S'(x) > 0$ on $[1, 8]$. Also

$$S(1) = 343 \left(\frac{\pi^2}{6} - 1 \right) - 539(1 - \gamma) < 0,$$

while $S(8) > 0$. Hence r' has exactly one zero on $[1, 8]$.

Now put $I(x) = D(x)r(x) - G(x)$. Then $N_0(x) = D'(x)I(x)$ and

$$I'(x) = D(x)r'(x).$$

Since $D(x) > 0$ for $x > 1$, I decreases once and then increases once. The removable endpoint has $\lim_{x \downarrow 1} I(x) = 0$, and the quadratic leading coefficient of N_0 at 1 is negative. Finally,

$$N_0(8) = (H_8 - \gamma) \log 5 - \frac{11}{70} \log(40320) > 0.$$

Thus N_0 , and hence F' , has exactly one zero on $(1, 8]$. Combining this with the source interval $(-1, 1)$ and the right-tail estimate proves the theorem. $\square$

Proposition 11.39 (BZ gamma quotient N single crossing one eight). *Let $N(x)$ be the Bulboaca–Zayed derivative-sign numerator from the Bulboaca–Zayed gamma-quotient derivative normal form. On $[1, 8]$, N has exactly one zero ξ , with $N(x) < 0$ for $1 \leq x < \xi$ and $N(x) > 0$ for $\xi < x \leq 8$.*

Proof. This proof is the component of Proposition 11.38 corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. $\square$

Corollary 11.40 (APP-0029: Bulboaca–Zayed Gamma quotient one-critical-point theorem). *Let*

$$F(x) = \frac{\log \Gamma(x+1)}{\log(x^2+6) - \log(x+6)}$$

with the removable values filled in on $(-1, \infty)$. Then F' has a unique zero x_m on $(-1, \infty)$. Moreover

$$F'(x) < 0 \quad (-1 < x < x_m), \quad F'(x) > 0 \quad (x > x_m),$$

and numerically $x_m = 1.126207061051643 \dots$

Proof. We use Proposition 11.39 and Proposition 11.38 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

proof.

The raw numerator has a removable zero at $x = 1$, since $G(1) = D(1) = 0$. Its quadratic leading coefficient is

$$\lim_{x \downarrow 1} \frac{N_0(x)}{(x-1)^2} = \frac{7(\pi^2/6 - 1) - 11(1 - \gamma)}{98} < 0,$$

again by $\pi^2 < 987/100$ and $\gamma < 5773/10000$, which gives $-1347/980000 < 0$. Therefore any normalization dividing out the removable $(x-1)^2$ factor is negative at the left endpoint.

The single-crossing candidate is true:

$$\text{theBulboaca} - \text{Zayedgamma} - \text{quotientNsinglecrossingoneeight}.$$

Together with the source's theorem on $(-1, 1)$ and the previous local right-tail theorem for $x \geq 8$, this proves the critical-window reduction

$$\text{theBulboaca} - \text{Zayedgamma} - \text{quotientcriticalwindowreduction}$$

and then the source problem

$$\text{theBulboacaZayedgammaquotientfullmonotonicity}.$$

The finite-envelope candidate is not separately promoted: the proof above is a clean analytic ratio-kernel argument rather than the finite rational cover specified in that candidate. The diagnostic obstruction-map candidate is also not needed after the single-crossing proof. $\square$

Proposition 11.41 (Ramanujan Turan n2 upper window counterfamily). *For the Ramanujan integral Turan gap, there are parameters $\alpha \in (0, 1/2)$ such that $H_2(x; \alpha) < 0$ for some $x > 0$; specifically, with $x = e^{-L}$ and $\alpha_L = 1/2 - 1/(4L)$, this holds for all sufficiently large L .*

Proof. Put

$$A_k(x) = (-1)^k I_R^{(k)}(x) = \int_0^\infty e^{-xt} \frac{t^{k-1}}{\pi^2 + \log^2 t} dt.$$

Then

$$H_2(x; \alpha) = A_2(x)^2 - \alpha A_1(x) A_3(x).$$

Let $x = e^{-L}$. After the substitution $y = xt$,

$$A_k(e^{-L}) = e^{kL} \int_0^\infty \frac{e^{-y} y^{k-1}}{\pi^2 + (L + \log y)^2} dy.$$

Splitting at $|\log y| \leq L^{2/3}$ and expanding the denominator uniformly on the central region gives, for fixed $k = 1, 2, 3$,

$$A_k(e^{-L}) = e^{kL} L^{-2} \Gamma(k) \left(1 - \frac{2\psi(k)}{L} + O(L^{-2}) \right).$$

Consequently

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L}) A_3(e^{-L})} = \frac{1}{2} \left(1 + \frac{2(\psi(1) + \psi(3) - 2\psi(2))}{L} + O(L^{-2}) \right).$$

Since

$$\psi(1) = -\gamma, \quad \psi(2) = 1 - \gamma, \quad \psi(3) = \frac{3}{2} - \gamma,$$

the bracketed correction is $-1/L + O(L^{-2})$, and hence

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L}) A_3(e^{-L})} = \frac{1}{2} - \frac{1}{2L} + O(L^{-2}).$$

Choose $\alpha_L = 1/2 - 1/(4L)$. For all sufficiently large L , $\alpha_L \in (0, 1/2)$ and

$$A_2(e^{-L})^2 - \alpha_L A_1(e^{-L}) A_3(e^{-L}) < 0.$$

Thus $H_2(e^{-L}; \alpha_L) < 0$. Complete monotonicity would imply nonnegativity at order zero, so the source's universal interval statement is false. $\square$

Counterexample 11.42 (APP-0030: Ramanujan Turan window has a negative answer). Let

$$I_R(x) = \int_0^\infty e^{-xt} \frac{dt}{t(\pi^2 + \log^2 t)}$$

and, for $n \geq 2$,

$$H_n(x; \alpha) = (I_R^{(n)}(x))^2 - \alpha I_R^{(n-1)}(x) I_R^{(n+1)}(x).$$

The proposed complete-monotonicity window

$$\frac{n-2}{n-1} < \alpha < \frac{n-1}{n}$$

is false. It already fails for $n = 2$.

Proof. We use Proposition 8.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Put

$$A_k(x) = (-1)^k I_R^{(k)}(x) = \int_0^\infty e^{-xt} \frac{t^{k-1}}{\pi^2 + \log^2 t} dt \quad (k \geq 1).$$

Then

$$H_2(x; \alpha) = A_2(x)^2 - \alpha A_1(x) A_3(x).$$

We study $x = e^{-L}$ as $L \rightarrow \infty$. With $y = xt$,

$$A_k(e^{-L}) = e^{kL} \int_0^\infty \frac{e^{-y} y^{k-1}}{\pi^2 + (L + \log y)^2} dy.$$

For fixed $k = 1, 2, 3$,

$$A_k(e^{-L}) = e^{kL} L^{-2} \Gamma(k) \left(1 - \frac{2\psi(k)}{L} + O(L^{-2}) \right),$$

where ψ is the digamma function. To justify the expansion, split the y -integral into the central region $|\log y| \leq L^{2/3}$ and its complement. On the central interval, with $z = \log y$, expand uniformly:

$$\frac{1}{\pi^2 + (L + \log y)^2} = L^{-2} \left(1 - \frac{2 \log y}{L} + O\left(\frac{1 + \log^2 y}{L^2}\right) \right),$$

and integrate against $e^{-y} y^{k-1}$. The lower complement $0 < y < e^{-L^{2/3}}$ is $O(e^{-cL^{2/3}})$ for $k = 1, 2, 3$ after the change $y = e^{-u}$, and the upper complement $y > e^{L^{2/3}}$ is super-exponentially small because of e^{-y} . These errors are $o(L^{-N})$ for every fixed N , which is more than needed for the displayed expansion.

Therefore

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L})A_3(e^{-L})} = \frac{\Gamma(2)^2}{\Gamma(1)\Gamma(3)} \left(1 + \frac{2(\psi(1) + \psi(3) - 2\psi(2))}{L} + O(L^{-2}) \right).$$

Using

$$\psi(1) = -\gamma, \quad \psi(2) = 1 - \gamma, \quad \psi(3) = \frac{3}{2} - \gamma,$$

we get

$$\psi(1) + \psi(3) - 2\psi(2) = -\frac{1}{2},$$

so

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L})A_3(e^{-L})} = \frac{1}{2} - \frac{1}{2L} + O(L^{-2}).$$

Choose

$$\alpha_L = \frac{1}{2} - \frac{1}{4L}.$$

For all sufficiently large L , $\alpha_L \in (0, 1/2)$ and

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L})A_3(e^{-L})} < \alpha_L.$$

Consequently

$$H_2(e^{-L}; \alpha_L) = A_2(e^{-L})^2 - \alpha_L A_1(e^{-L}) A_3(e^{-L}) < 0.$$

Complete monotonicity would imply nonnegativity at order zero. Hence $H_2(\cdot; \alpha_L)$ is not completely monotone for these fixed parameters $\alpha_L \in (0, 1/2)$. The source's universal Turan-window statement is false already in the $n = 2$ upper window.

the Ramanujan Turan n2 upper window counterfamily the Ramanujan Turan window negative answer

The new mechanism is a logarithmic-density moment-ratio asymptotic obstruction for quadratic Laplace-moment Turan gaps. It is distinct from the public APP-0014 Stieltjes/complete-Bernstein classification of I_R . $\square$

12 Moment-ratio and endpoint method applications

The following applications are source-facing consequences of the moment-ratio, Bessel zero, and endpoint-obstruction mechanisms developed earlier.

Counterexample 12.1 (APP-0039: Ramanujan Turan-window complete-monotonicity interval is false). The Mishra–Swaminathan Ramanujan integral Turan-window complete-monotonicity statement has a negative answer: the proposed interval already fails for $n = 2$ and suitable $\alpha \in (0, 1/2)$.

This resolves the source problem recorded as **APP-0039** [13].

Proof. We use Proposition 8.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Put

$$A_k(x) = (-1)^k I_R^{(k)}(x) = \int_0^\infty e^{-xt} \frac{t^{k-1}}{\pi^2 + \log^2 t} dt \quad (k \geq 1).$$

Then

$$H_2(x; \alpha) = A_2(x)^2 - \alpha A_1(x) A_3(x).$$

We study $x = e^{-L}$ as $L \rightarrow \infty$. With $y = xt$,

$$A_k(e^{-L}) = e^{kL} \int_0^\infty \frac{e^{-y} y^{k-1}}{\pi^2 + (L + \log y)^2} dy.$$

For fixed $k = 1, 2, 3$,

$$A_k(e^{-L}) = e^{kL} L^{-2} \Gamma(k) \left(1 - \frac{2\psi(k)}{L} + O(L^{-2}) \right),$$

where ψ is the digamma function. To justify the expansion, split the y -integral into the central region $|\log y| \leq L^{2/3}$ and its complement. On the central interval, with $z = \log y$, expand uniformly:

$$\frac{1}{\pi^2 + (L + \log y)^2} = L^{-2} \left(1 - \frac{2 \log y}{L} + O\left(\frac{1 + \log^2 y}{L^2}\right) \right),$$

and integrate against $e^{-y} y^{k-1}$. The lower complement $0 < y < e^{-L^{2/3}}$ is $O(e^{-cL^{2/3}})$ for $k = 1, 2, 3$ after the change $y = e^{-u}$, and the upper complement $y > e^{L^{2/3}}$ is super-exponentially small because of e^{-y} . These errors are $o(L^{-N})$ for every fixed N , which is more than needed for the displayed expansion.

Therefore

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L}) A_3(e^{-L})} = \frac{\Gamma(2)^2}{\Gamma(1) \Gamma(3)} \left(1 + \frac{2(\psi(1) + \psi(3) - 2\psi(2))}{L} + O(L^{-2}) \right).$$

Using

$$\psi(1) = -\gamma, \quad \psi(2) = 1 - \gamma, \quad \psi(3) = \frac{3}{2} - \gamma,$$

we get

$$\psi(1) + \psi(3) - 2\psi(2) = -\frac{1}{2},$$

so

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L})A_3(e^{-L})} = \frac{1}{2} - \frac{1}{2L} + O(L^{-2}).$$

Choose

$$\alpha_L = \frac{1}{2} - \frac{1}{4L}.$$

For all sufficiently large L , $\alpha_L \in (0, 1/2)$ and

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L})A_3(e^{-L})} < \alpha_L.$$

Consequently

$$H_2(e^{-L}; \alpha_L) = A_2(e^{-L})^2 - \alpha_L A_1(e^{-L})A_3(e^{-L}) < 0.$$

Complete monotonicity would imply nonnegativity at order zero. Hence $H_2(\cdot; \alpha_L)$ is not completely monotone for these fixed parameters $\alpha_L \in (0, 1/2)$. The source's universal Turan-window statement is false already in the $n = 2$ upper window.

the Ramanujan Turan n2 upper window counterfamily the Ramanujan Turan window negative answer

The new mechanism is a logarithmic-density moment-ratio asymptotic obstruction for quadratic Laplace-moment Turan gaps. It is distinct from the public APP-0014 Stieltjes/complete-Bernstein classification of I_R . $\square$

Proposition 12.2 (YT Bessel W general zero partial fraction). *For every $\nu > -1$, if $j_{\nu+1,n}$ are the positive zeros of $J_{\nu+1}$, then $G_\nu(s) = W_\nu(\sqrt{s})$ satisfies $G_\nu(s) = 2(\nu+1) + 2 \sum_{n \geq 1} s/(s + j_{\nu+1,n}^2)$ and $G'_\nu(s) = 2 \sum_{n \geq 1} j_{\nu+1,n}^2/(s + j_{\nu+1,n}^2)^2$. Consequently G_ν is a Bernstein function on $(0, \infty)$.*

Proof. Put $\mu = \nu + 1 > 0$. The canonical product for the modified Bessel function gives, locally uniformly in z ,

$$I_\mu(z) = \frac{(z/2)^\mu}{\Gamma(\mu+1)} \prod_{n=1}^{\infty} \left(1 + \frac{z^2}{j_{\mu,n}^2}\right),$$

where $j_{\mu,n}$ are the positive zeros of J_μ . Therefore

$$z \frac{I'_\mu(z)}{I_\mu(z)} = \mu + 2 \sum_{n=1}^{\infty} \frac{z^2}{z^2 + j_{\mu,n}^2}.$$

The recurrence

$$I'_\mu(z) = I_{\mu-1}(z) - \frac{\mu}{z} I_\mu(z)$$

then yields

$$W_\nu(z) = z \frac{I_{\mu-1}(z)}{I_\mu(z)} = 2\mu + 2 \sum_{n=1}^{\infty} \frac{z^2}{z^2 + j_{\mu,n}^2}.$$

Set $s = z^2$. Then

$$G_\nu(s) := W_\nu(\sqrt{s}) = 2(\nu + 1) + 2 \sum_{n=1}^{\infty} \frac{s}{s + j_{\nu+1,n}^2}.$$

Termwise differentiation is justified because $j_{\nu+1,n} \sim \pi n$, so the differentiated sums converge locally uniformly on $(0, \infty)$. Thus

$$G'_\nu(s) = 2 \sum_{n=1}^{\infty} \frac{j_{\nu+1,n}^2}{(s + j_{\nu+1,n}^2)^2}.$$

For every $k \geq 0$,

$$(-1)^k \frac{d^k}{ds^k} \frac{j_{\nu+1,n}^2}{(s + j_{\nu+1,n}^2)^2} = (k+1)! \frac{j_{\nu+1,n}^2}{(s + j_{\nu+1,n}^2)^{k+2}} \geq 0.$$

The locally uniformly convergent sum preserves these inequalities, hence G'_ν is completely monotone and G_ν is a Bernstein function.

If $0 < \tau \leq 1/2$, then $x^{2\tau}$ is a Bernstein function. By Bernstein-function composition closure,

$$x \mapsto W_\nu(x^\tau) = G_\nu(x^{2\tau})$$

is a Bernstein function on $(0, \infty)$. □

Corollary 12.3 (APP-0040: Yang-Tian Bessel- W power-Bernstein conjecture is true). *For $\tau \in (0, 1/2]$ and $\nu > -1$, $x \mapsto W_\nu(x^\tau)$ is a Bernstein function on $(0, \infty)$.*

*This resolves the source problem recorded as **APP-0040** [34].*

Proof. We use Proposition 12.2 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Put $\mu = \nu + 1 > 0$. The canonical product for the modified Bessel function gives, locally uniformly in z ,

$$I_\mu(z) = \frac{(z/2)^\mu}{\Gamma(\mu + 1)} \prod_{n=1}^{\infty} \left(1 + \frac{z^2}{j_{\mu,n}^2}\right),$$

where $j_{\mu,n}$ are the positive zeros of J_μ . Therefore

$$z \frac{I'_\mu(z)}{I_\mu(z)} = \mu + 2 \sum_{n=1}^{\infty} \frac{z^2}{z^2 + j_{\mu,n}^2}.$$

The recurrence

$$I'_\mu(z) = I_{\mu-1}(z) - \frac{\mu}{z} I_\mu(z)$$

then yields

$$W_\nu(z) = z \frac{I_{\mu-1}(z)}{I_\mu(z)} = 2\mu + 2 \sum_{n=1}^{\infty} \frac{z^2}{z^2 + j_{\mu,n}^2}.$$

Set $s = z^2$. Then

$$G_\nu(s) := W_\nu(\sqrt{s}) = 2(\nu + 1) + 2 \sum_{n=1}^{\infty} \frac{s}{s + j_{\nu+1,n}^2}.$$

Termwise differentiation is justified because $j_{\nu+1,n} \sim \pi n$, so the differentiated sums converge locally uniformly on $(0, \infty)$. Thus

$$G'_\nu(s) = 2 \sum_{n=1}^{\infty} \frac{j_{\nu+1,n}^2}{(s + j_{\nu+1,n}^2)^2}.$$

For every $k \geq 0$,

$$(-1)^k \frac{d^k}{ds^k} \frac{j_{\nu+1,n}^2}{(s + j_{\nu+1,n}^2)^2} = (k+1)! \frac{j_{\nu+1,n}^2}{(s + j_{\nu+1,n}^2)^{k+2}} \geq 0.$$

The locally uniformly convergent sum preserves these inequalities, hence G'_ν is completely monotone and G_ν is a Bernstein function.

If $0 < \tau \leq 1/2$, then $x^{2\tau}$ is a Bernstein function. By Bernstein-function composition closure,

$$x \mapsto W_\nu(x^\tau) = G_\nu(x^{2\tau})$$

is a Bernstein function on $(0, \infty)$. □

Counterexample 12.4 (APP-0041: Qi h_λ degree-four complete-monotonicity conjecture is false).
Let

$$\Psi(x) = [\psi'(x)]^2 + \psi''(x), \quad h_\lambda(x) = \Psi(x) - \frac{x^2 + \lambda x + 12}{12x^4(x+1)^2}.$$

The conjecture that $\deg_x^{\text{cm}} h_\lambda = \deg_x^{\text{cm}}(-h_\mu) = 4$ exactly for $\lambda \leq 0$ and $\mu \geq 4$ is false.

This resolves the source problem recorded as **APP-0041** [35].

Proof. We use Proposition 8.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

A completely monotone function must be nonincreasing, so its first derivative cannot be positive at any point.

Use the recurrence expansions at $x = 0^+$:

$$\psi'(x) = \frac{1}{x^2} + \zeta(2) - 2\zeta(3)x + 3\zeta(4)x^2 + O(x^3),$$

and

$$\psi''(x) = -\frac{2}{x^3} - 2\zeta(3) + 6\zeta(4)x + O(x^2).$$

Therefore

$$\Psi(x) = \frac{1}{x^4} - \frac{2}{x^3} + \frac{\pi^2}{3x^2} - \frac{4\zeta(3)}{x} + O(1).$$

Also

$$\frac{x^2 + \lambda x + 12}{12x^4(1+x)^2} = \frac{1}{x^4} + \left(\frac{\lambda}{12} - 2\right) \frac{1}{x^3} + \left(\frac{37}{12} - \frac{\lambda}{6}\right) \frac{1}{x^2} + O(x^{-1}).$$

Subtracting gives

$$h_\lambda(x) = -\frac{\lambda}{12x^3} + \left(\frac{\lambda}{6} + \frac{\pi^2}{3} - \frac{37}{12}\right) \frac{1}{x^2} + O(x^{-1}).$$

If $\lambda < 0$, then

$$x^4 h_\lambda(x) = -\frac{\lambda}{12}x + O(x^2),$$

so

$$\frac{d}{dx} [x^4 h_\lambda(x)] = -\frac{\lambda}{12} + O(x) > 0$$

for all sufficiently small $x > 0$. Hence $x^4 h_\lambda$ is not completely monotone.

If $\lambda = 0$, then

$$x^4 h_0(x) = \left(\frac{\pi^2}{3} - \frac{37}{12} \right) x^2 + O(x^3).$$

Since $4\pi^2 > 37$, the leading coefficient is positive, and

$$\frac{d}{dx} [x^4 h_0(x)] = 2 \left(\frac{\pi^2}{3} - \frac{37}{12} \right) x + O(x^2) > 0$$

for all sufficiently small $x > 0$. Hence $x^4 h_0$ is not completely monotone.

Similarly, for $\mu \geq 4$,

$$x^4 (-h_\mu(x)) = \frac{\mu}{12} x + O(x^2),$$

so

$$\frac{d}{dx} [x^4 (-h_\mu(x))] = \frac{\mu}{12} + O(x) > 0$$

for all sufficiently small $x > 0$. Hence $x^4 (-h_\mu)$ is not completely monotone for every $\mu \geq 4$.

Thus the degree-four transforms required by Qi's conjecture are not completely monotone on the conjectured source ranges. The conjecture in Remark 7.4 is false. $\square$

13 Bessel endpoint obstruction

The next group belongs to the same endpoint-obstruction mechanism used throughout the paper: a universal positivity or concavity statement is reduced to a one-point Riccati certificate, and exact rational bounds then decide the sign. The same certificate resolves the square-root log-concavity problem, the Riccati form of that problem, a related quadratic ratio bound, and the proposed three-regime certificate route.

Counterexample 13.1 (Not Bessel I sqrt log concavity nu ge 0). not(For every $\nu \geq 0$, the function $u \mapsto \sqrt{u} I_\nu(u)$ is strictly log-concave on $(0, \infty)$.)

Proof. Let

$$g_\nu(u) = \log(\sqrt{u} I_\nu(u)), \quad r_\nu(u) = \frac{I'_\nu(u)}{I_\nu(u)}.$$

Then

$$g''_\nu(u) = r'_\nu(u) - \frac{1}{2u^2}.$$

The modified Bessel equation

$$u^2 I''_\nu(u) + u I'_\nu(u) - (u^2 + \nu^2) I_\nu(u) = 0$$

gives

$$\frac{I''_\nu(u)}{I_\nu(u)} = 1 + \frac{\nu^2}{u^2} - \frac{r_\nu(u)}{u}.$$

Since

$$r'_\nu(u) = \frac{I''_\nu(u)}{I_\nu(u)} - r_\nu(u)^2,$$

we get

$$g''_\nu(u) = 1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2.$$

Thus the Bessel I Riccati log concavity inequality is exactly the strict log-concavity condition $g''_\nu(u) < 0$.

For $\nu = 0$,

$$r_0(u) = \frac{I'_0(u)}{I_0(u)} = \frac{I_1(u)}{I_0(u)}.$$

At $u = 10$, the Riccati expression is

$$g''_0(10) = \frac{199}{200} - \frac{r_0(10)}{10} - r_0(10)^2.$$

It is enough to prove

$$r_0(10) < \frac{9487}{10000},$$

because

$$\frac{199}{200} - \frac{1}{10} \frac{9487}{10000} - \left(\frac{9487}{10000} \right)^2 = \frac{9831}{100000000} > 0.$$

$$I_0(10) = \sum_{k=0}^{\infty} \frac{25^k}{(k!)^2}, \quad I_1(10) = \sum_{k=0}^{\infty} \frac{5 \cdot 25^k}{k!(k+1)!}.$$

With $N = 20$, set

$$L_0 = \sum_{k=0}^{20} \frac{25^k}{(k!)^2} < I_0(10).$$

For the I_1 tail after $N = 20$, the term ratio is bounded by

$$q = \frac{25}{22 \cdot 23} < 1,$$

so the script computes an exact rational upper bound $U_1 > I_1(10)$ and verifies

$$10000 U_1 < 9487 L_0.$$

Therefore $I_1(10)/I_0(10) < 9487/10000$, and hence

$$g''_0(10) > 0.$$

This proves the Bessel- I log-concavity counterexample to T - I sqrt-log concavity for $\nu > 0$, and the counterexample to T - I Riccati log-concavity inequality. $\square$

Counterexample 13.2 (APP-0043: Riccati log-concavity inequality for I_ν is false). The inequality

$$1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2 < 0, \quad r_\nu(u) = \frac{I'_\nu(u)}{I_\nu(u)},$$

does not hold for every $\nu \geq 0$ and $u > 0$.

This resolves the source problem recorded as **APP-0043** [36].

Proof. We use Proposition 8.1 and Proposition 13.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$g_\nu(u) = \log(\sqrt{u} I_\nu(u)), \quad r_\nu(u) = \frac{I'_\nu(u)}{I_\nu(u)}.$$

Then

$$g''_\nu(u) = r'_\nu(u) - \frac{1}{2u^2}.$$

The modified Bessel equation

$$u^2 I''_\nu(u) + u I'_\nu(u) - (u^2 + \nu^2) I_\nu(u) = 0$$

gives

$$\frac{I''_\nu(u)}{I_\nu(u)} = 1 + \frac{\nu^2}{u^2} - \frac{r_\nu(u)}{u}.$$

Since

$$r'_\nu(u) = \frac{I''_\nu(u)}{I_\nu(u)} - r_\nu(u)^2,$$

we get

$$g''_\nu(u) = 1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2.$$

Thus the Bessel I Riccati log concavity inequality is exactly the strict log-concavity condition $g''_\nu(u) < 0$.

For $\nu = 0$,

$$r_0(u) = \frac{I'_0(u)}{I_0(u)} = \frac{I_1(u)}{I_0(u)}.$$

At $u = 10$, the Riccati expression is

$$g''_0(10) = \frac{199}{200} - \frac{r_0(10)}{10} - r_0(10)^2.$$

It is enough to prove

$$r_0(10) < \frac{9487}{10000},$$

because

$$\frac{199}{200} - \frac{1}{10} \frac{9487}{10000} - \left(\frac{9487}{10000} \right)^2 = \frac{9831}{100000000} > 0.$$

$$I_0(10) = \sum_{k=0}^{\infty} \frac{25^k}{(k!)^2}, \quad I_1(10) = \sum_{k=0}^{\infty} \frac{5 \cdot 25^k}{k!(k+1)!}.$$

With $N = 20$, set

$$L_0 = \sum_{k=0}^{20} \frac{25^k}{(k!)^2} < I_0(10).$$

For the I_1 tail after $N = 20$, the term ratio is bounded by

$$q = \frac{25}{22 \cdot 23} < 1,$$

so the script computes an exact rational upper bound $U_1 > I_1(10)$ and verifies

$$10000 U_1 < 9487 L_0.$$

Therefore $I_1(10)/I_0(10) < 9487/10000$, and hence

$$g''_0(10) > 0.$$

This proves the Bessel- I log-concavity counterexample to T - I sqrt-log concavity for $\nu > 0$, and the counterexample to T - I Riccati log-concavity inequality. $\square$

Counterexample 13.3 (APP-0044: Bessel ratio quadratic lower bound is false). The claimed inequality

$$q_\nu(u)^2 + \frac{(2\nu+1)q_\nu(u)}{u} + \frac{\nu + \frac{1}{2}}{u^2} > 1, \quad q_\nu(u) = \frac{I_{\nu+1}(u)}{I_\nu(u)},$$

is false for all $\nu \geq 0$ and $u > 0$.

This resolves the source problem recorded as **APP-0044** [36].

Proof. We use Proposition 8.1 and Proposition 13.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$g_\nu(u) = \log(\sqrt{u} I_\nu(u)), \quad r_\nu(u) = \frac{I'_\nu(u)}{I_\nu(u)}.$$

Then

$$g''_\nu(u) = r'_\nu(u) - \frac{1}{2u^2}.$$

The modified Bessel equation

$$u^2 I''_\nu(u) + u I'_\nu(u) - (u^2 + \nu^2) I_\nu(u) = 0$$

gives

$$\frac{I''_\nu(u)}{I_\nu(u)} = 1 + \frac{\nu^2}{u^2} - \frac{r_\nu(u)}{u}.$$

Since

$$r'_\nu(u) = \frac{I''_\nu(u)}{I_\nu(u)} - r_\nu(u)^2,$$

we get

$$g''_\nu(u) = 1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2.$$

Thus the Bessel I Riccati log concavity inequality is exactly the strict log-concavity condition $g''_\nu(u) < 0$.

For $\nu = 0$,

$$r_0(u) = \frac{I'_0(u)}{I_0(u)} = \frac{I_1(u)}{I_0(u)}.$$

At $u = 10$, the Riccati expression is

$$g''_0(10) = \frac{199}{200} - \frac{r_0(10)}{10} - r_0(10)^2.$$

It is enough to prove

$$r_0(10) < \frac{9487}{10000},$$

because

$$\frac{199}{200} - \frac{1}{10} \frac{9487}{10000} - \left(\frac{9487}{10000} \right)^2 = \frac{9831}{100000000} > 0.$$

$$I_0(10) = \sum_{k=0}^{\infty} \frac{25^k}{(k!)^2}, \quad I_1(10) = \sum_{k=0}^{\infty} \frac{5 \cdot 25^k}{k!(k+1)!}.$$

With $N = 20$, set

$$L_0 = \sum_{k=0}^{20} \frac{25^k}{(k!)^2} < I_0(10).$$

For the I_1 tail after $N = 20$, the term ratio is bounded by

$$q = \frac{25}{22 \cdot 23} < 1,$$

so the script computes an exact rational upper bound $U_1 > I_1(10)$ and verifies

$$10000 U_1 < 9487 L_0.$$

Therefore $I_1(10)/I_0(10) < 9487/10000$, and hence

$$g_0''(10) > 0.$$

This proves the Bessel- I log-concavity counterexample to T - I sqrt-log concavity for $\nu > 0$, and the counterexample to T - I Riccati log-concavity inequality. $\square$

Counterexample 13.4 (APP-0045: Three-regime Bessel log-concavity certificate route is refuted). No valid three-regime certificate can prove the universal Riccati log-concavity inequality for I_ν , because the universal inequality itself is false.

This resolves the source problem recorded as **APP-0045** [36].

Proof. We use Proposition 8.1 and Proposition 13.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$g_\nu(u) = \log(\sqrt{u} I_\nu(u)), \quad r_\nu(u) = \frac{I'_\nu(u)}{I_\nu(u)}.$$

Then

$$g_\nu''(u) = r'_\nu(u) - \frac{1}{2u^2}.$$

The modified Bessel equation

$$u^2 I''_\nu(u) + u I'_\nu(u) - (u^2 + \nu^2) I_\nu(u) = 0$$

gives

$$\frac{I''_\nu(u)}{I_\nu(u)} = 1 + \frac{\nu^2}{u^2} - \frac{r_\nu(u)}{u}.$$

Since

$$r'_\nu(u) = \frac{I''_\nu(u)}{I_\nu(u)} - r_\nu(u)^2,$$

we get

$$g_\nu''(u) = 1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2.$$

Thus the Bessel I Riccati log concavity inequality is exactly the strict log-concavity condition $g_\nu''(u) < 0$.

For $\nu = 0$,

$$r_0(u) = \frac{I'_0(u)}{I_0(u)} = \frac{I_1(u)}{I_0(u)}.$$

At $u = 10$, the Riccati expression is

$$g_0''(10) = \frac{199}{200} - \frac{r_0(10)}{10} - r_0(10)^2.$$

It is enough to prove

$$r_0(10) < \frac{9487}{10000},$$

because

$$\frac{199}{200} - \frac{1}{10} \frac{9487}{10000} - \left(\frac{9487}{10000} \right)^2 = \frac{9831}{100000000} > 0.$$

$$I_0(10) = \sum_{k=0}^{\infty} \frac{25^k}{(k!)^2}, \quad I_1(10) = \sum_{k=0}^{\infty} \frac{5 \cdot 25^k}{k!(k+1)!}.$$

With $N = 20$, set

$$L_0 = \sum_{k=0}^{20} \frac{25^k}{(k!)^2} < I_0(10).$$

For the I_1 tail after $N = 20$, the term ratio is bounded by

$$q = \frac{25}{22 \cdot 23} < 1,$$

so the script computes an exact rational upper bound $U_1 > I_1(10)$ and verifies

$$10000 U_1 < 9487 L_0.$$

Therefore $I_1(10)/I_0(10) < 9487/10000$, and hence

$$g_0''(10) > 0.$$

This proves the Bessel- I log-concavity counterexample to T - I sqrt-log concavity for $\nu > 0$, and the counterexample to T - I Riccati log-concavity inequality. $\square$

14 Endpoint obstruction certificates

This section collects source problems whose solutions are obtained by reducing the proposed universal statement to an explicit obstruction. Each result is placed with the certificate or counterexample that supplies its proof, so the source question appears as a consequence of the surrounding method.

Counterexample 14.1 (APP-0031: Baricz gamma-quotient Bernstein test is false). Baricz's question asking whether $x \mapsto \Gamma(x)\Gamma(x-a+b)/(\Gamma(x-a)\Gamma(x+b))$ is a Bernstein function on (a, ∞) for every $a, b > 0$ has a negative answer.

This resolves the source problem recorded as **APP-0031** [16].

Proof. We use Proposition 8.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Take $a = 2$, $b = 3$, and set $y = x - 2 > 0$. Then by the Gamma recurrence,

$$\frac{\Gamma(x)\Gamma(x-a+b)}{\Gamma(x-a)\Gamma(x+b)} = \frac{\Gamma(x)\Gamma(x+1)}{\Gamma(x-2)\Gamma(x+3)} = \frac{(x-2)(x-1)}{(x+1)(x+2)} = \frac{y(y+1)}{(y+3)(y+4)}.$$

Let

$$g(y) = \frac{y(y+1)}{(y+3)(y+4)}, \quad y > 0.$$

If g were a Bernstein function, then g' would be completely monotone. In particular $g''' \geq 0$ would be necessary.

Exact symbolic differentiation gives

$$g'(y) = \frac{6(y^2 + 4y + 2)}{(y+3)^2(y+4)^2},$$

$$g''(y) = -\frac{12(y^3 + 6y^2 + 6y - 10)}{(y+3)^3(y+4)^3},$$

and

$$g'''(y) = \frac{36(y^4 + 8y^3 + 12y^2 - 40y - 94)}{(y+3)^4(y+4)^4}.$$

At $y = 1$,

$$g'''(1) = -\frac{1017}{40000} < 0.$$

Therefore g' is not completely monotone, g is not Bernstein, and the Baricz for-all- a, b Bernstein question has a negative answer. $\square$

Proposition 14.2 (From Mills all L moment ratio quadratic bound). *Let $M_L(t) = (-1)^L r^{(L)}(t) = \int_0^\infty u^L e^{-tu-u^2/2} du$ for the standard normal Mills ratio. For every $L \geq 0$ and $t \geq 0$, with $m_L = M_{L+1}/M_L$, one has $m_L(t) < U_L(t)$, where $U_L(t)$ is the positive root $\frac{\sqrt{t^2(t^2+4L+7)^2+4(L+1)(t^2+4L+8)(t^2+4L+6)}-t(t^2+4L+7)}{2(t^2+4L+8)}$*

Proof. Set

$$m_L(t) = \frac{M_{L+1}(t)}{M_L(t)}.$$

The recurrence gives

$$\frac{M_{L+2}}{M_L} = L + 1 - tm_L,$$

$$\frac{M_{L+3}}{M_L} = (t^2 + L + 2)m_L - t(L + 1),$$

and

$$\frac{M_{L+4}}{M_L} = (L + 1)(t^2 + L + 3) - t(t^2 + 2L + 5)m_L.$$

Substitution into the determinant inequality and expansion give

$$(L + 1)(t^2 + 4L + 6) - t(t^2 + 4L + 7)m_L - (t^2 + 4L + 8)m_L^2 > 0.$$

Equivalently,

$$(t^2 + 4L + 8)m_L^2 + t(t^2 + 4L + 7)m_L - (L + 1)(t^2 + 4L + 6) < 0.$$

Since $m_L > 0$, the positive root yields the uniform strict bound

$$m_L(t) < U_L(t),$$

where

$$U_L(t) = \frac{\sqrt{t^2(t^2 + 4L + 7)^2 + 4(L + 1)(t^2 + 4L + 8)(t^2 + 4L + 6)} - t(t^2 + 4L + 7)}{2(t^2 + 4L + 8)}.$$

This is the determinant-to-bound extractor. It is a single theorem for every $L \geq 0$.

Define polynomials $P_n(t)$ by

$$P_0 = 1, \quad P_1 = t, \quad P_{n+1} = tP_n + nP_{n-1}.$$

Define $B_n(t)$ by

$$B_0 = 0, \quad B_1 = 1, \quad B_{n+1} = nB_{n-1} - tB_n.$$

Then the same recurrence gives

$$M_n(t) = (-1)^n P_n(t) r(t) + B_n(t).$$

The ratio bound $M_{L+1} < U_L M_L$ becomes

$$((-1)^{L+1} P_{L+1} - (-1)^L U_L P_L) r < U_L B_L - B_{L+1}.$$

Since $P_n(t) > 0$ for $t > 0$ and $P_{2j}(0) > 0$, the coefficient has sign $(-1)^{L+1}$. Thus the bounds alternate:

For even $L \geq 0$,

$$r(t) > \frac{B_{L+1}(t) - U_L(t) B_L(t)}{P_{L+1}(t) + U_L(t) P_L(t)}.$$

For odd $L \geq 1$,

$$r(t) < \frac{U_L(t) B_L(t) - B_{L+1}(t)}{P_{L+1}(t) + U_L(t) P_L(t)}.$$

These formulas are valid at $t = 0$ by direct evaluation of the recurrences and for $t > 0$ by the positivity of M_L . $\square$

Corollary 14.3 (APP-0032: From's all- L Mills-ratio bound family is explicit). *Let $P_0 = 1$, $P_1 = t$, $P_{n+1} = tP_n + nP_{n-1}$, and $B_0 = 0$, $B_1 = 1$, $B_{n+1} = nB_{n-1} - tB_n$. With $U_L(t)$ as in the all- L moment-ratio bound, the standard normal Mills ratio satisfies, for even $L \geq 0$, $r(t) > [B_{L+1} - U_L B_L] / [P_{L+1} + U_L P_L]$, and for odd $L \geq 1$, $r(t) < [U_L B_L - B_{L+1}] / [P_{L+1} + U_L P_L]$, for all $t \geq 0$.*

*This resolves the source problem recorded as **APP-0032** [15].*

Proof. We use Proposition 11.6 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Set

$$m_L(t) = \frac{M_{L+1}(t)}{M_L(t)}.$$

The recurrence gives

$$\begin{aligned} \frac{M_{L+2}}{M_L} &= L + 1 - t m_L, \\ \frac{M_{L+3}}{M_L} &= (t^2 + L + 2) m_L - t(L + 1), \end{aligned}$$

and

$$\frac{M_{L+4}}{M_L} = (L + 1)(t^2 + L + 3) - t(t^2 + 2L + 5) m_L.$$

Substitution into the determinant inequality and expansion give

$$(L+1)(t^2+4L+6) - t(t^2+4L+7)m_L - (t^2+4L+8)m_L^2 > 0.$$

Equivalently,

$$(t^2+4L+8)m_L^2 + t(t^2+4L+7)m_L - (L+1)(t^2+4L+6) < 0.$$

Since $m_L > 0$, the positive root yields the uniform strict bound

$$m_L(t) < U_L(t),$$

where

$$U_L(t) = \frac{\sqrt{t^2(t^2+4L+7)^2 + 4(L+1)(t^2+4L+8)(t^2+4L+6)} - t(t^2+4L+7)}{2(t^2+4L+8)}.$$

This is the determinant-to-bound extractor. It is a single theorem for every $L \geq 0$.

Define polynomials $P_n(t)$ by

$$P_0 = 1, \quad P_1 = t, \quad P_{n+1} = tP_n + nP_{n-1}.$$

Define $B_n(t)$ by

$$B_0 = 0, \quad B_1 = 1, \quad B_{n+1} = nB_{n-1} - tB_n.$$

Then the same recurrence gives

$$M_n(t) = (-1)^n P_n(t)r(t) + B_n(t).$$

The ratio bound $M_{L+1} < U_L M_L$ becomes

$$((-1)^{L+1}P_{L+1} - (-1)^L U_L P_L) r < U_L B_L - B_{L+1}.$$

Since $P_n(t) > 0$ for $t > 0$ and $P_{2j}(0) > 0$, the coefficient has sign $(-1)^{L+1}$. Thus the bounds alternate:

For even $L \geq 0$,

$$r(t) > \frac{B_{L+1}(t) - U_L(t)B_L(t)}{P_{L+1}(t) + U_L(t)P_L(t)}.$$

For odd $L \geq 1$,

$$r(t) < \frac{U_L(t)B_L(t) - B_{L+1}(t)}{P_{L+1}(t) + U_L(t)P_L(t)}.$$

These formulas are valid at $t = 0$ by direct evaluation of the recurrences and for $t > 0$ by the positivity of M_L . $\square$

Proposition 14.4 (Ramanujan density logsquare CM). *The function $\phi_0(t) = 1/[t(\pi^2 + \log^2 t)]$ is completely monotone on $(0, \infty)$.*

Proof. The density

$$\phi_0(t) = \frac{1}{t(\pi^2 + \log^2 t)}$$

is completely monotone on $(0, \infty)$. Consequently I_R is a Stieltjes function. Moreover, the antiderivative

$$\tilde{I}_R(x) = a + \int_0^\infty (1 - e^{-xt})\phi_0(t) dt$$

is a complete Bernstein function.

For $u = \log t$,

$$\int_0^1 e^{-au} \sin(\pi a) da = \frac{\pi(1 + e^{-u})}{u^2 + \pi^2}.$$

Substituting $u = \log t$ gives

$$\int_0^1 t^{-a} \sin(\pi a) da = \frac{\pi(1 + t^{-1})}{\pi^2 + \log^2 t}.$$

Therefore

$$\phi_0(t) = \frac{1}{t(\pi^2 + \log^2 t)} = \frac{1}{\pi(1 + t)} \int_0^1 t^{-a} \sin(\pi a) da.$$

For $0 < a < 1$, t^{-a} is completely monotone because

$$t^{-a} = \frac{1}{\Gamma(a)} \int_0^\infty e^{-ts} s^{a-1} ds.$$

Also $t \mapsto (1 + t)^{-1}$ is completely monotone. Complete monotonicity is closed under products and positive mixtures, and $\sin(\pi a) > 0$ on $(0, 1)$. Hence ϕ_0 is completely monotone.

If ϕ is completely monotone and

$$F(x) = \int_0^\infty e^{-xt} \phi(t) dt$$

is finite for $x > 0$, then F is Stieltjes: write $\phi(t) = \int_0^\infty e^{-ts} d\mu(s)$ and use Tonelli to obtain

$$F(x) = \int_0^\infty \frac{d\mu(s)}{x + s}.$$

Applying this to ϕ_0 proves that I_R is Stieltjes.

Finally, the source already proves that

$$\tilde{I}_R(x) = a + \int_0^\infty (1 - e^{-xt}) \phi_0(t) dt$$

is a Bernstein function. The Levy density ϕ_0 is completely monotone, and it satisfies the Bernstein integrability condition

$$\int_0^\infty (1 \wedge t) \phi_0(t) dt < \infty.$$

The standard complete-Bernstein criterion for Levy densities therefore gives that $\tilde{I}_R$ is a complete Bernstein function. $\square$

Corollary 14.5 (APP-0033: Ramanujan antiderivative is complete Bernstein). *The antiderivative $\tilde{I}_R(x) = a + \int_0^\infty (1 - e^{-xt}) dt / [t(\pi^2 + \log^2 t)]$ of the Ramanujan integral is a complete Bernstein function.*

*This resolves the source problem recorded as **APP-0033** [13].*

Proof. We use Proposition 11.2 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

For $u = \log t$,

$$\int_0^1 e^{-au} \sin(\pi a) da = \frac{\pi(1 + e^{-u})}{u^2 + \pi^2}.$$

Substituting $u = \log t$ gives

$$\int_0^1 t^{-a} \sin(\pi a) da = \frac{\pi(1+t^{-1})}{\pi^2 + \log^2 t}.$$

Therefore

$$\phi_0(t) = \frac{1}{t(\pi^2 + \log^2 t)} = \frac{1}{\pi(1+t)} \int_0^1 t^{-a} \sin(\pi a) da.$$

For $0 < a < 1$, t^{-a} is completely monotone because

$$t^{-a} = \frac{1}{\Gamma(a)} \int_0^\infty e^{-ts} s^{a-1} ds.$$

Also $t \mapsto (1+t)^{-1}$ is completely monotone. Complete monotonicity is closed under products and positive mixtures, and $\sin(\pi a) > 0$ on $(0, 1)$. Hence ϕ_0 is completely monotone.

If ϕ is completely monotone and

$$F(x) = \int_0^\infty e^{-xt} \phi(t) dt$$

is finite for $x > 0$, then F is Stieltjes: write $\phi(t) = \int_0^\infty e^{-ts} d\mu(s)$ and use Tonelli to obtain

$$F(x) = \int_0^\infty \frac{d\mu(s)}{x+s}.$$

Applying this to ϕ_0 proves that I_R is Stieltjes.

Finally, the source already proves that

$$\tilde{I}_R(x) = a + \int_0^\infty (1 - e^{-xt}) \phi_0(t) dt$$

is a Bernstein function. The Levy density ϕ_0 is completely monotone, and it satisfies the Bernstein integrability condition

$$\int_0^\infty (1 \wedge t) \phi_0(t) dt < \infty.$$

The standard complete-Bernstein criterion for Levy densities therefore gives that $\tilde{I}_R$ is a complete Bernstein function.

the CM closure product positive mixture the Stieltjes density criterion S1 the CBF Levy density CM criterion the Ramanujan density logsquare CM the Ramanujan integral Stieltjes the Ramanujan antiderivative complete Bernstein

the Ramanujan Turan window CM open problem □

Corollary 14.6 (APP-0034: Bulboaca-Zayed gamma-quotient monotonicity solved). *For the continuous extension $\tilde{F}$ of $F(x) = \log \Gamma(x+1)/(\log(x^2+6) - \log(x+6))$ on $(-1, \infty)$, prove analytically that $\tilde{F}'$ has a unique zero $x_m \simeq 1.126207061\dots$, with $\tilde{F}' < 0$ on $(-1, x_m)$ and $\tilde{F}' > 0$ on (x_m, ∞) .*

*This resolves the source problem recorded as **APP-0034** [5].*

Proof. We use Proposition 11.39 and Proposition 11.38 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

proof.

The raw numerator has a removable zero at $x = 1$, since $G(1) = D(1) = 0$. Its quadratic leading coefficient is

$$\lim_{x \downarrow 1} \frac{N_0(x)}{(x-1)^2} = \frac{7(\pi^2/6 - 1) - 11(1 - \gamma)}{98} < 0,$$

again by $\pi^2 < 987/100$ and $\gamma < 5773/10000$, which gives $-1347/980000 < 0$. Therefore any normalization dividing out the removable $(x-1)^2$ factor is negative at the left endpoint.

The single-crossing candidate is true:

$$\text{theBulboaca} - \text{Zayedgamma} - \text{quotientNsinglecrossingoneeight}.$$

Together with the source's theorem on $(-1, 1)$ and the previous local right-tail theorem for $x \geq 8$, this proves the critical-window reduction

$$\text{theBulboaca} - \text{Zayedgamma} - \text{quotientcriticalwindowreduction}$$

and then the source problem

$$\text{theBulboacaZayedgammaquotientfullmonotonicity}.$$

The finite-envelope candidate is not separately promoted: the proof above is a clean analytic ratio-kernel argument rather than the finite rational cover specified in that candidate. The diagnostic obstruction-map candidate is also not needed after the single-crossing proof. $\square$

Proposition 14.7 (Keady self bijection inverse CM negative example). *The function $f(x) = x^{-1} + 100e^{-x}$ is a completely monotone decreasing bijection $(0, \infty) \rightarrow (0, \infty)$, but f^{-1} is not completely monotone.*

Proof. This proof is the component of Proposition 14.8 corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. $\square$

Proposition 14.8 (Keady inverse third derivative sign certificate). *For $f(x) = x^{-1} + 100e^{-x}$ and $g = f^{-1}$, the inverse-derivative certificate gives $g'''(f(1/8)) > 0$.*

Proof. There exists a completely monotone decreasing bijection

$$f : (0, \infty) \rightarrow (0, \infty)$$

whose inverse is not completely monotone. One explicit example is

$$f(x) = \frac{1}{x} + 100e^{-x}.$$

For every $n \geq 0$,

$$(-1)^n f^{(n)}(x) = \frac{n!}{x^{n+1}} + 100e^{-x} > 0.$$

Thus f is strictly completely monotone. Also

$$f'(x) = -x^{-2} - 100e^{-x} < 0,$$

so f is strictly decreasing. Finally,

$$\lim_{x \downarrow 0} f(x) = \infty, \quad \lim_{x \rightarrow \infty} f(x) = 0.$$

Therefore f is a decreasing bijection from $(0, \infty)$ onto $(0, \infty)$.

Let $g = f^{-1}$ and write $y = f(x)$. Inverse differentiation gives

$$g'''(f(x)) = \frac{3(f''(x))^2 - f'''(x)f'(x)}{(f'(x))^5}.$$

For $f_a(x) = x^{-1} + ae^{-x}$, define

$$N_a(x) = 3(f_a''(x))^2 - f_a'''(x)f_a'(x).$$

A direct calculation gives

$$N_a(x) = \frac{6}{x^6} + ae^{-x} \left(-\frac{6}{x^4} + \frac{12}{x^3} - \frac{1}{x^2} \right) + 2a^2e^{-2x}.$$

At $a = 100$ and $x_0 = 1/8$,

$$N_{100}(1/8) = 1,572,864 - 1,849,600e^{-1/8} + 20,000e^{-1/4}.$$

Using $e^{-1/8} > 7/8$ and $e^{-1/4} < 1$,

$$N_{100}(1/8) < 1,572,864 - 1,849,600 \cdot \frac{7}{8} + 20,000 = -25,536 < 0.$$

Since $f'(1/8) < 0$, the denominator $(f'(1/8))^5$ is negative. Hence

$$g'''(f(1/8)) = \frac{N_{100}(1/8)}{(f'(1/8))^5} > 0.$$

A completely monotone function must satisfy $(-1)^3 g''' \geq 0$, i.e. $g''' \leq 0$. Therefore $g = f^{-1}$ is not completely monotone. $\square$

Counterexample 14.9 (APP-0035: Keady self-bijection inverse-CM question is negative). Keady Question 3 has a negative answer for the self-range case: there is a completely monotone decreasing bijection $(0, \infty) \rightarrow (0, \infty)$ whose inverse is not completely monotone.

This resolves the source problem recorded as **APP-0035** [33].

Proof. We use Proposition 14.8 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

There exists a completely monotone decreasing bijection

$$f : (0, \infty) \rightarrow (0, \infty)$$

whose inverse is not completely monotone. One explicit example is

$$f(x) = \frac{1}{x} + 100e^{-x}.$$

For every $n \geq 0$,

$$(-1)^n f^{(n)}(x) = \frac{n!}{x^{n+1}} + 100e^{-x} > 0.$$

Thus f is strictly completely monotone. Also

$$f'(x) = -x^{-2} - 100e^{-x} < 0,$$

so f is strictly decreasing. Finally,

$$\lim_{x \downarrow 0} f(x) = \infty, \quad \lim_{x \rightarrow \infty} f(x) = 0.$$

Therefore f is a decreasing bijection from $(0, \infty)$ onto $(0, \infty)$.

Let $g = f^{-1}$ and write $y = f(x)$. Inverse differentiation gives

$$g'''(f(x)) = \frac{3(f''(x))^2 - f'''(x)f'(x)}{(f'(x))^5}.$$

For $f_a(x) = x^{-1} + ae^{-x}$, define

$$N_a(x) = 3(f_a''(x))^2 - f_a'''(x)f_a'(x).$$

A direct calculation gives

$$N_a(x) = \frac{6}{x^6} + ae^{-x} \left(-\frac{6}{x^4} + \frac{12}{x^3} - \frac{1}{x^2} \right) + 2a^2e^{-2x}.$$

At $a = 100$ and $x_0 = 1/8$,

$$N_{100}(1/8) = 1,572,864 - 1,849,600e^{-1/8} + 20,000e^{-1/4}.$$

Using $e^{-1/8} > 7/8$ and $e^{-1/4} < 1$,

$$N_{100}(1/8) < 1,572,864 - 1,849,600 \cdot \frac{7}{8} + 20,000 = -25,536 < 0.$$

Since $f'(1/8) < 0$, the denominator $(f'(1/8))^5$ is negative. Hence

$$g'''(f(1/8)) = \frac{N_{100}(1/8)}{(f'(1/8))^5} > 0.$$

A completely monotone function must satisfy $(-1)^3 g''' \geq 0$, i.e. $g''' \leq 0$. Therefore $g = f^{-1}$ is not completely monotone. $\square$

Counterexample 14.10 (APP-0036: Baskakov even-line complete-monotonicity fails at $r = 8$). Abel–Gawronski–Neuschel’s even-power Baskakov complete-monotonicity conjecture is false; it fails at $\alpha = 1$, $r = 8$.

This resolves the source problem recorded as **APP-0036** [25].

Proof. We use Proposition 8.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Set

$$D_m(x) = (1+x)^{2m} - x^{2m}.$$

The roots are obtained from

$$\frac{1+x}{x} = \omega_k, \quad \omega_k = e^{\pi i k / m}, \quad k = 1, \dots, 2m-1.$$

Thus

$$\lambda_{k,m} = \frac{1}{\omega_k - 1} = -\frac{1}{2} - \frac{i}{2} \cot \frac{\pi k}{2m}.$$

At this root,

$$D'_m(\lambda_{k,m}) = 2m\lambda_{k,m}^{2m-1}(\omega_k^{-1} - 1),$$

and direct simplification gives the real residue

$$R_{k,m} = \frac{1}{D'_m(\lambda_{k,m})} = \frac{(-1)^{m+k}}{2m} \left(2 \sin \frac{\pi k}{2m} \right)^{2m-2}.$$

Consequently the inverse-Laplace density is

$$\rho_m(t) = \sum_{k=1}^{2m-1} R_{k,m} e^{\lambda_{k,m} t}.$$

Pairing conjugate roots gives

$$\rho_m(t) = \frac{e^{-t/2}}{2m} \left[2^{2m-2} + 2 \sum_{k=1}^{m-1} (-1)^{m+k} \left(2 \sin \frac{\pi k}{2m} \right)^{2m-2} \cos \left(\frac{t}{2} \cot \frac{\pi k}{2m} \right) \right].$$

This proves the finite trigonometric density formula used by the diagnostic candidate.

For $m = 2$, this gives

$$\rho_2(t) = e^{-t/2} \left(1 - \cos \frac{t}{2} \right) = 2e^{-t/2} \sin^2 \frac{t}{4} \geq 0,$$

recovering the previously admitted $r = 4, \alpha = 1$ seed.

For $m = 3$,

$$\rho_3(t) = \frac{e^{-t/2}}{6} \left[16 + 2 \cos \left(\frac{\sqrt{3}t}{2} \right) - 18 \cos \left(\frac{t}{2\sqrt{3}} \right) \right].$$

Set $u = t/(2\sqrt{3})$. Since $\cos(3u) = 4 \cos^3 u - 3 \cos u$,

$$\rho_3(t) = \frac{4}{3} e^{-t/2} (1 - \cos u)^2 (2 + \cos u) \geq 0.$$

Thus $r = 6, \alpha = 1$ is completely monotone by an explicit positive density.

For $m = 4$, write $h_4(t) = e^{t/2} \rho_4(t)$. The density formula gives

$$h_4(t) = 8 + 2 \cos \frac{t}{2} - \frac{(2 - \sqrt{2})^3}{4} \cos \left(\frac{1 + \sqrt{2}}{2} t \right) - \frac{(2 + \sqrt{2})^3}{4} \cos \left(\frac{\sqrt{2} - 1}{2} t \right).$$

At $t = 10\pi$,

$$\cos(t/2) = \cos(5\pi) = -1,$$

and both other cosine terms equal $-\cos(5\pi\sqrt{2})$. Since

$$(2 - \sqrt{2})^3 + (2 + \sqrt{2})^3 = 40,$$

we obtain

$$h_4(10\pi) = 6 + 10 \cos(5\pi\sqrt{2}).$$

Now

$$0 < 5\sqrt{2} - 7 < \frac{1}{4}$$

because $7/5 < \sqrt{2} < 29/20$. Hence, with $\delta = 5\sqrt{2} - 7$,

$$\cos(5\pi\sqrt{2}) = \cos(7\pi + \pi\delta) = -\cos(\pi\delta) < -\cos\frac{\pi}{4} = -\frac{\sqrt{2}}{2} < -\frac{3}{5}.$$

Therefore

$$h_4(10\pi) < 6 + 10\left(-\frac{3}{5}\right) = 0,$$

and

$$\rho_4(10\pi) = e^{-5\pi}h_4(10\pi) < 0.$$

Since the inverse-Laplace transform is unique for measures on $[0, \infty)$, $f_1^{[8]}(x) = 1/((1+x)^8 - x^8)$ cannot be completely monotone. $\square$

Counterexample 14.11 (APP-0048: BMR tau-Gauss ordinary concavity is false). The Bansal–Mehrez–Raina tau-Gauss source function $a \mapsto {}_2\phi_1^\tau(a, c-a; c; z)$ is not concave in general on $(0, c)$. In the subcase $c = 2$, $\tau = 1$, and $z = 1 - e^{-4}$, one has $F''(a, z) > 0$ for all sufficiently small $a > 0$.

This resolves the source problem recorded as **APP-0048** [39].

Proof. We use Proposition 8.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

For $0 < z < 1$, define probability weights

$$w_k(a, z) = \frac{A_k(a)z^k/k!}{F(a, z)}.$$

Let

$$g_k(a) = \partial_a \log A_k(a), \quad h_k(a) = \partial_a^2 \log A_k(a).$$

Then direct differentiation of the log-sum gives

$$\partial_a^2 \log F(a, z) = \sum_{k \geq 0} w_k(a, z) h_k(a) + \text{Var}_{w(a, z)}(g_k(a)).$$

Thus global log-concavity is equivalent to the variance-domination inequality

$$\text{Var}_{w(a, z)}(g_k(a)) \leq \sum_{k \geq 0} w_k(a, z) (-h_k(a))$$

for all $0 < a < c$, $c, \tau > 0$, and $0 < z < 1$. This is the sharp remaining gate; coefficientwise log-concavity alone is insufficient.

Take the classical subcase

$$c = 2, \quad \tau = 1, \quad z = 1 - e^{-4}.$$

Then

$$F(a, z) = \sum_{k=0}^{\infty} \frac{(a)_k (2-a)_k}{(2)_k} \frac{z^k}{k!}.$$

For $k \geq 1$, as $a \downarrow 0$,

$$\frac{(a)_k (2-a)_k}{(2)_k k!} = \frac{a}{k} + a^2 \left(\frac{1}{k} - \frac{1}{k^2} - \frac{1}{k(k+1)} \right) + O(a^3),$$

with locally uniform summability for fixed $0 < z < 1$. Hence

$$F(a, z) = 1 + aL(z) + a^2M(z) + O(a^3),$$

where

$$L(z) = \sum_{k \geq 1} \frac{z^k}{k} = -\log(1 - z)$$

and

$$M(z) = \sum_{k \geq 1} z^k \left(\frac{1}{k} - \frac{1}{k^2} - \frac{1}{k(k+1)} \right) = \frac{L(z)}{z} - \text{Li}_2(z) - 1.$$

For $z = 1 - e^{-4}$, $L(z) = 4$, $L(z)/z > 4$, and $\text{Li}_2(z) < \pi^2/6 < 2$. Therefore

$$M(z) > 4 - 2 - 1 = 1 > 0.$$

It follows that

$$F''(a, z) \rightarrow 2M(z) > 0 \quad (a \downarrow 0),$$

so $F''(a, z) > 0$ for all sufficiently small $a > 0$. Therefore the source function is not concave in a in general. $\square$

Corollary 14.12 (APP-0049: Baricz arithmetic/arithmetic zero-balanced hypergeometric threshold). *For $c > 0$, define $F_a(r) = {}_2F_1(a, c - a; c; r)$ on $0 < a < c$, $0 < r < 1$. The uniform arithmetic/arithmetic inequality $(F_{a_1}(r) + F_{a_2}(r))/2 \leq F_{(a_1+a_2)/2}(r)$ for all $a_1, a_2 \in (0, c)$ and all $r \in (0, 1)$ holds if and only if $0 < c \leq 1$.*

*This resolves the source problem recorded as **APP-0049** [40].*

Proof. We use Proposition 8.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

For $c > 0$, set

$$F_a(r) = {}_2F_1(a, c - a; c; r), \quad 0 < a < c, \quad 0 < r < 1.$$

The Baricz arithmetic/arithmetic source slice asks when

$$\frac{F_{a_1}(r) + F_{a_2}(r)}{2} \leq F_{(a_1+a_2)/2}(r)$$

holds for all $a_1, a_2 \in (0, c)$ and all $r \in (0, 1)$.

The answer is exactly $0 < c \leq 1$.

The existing local node the Hypergeometric Fa geometric mean logconvex slice imports Baricz's theorem:

$$\sqrt{F_{a_1}(r)F_{a_2}(r)} \leq \frac{F_{a_1}(r) + F_{a_2}(r)}{2} \leq F_{(a_1+a_2)/2}(r)$$

for $0 < c \leq 1$, $a_1, a_2 \in (0, c)$, and $0 < r < 1$. This proves the A/A inequality on that range.

Fix $c > 1$. For $0 < r < 1$,

$$F_a(r) = \sum_{k=0}^{\infty} \frac{(a)_k (c-a)_k}{(c)_k} \frac{r^k}{k!}.$$

For $k \geq 1$, as $a \rightarrow 0$,

$$(a)_k = a(k-1)! (1 + aH_{k-1} + O(a^2))$$

and

$$\frac{(c-a)_k}{(c)_k} = 1 - a \sum_{j=0}^{k-1} \frac{1}{c+j} + O(a^2).$$

Hence

$$\frac{(a)_k(c-a)_k}{(c)_k k!} = \frac{a}{k} + \frac{a^2}{k} \left(H_{k-1} - \sum_{j=0}^{k-1} \frac{1}{c+j} \right) + O(a^3).$$

For fixed $r < 1$, normal convergence of the differentiated hypergeometric series justifies termwise expansion, so

$$F_a(r) = 1 + aL(r) + a^2 M_c(r) + O(a^3),$$

where

$$L(r) = \sum_{k \geq 1} \frac{r^k}{k} = -\log(1-r)$$

and

$$M_c(r) = \sum_{k \geq 1} \left(H_{k-1} - \sum_{j=0}^{k-1} \frac{1}{c+j} \right) \frac{r^k}{k}.$$

The bracket is

$$B_{c,k} = \psi(k) + \gamma - \psi(c+k) + \psi(c),$$

so

$$B_{c,k} \rightarrow \gamma + \psi(c).$$

Because $c > 1$, $\psi(c) > \psi(1) = -\gamma$, and therefore $\gamma + \psi(c) > 0$. Thus $B_{c,k} \geq \eta > 0$ for all sufficiently large k , and

$$M_c(r) \rightarrow +\infty \quad (r \uparrow 1).$$

Choose $r_0 \in (0, 1)$ with $M_c(r_0) > 0$. Then

$$\partial_a^2 F_a(r_0) \rightarrow 2M_c(r_0) > 0 \quad (a \downarrow 0).$$

For some sufficiently small interior $a_0 \in (0, c)$, $\partial_a^2 F_{a_0}(r_0) > 0$. By continuity, $\partial_a^2 F_a(r_0) > 0$ on a small interval around a_0 . Taking $h > 0$ small enough that $a_0 \pm h \in (0, c)$, Taylor's theorem gives

$$\frac{F_{a_0-h}(r_0) + F_{a_0+h}(r_0)}{2} > F_{a_0}(r_0).$$

Since $a_0 = ((a_0 - h) + (a_0 + h))/2$, the A/A inequality fails.

Therefore the source A/A inequality holds uniformly if and only if $0 < c \leq 1$. $\square$

Proposition 14.13 (Stolarsky power mean pgt1 asymptotic obstruction). *For $p > 1$, $d \neq 0$, and $y > |d|$, the shifted power mean $G(y) = H_p(y + d, y - d)$ satisfies $G''(y) > 0$. In particular, $G(y) = y + \frac{(p-1)d^2}{2y} + O(y^{-3})$ as $y \rightarrow \infty$, so G' is eventually increasing.*

Proof. Let

$$c = \frac{a+b}{2}, \quad d = \frac{a-b}{2}.$$

Since $a, b > 0$ and $a \neq b$, we have $c > |d| > 0$. Put $y = x + c$. Then on the natural half-line $y > |d|$,

$$F(x) = G(y), \quad G(y) = \left(\frac{(y+d)^p + (y-d)^p}{2} \right)^{1/p}.$$

Translation does not affect the sign pattern of derivatives, so it is enough to show that G is not Bernstein.

For $|t| < 1$, define

$$\phi_p(t) = H_p(1+t, 1-t) = 2^{-1/p} ((1+t)^p + (1-t)^p)^{1/p}.$$

By homogeneity,

$$G(y) = y\phi_p(d/y).$$

Let

$$A = 1+t, \quad B = 1-t, \quad N = A^p + B^p.$$

Then $\phi_p(t) = 2^{-1/p} N^{1/p}$. Differentiating twice gives

$$\phi_p''(t) = 2^{-1/p} (p-1) N^{1/p-2} (N(A^{p-2} + B^{p-2}) - (A^{p-1} - B^{p-1})^2).$$

The bracket simplifies to

$$A^{p-2} B^{p-2} (A+B)^2,$$

because

$$(A^p + B^p)(A^{p-2} + B^{p-2}) - (A^{p-1} - B^{p-1})^2 = A^{p-2} B^{p-2} (A^2 + 2AB + B^2).$$

For $p > 1$ and $|t| < 1$, all factors are positive, so

$$\phi_p''(t) > 0.$$

Since $t = d/y$,

$$G'(y) = \phi_p(t) - t\phi_p'(t)$$

and

$$G''(y) = \frac{t^2}{y} \phi_p''(t) = \frac{d^2}{y^3} \phi_p''(d/y) > 0$$

for every $y > |d|$.

For $|u| < 1$, define

$$A(u) = \frac{(1+u)^p + (1-u)^p}{2}.$$

The odd terms cancel in the binomial expansion, giving

$$A(u) = 1 + \frac{p(p-1)}{2} u^2 + O(u^4) \quad (u \rightarrow 0).$$

Since $z \mapsto z^{1/p}$ is analytic near $z = 1$,

$$A(u)^{1/p} = 1 + \frac{1}{p} \cdot \frac{p(p-1)}{2} u^2 + O(u^4) = 1 + \frac{p-1}{2} u^2 + O(u^4).$$

With $u = d/y$, this yields

$$G(y) = yA(d/y)^{1/p} = y + \frac{(p-1)d^2}{2y} + O(y^{-3}) \quad (y \rightarrow \infty).$$

Differentiating the same analytic expansion gives

$$G'(y) = 1 - \frac{(p-1)d^2}{2y^2} + O(y^{-4}),$$

and hence

$$G''(y) = \frac{(p-1)d^2}{y^3} + O(y^{-5}).$$

Because $p > 1$ and $d \neq 0$, the leading coefficient is positive. This recovers the tail obstruction from the exact curvature certificate above.

If G were a Bernstein function, then G' would be completely monotone. In particular,

$$(G')'(y) = G''(y) \leq 0$$

throughout the interval. This contradicts the positivity of $G''(y)$ proved above.

Thus G , and equivalently F , is not a Bernstein function. $\square$

Counterexample 14.14 (APP-0050: Stolarsky shifted power means with $p > 1$ are not Bernstein).
 not (For some $p > 1$, the shifted power mean $x \mapsto H_p(x+a, x+b)$, where $H_p(a, b) = ((a^p + b^p)/2)^{1/p}$, is a Bernstein function for every positive pair of shifts a, b .)

This resolves the source problem recorded as **APP-0050** [41].

Proof. We use Proposition 14.13 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$c = \frac{a+b}{2}, \quad d = \frac{a-b}{2}.$$

Since $a, b > 0$ and $a \neq b$, we have $c > |d| > 0$. Put $y = x + c$. Then on the natural half-line $y > |d|$,

$$F(x) = G(y), \quad G(y) = \left(\frac{(y+d)^p + (y-d)^p}{2} \right)^{1/p}.$$

Translation does not affect the sign pattern of derivatives, so it is enough to show that G is not Bernstein.

For $|t| < 1$, define

$$\phi_p(t) = H_p(1+t, 1-t) = 2^{-1/p} ((1+t)^p + (1-t)^p)^{1/p}.$$

By homogeneity,

$$G(y) = y\phi_p(d/y).$$

Let

$$A = 1+t, \quad B = 1-t, \quad N = A^p + B^p.$$

Then $\phi_p(t) = 2^{-1/p} N^{1/p}$. Differentiating twice gives

$$\phi_p''(t) = 2^{-1/p} (p-1) N^{1/p-2} (N(A^{p-2} + B^{p-2}) - (A^{p-1} - B^{p-1})^2).$$

The bracket simplifies to

$$A^{p-2} B^{p-2} (A+B)^2,$$

because

$$(A^p + B^p)(A^{p-2} + B^{p-2}) - (A^{p-1} - B^{p-1})^2 = A^{p-2} B^{p-2} (A^2 + 2AB + B^2).$$

For $p > 1$ and $|t| < 1$, all factors are positive, so

$$\phi_p''(t) > 0.$$

Since $t = d/y$,

$$G'(y) = \phi_p(t) - t\phi_p'(t)$$

and

$$G''(y) = \frac{t^2}{y} \phi_p''(t) = \frac{d^2}{y^3} \phi_p''(d/y) > 0$$

for every $y > |d|$.

For $|u| < 1$, define

$$A(u) = \frac{(1+u)^p + (1-u)^p}{2}.$$

The odd terms cancel in the binomial expansion, giving

$$A(u) = 1 + \frac{p(p-1)}{2} u^2 + O(u^4) \quad (u \rightarrow 0).$$

Since $z \mapsto z^{1/p}$ is analytic near $z = 1$,

$$A(u)^{1/p} = 1 + \frac{1}{p} \cdot \frac{p(p-1)}{2} u^2 + O(u^4) = 1 + \frac{p-1}{2} u^2 + O(u^4).$$

With $u = d/y$, this yields

$$G(y) = yA(d/y)^{1/p} = y + \frac{(p-1)d^2}{2y} + O(y^{-3}) \quad (y \rightarrow \infty).$$

Differentiating the same analytic expansion gives

$$G'(y) = 1 - \frac{(p-1)d^2}{2y^2} + O(y^{-4}),$$

and hence

$$G''(y) = \frac{(p-1)d^2}{y^3} + O(y^{-5}).$$

Because $p > 1$ and $d \neq 0$, the leading coefficient is positive. This recovers the tail obstruction from the exact curvature certificate above.

If G were a Bernstein function, then G' would be completely monotone. In particular,

$$(G')'(y) = G''(y) \leq 0$$

throughout the interval. This contradicts the positivity of $G''(y)$ proved above.

Thus G , and equivalently F , is not a Bernstein function. □

15 Endpoint obstruction certificates

This section collects applications in which a universal positivity, convexity, or monotonicity claim is decided by reducing it to a local obstruction. The common method is to isolate a boundary, endpoint, or explicit witness where the proposed sign pattern cannot hold.

Counterexample 15.1 (APP-0042: Modified Bessel square-root log-concavity fails). not(For every $\nu \geq 0$, the function $u \mapsto \sqrt{u} I_\nu(u)$ is strictly log-concave on $(0, \infty)$.)

This resolves the source problem recorded as **APP-0042** [36].

Proof. We use Proposition 8.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$g_\nu(u) = \log(\sqrt{u} I_\nu(u)), \quad r_\nu(u) = \frac{I'_\nu(u)}{I_\nu(u)}.$$

Then

$$g''_\nu(u) = r'_\nu(u) - \frac{1}{2u^2}.$$

The modified Bessel equation

$$u^2 I''_\nu(u) + u I'_\nu(u) - (u^2 + \nu^2) I_\nu(u) = 0$$

gives

$$\frac{I''_\nu(u)}{I_\nu(u)} = 1 + \frac{\nu^2}{u^2} - \frac{r_\nu(u)}{u}.$$

Since

$$r'_\nu(u) = \frac{I''_\nu(u)}{I_\nu(u)} - r_\nu(u)^2,$$

we get

$$g''_\nu(u) = 1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2.$$

Thus the Bessel I Riccati log concavity inequality is exactly the strict log-concavity condition $g''_\nu(u) < 0$.

For $\nu = 0$,

$$r_0(u) = \frac{I'_0(u)}{I_0(u)} = \frac{I_1(u)}{I_0(u)}.$$

At $u = 10$, the Riccati expression is

$$g''_0(10) = \frac{199}{200} - \frac{r_0(10)}{10} - r_0(10)^2.$$

It is enough to prove

$$r_0(10) < \frac{9487}{10000},$$

because

$$\frac{199}{200} - \frac{1}{10} \frac{9487}{10000} - \left(\frac{9487}{10000} \right)^2 = \frac{9831}{100000000} > 0.$$

$$I_0(10) = \sum_{k=0}^{\infty} \frac{25^k}{(k!)^2}, \quad I_1(10) = \sum_{k=0}^{\infty} \frac{5 \cdot 25^k}{k!(k+1)!}.$$

With $N = 20$, set

$$L_0 = \sum_{k=0}^{20} \frac{25^k}{(k!)^2} < I_0(10).$$

For the I_1 tail after $N = 20$, the term ratio is bounded by

$$q = \frac{25}{22 \cdot 23} < 1,$$

so the script computes an exact rational upper bound $U_1 > I_1(10)$ and verifies

$$10000 U_1 < 9487 L_0.$$

Therefore $I_1(10)/I_0(10) < 9487/10000$, and hence

$$g_0''(10) > 0.$$

This proves the Bessel- I log-concavity counterexample to T - I sqrt-log concavity for $\nu > 0$, and the counterexample to T - I Riccati log-concavity inequality. $\square$

16 Determinant and moment-ratio applications

This section develops applications governed by determinant compression and moment-ratio comparisons. The method turns recursive derivative information into a finite or structured positivity problem, so bounds and counterexamples can be read from a smaller normal form.

Corollary 16.1 (APP-0038: Ma-Weigert derivative-sign regions form a chain). *In the Ma-Weigert log-function class $\mathcal{F}_{1,n}$, the derivative-sign regions $D_k = \{f : L^k f \geq 0\}$, with $L = -d/dx$, form a descending chain: $D_{k+1} \subseteq D_k$ for every k, n .*

*This resolves the source problem recorded as **APP-0038** [11].*

Proof. We use Proposition 11.32 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

We first prove that for each $k \geq 0$,

$$L^k f(x) = x^{-k-1} p_k(\log x)$$

for some polynomial p_k .

For $k = 0$, this is the definition with $p_0 = p$. Suppose

$$L^k f(x) = x^{-k-1} p_k(\log x).$$

Then

$$\begin{aligned} L^{k+1} f(x) &= -\frac{d}{dx} \left(x^{-k-1} p_k(\log x) \right) \\ &= x^{-k-2} \left((k+1) p_k(\log x) - p'_k(\log x) \right). \end{aligned}$$

Thus $p_{k+1}(y) = (k+1)p_k(y) - p'_k(y)$, which is again a polynomial.

Since every polynomial in $\log x$ grows slower than every positive power of x ,

$$x^{-k-1} p_k(\log x) \rightarrow 0 \quad (x \rightarrow \infty).$$

Therefore

$$\lim_{x \rightarrow \infty} L^k f(x) = 0$$

for every $k \geq 0$.

Assume $f \in D_{k+1}$. Then

$$L^{k+1} f(x) \geq 0$$

on $(0, \infty)$. Put

$$g(x) = L^k f(x).$$

By definition of L ,

$$-g'(x) = L^{k+1}f(x) \geq 0.$$

Using $g(x) \rightarrow 0$ as $x \rightarrow \infty$, integrate from x to R :

$$g(x) - g(R) = \int_x^R L^{k+1}f(t) dt.$$

Letting $R \rightarrow \infty$ gives

$$L^k f(x) = g(x) = \int_x^\infty L^{k+1}f(t) dt \geq 0.$$

Hence $f \in D_k$. Therefore

$$D_{k+1} \subseteq D_k$$

for all $k \geq 0$ and all n . □

17 Dual-certificate and algebraic extremal applications

This section contains applications in which an algebraic extremum is identified by a dual certificate. The structural principle supplies the extremal norm or moment identity, and the source-specific computation verifies that the candidate object attains the certified value.

Definition 17.1 (Finite dimensional l1 dual certificate language). Let V be a real vector space and let $\mathcal{A} \subset V$ be a finite family. The elements of $\mathcal{A}$ are the coordinate atoms. A primal representation of $v \in V$ is a finite expansion $v = \sum_{a \in \mathcal{A}} c_a a$, and its ℓ^1 -norm is $\sum_{a \in \mathcal{A}} |c_a|$. A dual certificate for the lower bound M is a linear functional $\Lambda : V \rightarrow \mathbb{R}$ such that $\Lambda(v) = M$ and $|\Lambda(a)| \leq 1$ for every $a \in \mathcal{A}$.

Proposition 17.1 (Finite-dimensional dual certificate for exact l1 representation norm). *In the finite-dimensional ℓ^1 dual-certificate setting, if v has a primal representation of norm M and a dual certificate Λ with $\Lambda(v) = M$, then the exact minimal ℓ^1 -representation norm of v by the coordinate atoms is M .*

Proof. Let V be a real vector space, let $\mathcal{A} \subset V$ be a finite coordinate family, and let $v \in V$. Suppose there is a representation

$$v = \sum_{a \in \mathcal{A}} c_a a, \quad \sum_{a \in \mathcal{A}} |c_a| = M,$$

and a linear functional $\Lambda : V \rightarrow \mathbb{R}$ such that

$$\Lambda(v) = M, \quad |\Lambda(a)| \leq 1 \quad (a \in \mathcal{A}).$$

Then the least ℓ^1 -coefficient norm of a representation of v by elements of $\mathcal{A}$ is exactly M .

For any other representation $v = \sum_{a \in \mathcal{A}} b_a a$, linearity and the dual bound give

$$M = \Lambda(v) = \sum_{a \in \mathcal{A}} b_a \Lambda(a) \leq \sum_{a \in \mathcal{A}} |b_a| |\Lambda(a)| \leq \sum_{a \in \mathcal{A}} |b_a|.$$

Thus every representation has norm at least M . The displayed primal representation has norm M , so the minimum is M . □

Definition 17.2 (Weyl algebra balanced words and coefficient norm). The real Weyl algebra A_1 is generated by X and D with relation $DX - XD = 1$. A balanced word of degree (n, n) is a word containing exactly n copies of D and n copies of X . The coefficient ℓ^1 -norm of a finite word expansion $\sum_w a_w w$ is $\sum_w |a_w|$.

Corollary 17.2 (APP-0046: Tao–Sawin finite Weyl l1 minimum is exact). *In the Weyl algebra $DX - XD$ equals one, the least l1 coefficient norm of a balanced degree (n, n) word combination equal to one is two to the n divided by n factorial.*

$$L_n = \min \left\{ \sum_w |a_w| : \sum_w a_w w = 1, w \text{ has } n \text{ copies of } D \text{ and } n \text{ copies of } X \right\} = \frac{2^n}{n!}.$$

*This resolves the source problem recorded as **APP-0046** [37].*

Proof. We use Proposition 17.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let A_1 be the Weyl algebra over a field of characteristic 0, generated by X, D with

$$DX - XD = 1.$$

Let W_n be the set of words containing exactly n copies of D and n copies of X . For

$$P = \sum_{w \in W_n} a_w w$$

define

$$\|P\|_1 = \sum_{w \in W_n} |a_w|.$$

Let

$$L_n = \min \left\{ \|P\|_1 : \sum_{w \in W_n} a_w w = 1 \text{ in } A_1 \right\}.$$

Then

$$L_n = \frac{2^n}{n!}.$$

Set

$$P_n = \frac{1}{n!} \text{ad}_D^n(X^n).$$

Since $\text{ad}_D(X) = 1$, repeated derivation gives

$$\text{ad}_D^n(X^n) = n!.$$

Also

$$\text{ad}_D^n(X^n) = \sum_{a=0}^n (-1)^a \binom{n}{a} D^{n-a} X^n D^a.$$

Thus $P_n = 1$ in A_1 , and

$$\|P_n\|_1 = \frac{1}{n!} \sum_{a=0}^n \binom{n}{a} = \frac{2^n}{n!}.$$

Therefore $L_n \leq 2^n/n!$.

For $q \in \mathbb{C}$, define Φ_q on normal-ordered monomials by

$$\Phi_q(X^r D^s) = \begin{cases} q^r r!, & r = s, \\ 0, & r \neq s. \end{cases}$$

The functional relevant to the lower bound is

$$\Lambda_n = \Phi_{-1/2},$$

so

$$\Lambda_n(X^i D^i) = (-1)^i 2^{-i} i!.$$

In particular, $\Lambda_n(1) = 1$.

For letters $a_i \in \{X, D\}$,

$$\Phi_q(a_1 \cdots a_m) = \sum_{\pi} \prod_{\{i,j\} \in \pi, i < j} C_q(a_i, a_j),$$

where π runs over pairings pairing each X with a D , and

$$C_q(X, D) = q, \quad C_q(D, X) = q + 1, \quad C_q(X, X) = C_q(D, D) = 0.$$

To prove this, use formal exponentials. Since

$$e^{uD} e^{vX} = e^{uv} e^{vX} e^{uD},$$

normal ordering gives

$$e^{t_1 a_1} \cdots e^{t_m a_m} = \exp \left(\sum_{\substack{i < j \\ a_i = D, a_j = X}} t_i t_j \right) e^{UX} e^{VD},$$

where

$$U = \sum_{i: a_i = X} t_i, \quad V = \sum_{i: a_i = D} t_i.$$

Applying Φ_q gives

$$\Phi_q(e^{UX} e^{VD}) = e^{qUV}.$$

Hence

$$\Phi_q(e^{t_1 a_1} \cdots e^{t_m a_m}) = \exp \left(\sum_{i < j} C_q(a_i, a_j) t_i t_j \right).$$

Extracting the coefficient of $t_1 \cdots t_m$ gives the Wick pairing formula.

Take $q = -1/2$. Then

$$C_{-1/2}(X, D) = -\frac{1}{2}, \quad C_{-1/2}(D, X) = \frac{1}{2}.$$

Every nonzero contraction has absolute value $1/2$. If $w \in W_n$, then pairings of the n X 's with the n D 's are in bijection with permutations, so there are exactly $n!$ possible pairings. Therefore

$$|\Lambda_n(w)| = |\Phi_{-1/2}(w)| \leq n! \left(\frac{1}{2} \right)^n = \frac{n!}{2^n}.$$

Now let

$$P = \sum_{w \in W_n} a_w w$$

and suppose $P = 1$ in A_1 . Applying Λ_n ,

$$1 = |\Lambda_n(1)| = |\Lambda_n(P)| \leq \sum_{w \in W_n} |a_w| |\Lambda_n(w)| \leq \frac{n!}{2^n} \sum_{w \in W_n} |a_w|.$$

Thus

$$\|P\|_1 \geq \frac{2^n}{n!}.$$

Combining this with the upper bound proves

$$L_n = \frac{2^n}{n!}.$$

This solves the finite homogeneous Weyl algebra ℓ^1 coefficient model problem from the Tao-Sawin comment thread. It does not solve the full operator-norm problem for commutators close to the identity. $\square$

18 Gamma and polygamma applications

This section contains applications whose main mechanism is gamma-function curvature, polygamma sign control, or beta-window reduction. The recurring strategy is to rewrite parameter inequalities as logarithmic-derivative or endpoint statements and then propagate the resulting sign information.

Counterexample 18.1 (APP-0037: Du-Wang h_3 is increasing exactly on $\frac{1}{2} \leq a \leq 1$). For $0 < a < 2$, Du-Wang's function h_3 is increasing on $(0, \infty)$ exactly for $1/2 \leq a \leq 1$, and is not monotone for $0 < a < 1/2$ or $1 < a < 2$.

This resolves the source problem recorded as **APP-0037** [24].

Proof. We use Proposition 11.28 and Proposition 11.27 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The middle-window monotonicity theorem on $1/2 \leq a \leq 1$ and the outer-window nonmonotonicity theorem on $0 < a < 1/2$ and $1 < a < 2$ partition the domain, and together they give exactly the source classification. $\square$

19 Endpoint obstruction certificates

This section collects applications in which a universal positivity, convexity, or monotonicity claim is decided by reducing it to a local obstruction. The common method is to isolate a boundary, endpoint, or explicit witness where the proposed sign pattern cannot hold.

Definition 19.1 (Hyperbolically completely monotone density). A positive density f on $(0, \infty)$ is hyperbolically completely monotone if, for each fixed $u > 0$, the function $v \mapsto f(uv)f(u/v)$ is completely monotone as a function of $v + v^{-1}$, in the usual hyperbolic variable.

Proposition 19.1 (BPV noncentral chi square HCM leading Bell obstruction chain). *For the noncentral chi-square HCM problem, the leading small- u HCM sign constraints are generated by the Bell recurrence*

$$B_0 = 1, \quad B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_{n-k} c_{k+1}.$$

With

$$c_1 = \frac{\lambda/\mu - 1}{2}, \quad c_j = j! b_j(\mu) \lambda^j \quad (j \geq 2),$$

and

$$\log \sum_{j \geq 0} \frac{y^j}{4^j j! (\mu/2)_j} = \sum_{j \geq 1} b_j(\mu) y^j,$$

the HCM necessary condition is $(-1)^n B_n \geq 0$ for every $n \geq 1$.

Proof. The source asks for the optimal parameter range in which

$$\chi_{\mu,\lambda}(x) = \frac{1}{2} e^{-(x+\lambda)/2} \left(\frac{x}{\lambda} \right)^{\mu/4-1/2} I_{\mu/2-1}(\sqrt{\lambda x})$$

is hyperbolically completely monotone.

The local small- u hyperbolic-product expansion is

$$\log\{\chi_{\mu,\lambda}(uv)\chi_{\mu,\lambda}(u/v)\} = C_u + \frac{u}{2} \left(\frac{\lambda}{\mu} - 1 \right) w - \frac{\lambda^2 u^2}{4\mu^2(\mu+2)}(w^2 - 2) + O(u^3), \quad w = v + v^{-1}.$$

Consequences:

$$\lambda > \mu \implies \chi_{\mu,\lambda} \notin HCM,$$

and

$$(\lambda - \mu)^2 < \frac{2\lambda^2}{\mu + 2} \implies \chi_{\mu,\lambda} \notin HCM.$$

Equivalently, inside $0 < \lambda \leq \mu$, the second obstruction excludes

$$\frac{\lambda}{\mu} > \frac{1}{1 + \sqrt{2/(\mu+2)}}.$$

Remaining frontier:

$$0 < \lambda \leq \mu, \quad (\lambda - \mu)^2 \geq \frac{2\lambda^2}{\mu + 2}.$$

This note is partial theory growth only. Do not stage as a solved application unless the full optimal HCM range is proved or the source problem is explicitly narrowed. $\square$

Counterexample 19.2 (APP-0047: BPV noncentral chi-square HCM range has no positive non-centrality). For every $\mu > 0$ and every positive noncentrality $\lambda > 0$, the noncentral chi-square density $\chi_{\mu,\lambda}$ is not hyperbolically completely monotone. Equivalently, the BPV optimal HCM range for positive noncentrality is empty; if the central case $\lambda = 0$ is adjoined, the remaining HCM range is exactly the central gamma line.

This resolves the source problem recorded as **APP-0047** [38].

Proof. We use Proposition 19.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let $a = \mu/2 > 0$. The complete Bell generating function is

$$\sum_{n \geq 0} \frac{B_n z^n}{n!} = \exp \left(\sum_{j \geq 1} \frac{c_j z^j}{j!} \right) = e^{-z/2} S_\mu(\lambda z).$$

Indeed $b_1(\mu) = 1/(2\mu)$, so the $j = 1$ term of $\log S_\mu(\lambda z)$ contributes $\lambda z/(2\mu)$, and the exponential factor $e^{-z/2}$ supplies the remaining $-z/2$.

Define

$$A_n = \frac{(-1)^n B_n}{n!}.$$

Then

$$\sum_{n \geq 0} A_n t^n = e^{t/2} S_\mu(-\lambda t) = e^{t/2} {}_0F_1 \left(; a; -\frac{\lambda t}{4} \right).$$

Equivalently,

$$A_n = \frac{2^{-n}}{(a)_n} L_n^{a-1} \left(\frac{\lambda}{2} \right),$$

so the leading HCM signs are exactly the fixed-argument Laguerre signs.

Use the classical identity

$${}_0F_1 \left(; a; -\frac{x^2}{4} \right) = \Gamma(a) \left(\frac{x}{2} \right)^{1-a} J_{a-1}(x).$$

Since $a > 0$, the Bessel function J_{a-1} has a positive real zero $j_{a-1,1} > 0$. For $\lambda > 0$, set

$$t_0 = \frac{j_{a-1,1}^2}{\lambda} > 0.$$

Then

$$e^{t_0/2} {}_0F_1 \left(; a; -\frac{\lambda t_0}{4} \right) = 0.$$

If every leading HCM sign were nonnegative, then all coefficients A_n would be nonnegative and $A_0 = 1$, forcing

$$\sum_{n \geq 0} A_n t_0^n > 0,$$

contradicting the displayed zero. Therefore, for every $\mu > 0$ and $\lambda > 0$, there is an integer $n \geq 1$ such that

$$(-1)^n B_n < 0.$$

The leading expansion gives, for any fixed compact w -window,

$$\frac{1}{H_u(w)} \frac{d^n}{dw^n} H_u(w) = B_n u^n + O(u^{n+1}) \quad (u \downarrow 0).$$

Choosing $u > 0$ sufficiently small yields

$$(-1)^n \frac{d^n}{dw^n} H_u(w) < 0,$$

which violates complete monotonicity of $w \mapsto H_u(w)$.

Conclusion. For every $\mu > 0$ and every positive noncentrality $\lambda > 0$,

$$\chi_{\mu,\lambda} \notin HCM.$$

Thus the BPV optimal HCM range for positive noncentrality is empty. If one adjoins the central case $\lambda = 0$, the remaining density is a gamma density and is HCM; hence the extended closed range is exactly the central line $\lambda = 0$.

Strict APP status: candidate for the next APP slot after APP-0046 because the source explicitly asks the optimal range, the local theorem answers it, and the result is not among public APP-0001–APP-0045. $\square$

20 Conclusion

The theory developed here is organized around a single principle: positivity should be transported only through mechanisms that preserve the relevant analytic structure. Zeta-law compression turns Dirichlet sums into probabilistic identities, special-function curvature is tested through Laplace kernels, and application-level claims are resolved by moving between Stieltjes, Bernstein, determinant, ratio, and endpoint-obstruction forms.

This perspective gives the paper its hierarchy. Definitions fix the common objects, lemmas and propositions isolate reusable transformations, theorems state the main transport mechanisms, corollaries and counterexamples record their consequences, and the application ledger points each source problem to the result that resolves it. The result is a connected theory of positivity transport rather than a list of unrelated solved examples.

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